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AIRPORT MASTER PLAN

FOR

COTTAGE GROVE STATE AIRPORT

COTTAGE GROVE, OREGON



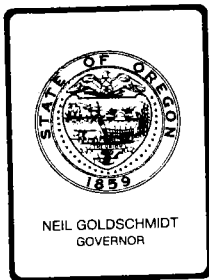
Prepared For The
STATE OF OREGON
DIVISION OF AERONAUTICS
SALEM, OREGON

BY


FORESITE GROUP

OR T/Ae8 .2A17

Airport master plan for
Cottage Grove State Airport



State of Oregon Aeronautics Division

3040 25th STREET S.E., SALEM, OREGON 97310 PHONE 378-4880

July 12, 1988

Don Larson, Aviation Planner
Federal Aviation Administration
Seattle Airports District Office, Boeing Field
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COTTAGE GROVE STATE AIRPORT Final Master Plan

Enclosed please find the final copies of the Cottage Grove State Airport Master Plan and appropriate drawings. In addition, please note the below corrections to the plan:

- Page 3-9: Please strike the last sentence of the first paragraph.
- Page 5-7: We have examined the feasibility of extending the runway to 3,700 feet and elect to keep the extension in mind as a future development alternative.
- Page 6-5: The third sentence of the last paragraph should read:
"with the north end positioned to the west approximately 100 feet from the existing runway end."

We are pleased to have completed this project and look forward to receiving FAA approval of the plan in the near future.

Sincerely,


PAUL E. BURKET
Aeronautics Administrator

PEB:FNM:fnm

cc: Jim Hawley, President, Foresite Group, Aviation Planners

AIRPORT MASTER PLAN

for

COTTAGE GROVE STATE AIRPORT

Cottage Grove, Oregon

April 1988

prepared for

**STATE OF OREGON
AERONAUTICS DIVISION**

by

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Airport master plan for
Cottage Grove State Airport

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REQUIRED STATEMENT
Cottage Grove State Airport Master Plan

The following statement is provided as required by Paragraph 429.a of Federal Aviation Order 5100.38, Airport Improvement Program (AIP) Handbook:

"The preparation of this document was financed in part through a planning grant from the Federal Aviation Administration as provided under Section 505 of the Airport and Airway Improvement Act of 1982. The contents do not necessarily reflect the official views or policy of the FAA. Acceptance of this report by the FAA does not in any way constitute a commitment on the part of the United States to participate in any development depicted therein nor does it indicate that the proposed development is environmentally acceptable in accordance with appropriate public laws."

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SOURCES OF DOCUMENTATION

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INTRODUCTION

Chapter 1 INTRODUCTION

The State of Oregon Aeronautics Division has undertaken this Master Plan Study for Cottage Grove State Airport to determine the existing and future role of the airport, and to provide the needed direction for its future development. This Plan will evaluate and recommend development of facilities that will be required to meet updated forecast demands as well as to establish procedures and priorities for implementation. The program derived from this study should serve as a guide in the overall management and operation of the airport. By providing a phased outline of indicated needs and giving responsible officials advance notice of those needs to aid in future scheduling and budgeting, this Plan will also facilitate orderly and timely development of the Airport.

The study project is being prepared by the airport consulting firm of FORESITE Group, Inc., of Corvallis, Oregon.

The 1974 Airport Layout Plan (ALP) had been revised in 1979 to reflect the additions of a shop hangar, the westerly taxiway and improvements in 1981, and new hangar construction in 1984. Changing conditions, limited availability of State-owned land for future airport uses, and development pressure upon adjacent lands necessitate the careful analysis of existing conditions and forecasted demand that is inherent in the Master Plan Study process.

OBJECTIVES

The primary objectives of this Master Planning process are to evaluate the existing Cottage Grove State Airport and determine the future actions required to maintain an adequate and reliable level of service for the area's aviation potential. The completed Plan will yield a safe, efficient, and economical air transportation facility. To accomplish these objectives, the Study will:

- o Inventory and analyze data pertinent to Cottage Grove State Airport and the surrounding area.
- o Analyze general socioeconomic factors and evaluate the area's potential to generate aviation activity.
- o Forecast levels of aviation activity through the year 2007.

- o Identify and analyze alternatives to meet the airport's needs.
- o Update the Airport Layout Plan and generate other plan drawings depicting facilities, airspace, and other factors.
- o Schedule priorities, phase proposed development, and estimate costs.

In addition to the objectives established by the State of Oregon Aeronautics Division and requirements of the Federal Aviation Administration (FAA), the following additional objectives of this Study are as indicated:

- o Facilitate community understanding of the airport.
- o Help formulate and update community aviation goals.
- o Document the critical planning assumptions and analysis to facilitate long-term utility of the final Plan.
- o Prepare the final Plans and Master Plan Report which will be working tools for the State of Oregon Aeronautics Division.

PERSPECTIVE ON AVIATION AND AIRPORTS

The expansion and technological advancement of the transportation industry has contributed greatly to the growth and economic development of the United States. This has included shipping, railroads, automobiles, and, most recently, aviation. In less than half a century, aeronautics advanced from primitive flying machines to supersonic jets. To accommodate the rapid increase in the quantity and quality of aircraft, facilities are required which can keep pace with their basing and operational needs as well as with the needs of users of the aviation system.

Air transportation problems and the considerations for airport development have become increasingly complex. Therefore, a systems approach has been instituted to account for demands and relationships. By definition, a system is an interacting and interdependent group of entities which form a unified whole. Cottage Grove State Airport (CGSA) is one of the 3,219 airports in the FAA National Plan of Integrated Airport Systems (NPIAS) and one of the 98 public-use airports in the Oregon Aviation System Plan. As such, it has a functional role as an important general aviation facility for the State of Oregon and the Pacific Northwest. At CGSA as with other components of this system, an

airport development program must be considered in relation to the total social, economic, and political system in which the airport exists, and not be relegated narrowly to aviation issues, nor even to the somewhat wider transportation concerns.

The FAA defines three broad categories of aviation activity: general aviation, certificated air carrier, and military. These three categories comprise the quarter-million aircraft that share the nation's airspace. By any measure, the United States' airspace and airports system is the busiest in the world. General aviation (GA) accounts for 98 percent of all civil aircraft and 74 percent of all hours flown. It also accounts for all of the aircraft and operations at Cottage Grove State Airport.

This Master Plan study, addressing the various roles of the airport's part of larger systems, is evidence that the Oregon State Aeronautics Division recognizes the importance of aviation in community and regional planning and the associated challenges inherent in providing for future needs. The cost of maintaining an airport is an investment which yields benefits for the State and the community it serves. A sound and realistic Master Plan will enable Cottage Grove State Airport to increase its contribution as a transportation resource and economic asset, while infusing a feeling of pride in the people it serves.

CONCLUSIONS AND RECOMMENDATIONS

Chapter 2 CONCLUSIONS AND RECOMMENDATIONS

CONCLUSIONS

Studies, analysis, and public meetings provided data for the Master Plan Study. This data has led to the following major conclusions by the Consultant team:

1. That Cottage Grove State Airport is a Basic Utility category, general aviation airport providing a vital service to the residents of the City of Cottage Grove and Lane County, Oregon.
2. In 1986, there were 32 based aircraft, which accounted for 11,200 operations.
3. Based aircraft are projected to increase to approximately 70 aircraft by the year 2007.
4. The total aircraft operations, based upon 350 operations per based aircraft, will total in excess of 24,500 by the year 2007.
5. The construction of a parallel taxiway on the west side of Runway 15-33 was determined to be the number one development project needed at the airport.
6. The critical aircraft to use the airport on a frequent basis is the twin-engine turboprop Beech King-Air type.
7. Runway length for a Basic Utility Stage II airport is recommended to be a minimum of 3,100 feet. The existing runway length is 3,200 feet. The maximum runway length obtainable, using the existing runway location and alignment, is 3,390 feet.
8. Additional land is required if new hangars and aircraft parking areas are to be developed. Several land-locked parcels of land to the east of the runway should be acquired.
9. Existing power lines crossing over the east-west taxiway are a potential hazard to aircraft, especially helicopters hover-taxiing from the runway to the extreme west transient apron.
10. The east-west taxiway width and clearances from adjacent hangars, which do not meet FAA standards, could be a limiting factor for future growth.
11. To enhance future activity at the Cottage Grove State Airport, an nonprecision instrument approach should be developed.

RECOMMENDATIONS

To provide for and foster aviation in the best interests of the public in the Cottage Grove area, the Master Plan study recommends that the State of Oregon, Aeronautics Division, take the following steps:

1. Purchase additional land and aviation easements to safeguard the alternative of lengthening Runway 15-33 to the north and allow adequate area to increase the number of hangars and tie-downs.
2. Construct a new parallel taxiway along the west side of Runway 15-33 to improve operating characteristics and safety for users of the airport.
3. Extend Runway 15-33 to the north by 190 feet to maximize its runway length, based on terrain and FAA standards.
4. Provide for the continued and orderly expansion of the general aviation facilities at the airport to provide for and foster future aviation demand. This expansion should follow the staged development plan, yet be accomplished within the constraints of prudent economic planning.
5. Remove or relocate the power lines that cross over the east-west taxiway to increase the safe utilization of the westside facilities.
6. Request funding assistance under FAA and other federal funding programs for all eligible capital improvements.
7. Through the appropriate governmental jurisdictions, accomplish the required permitting for airport development.
8. Expand the width of the east-west taxiway along the south side to accommodate larger transient aircraft and create a safer spacing between the taxiway and hangars.
9. The development of a nonprecision instrument approach through the installation of a new NDB transmitter located 1 to 2 nautical miles to the north on the extended runway centerline.
10. Revise appropriate lease policies to ensure the maximum possible financial return from airport operations consistent with a policy of promoting aviation and economically healthy entities on the airport, while working toward a more financially sound airport operation.
11. Adopt the Master Plan and initiate the development of the recommended improvements in a timely manner.

INVENTORY

Chapter 3 INVENTORY

Prior to developing the Master Plan for Cottage Grove State Airport (CGSA),¹ a series of meetings was held with State Aeronautics Division personnel, and with Lane County and Cottage Grove representatives. The purpose of these meetings was to collect information and insight relative to the airport's operation, and the factors influencing that operation. Data pertaining to the airport and surrounding area were gathered and studied. These data included:

- o The previous Airport Layout Plan, other area plans and studies, and any other such sources, reviewed for information or influence they might have upon the development or implementation of this study's recommendations.
- o Physical inventories and descriptions of the facilities and services obtained from the State Aeronautics Division, the general aviation service operators, and the Consultant's on-site inspections.
- o Background information on the Cottage Grove State Airport vicinity, including population, socioeconomic, and other indicators of development potential in the region.
- o Historical airport development and aviation activity indicators as a basis upon which to identify present and future facility needs.

As the assumptions and recommendations of this study are based in large part upon existing conditions on the airport and in the surrounding community, it is essential that the data base be accurate and comprehensive. The remainder of this chapter summarizes portions of the inventory and introduces topics to be analyzed in greater detail in subsequent chapters.

AIRPORT SETTING

The site of Cottage Grove State Airport is contiguous to the easterly boundary of the City of Cottage Grove, in Southwestern Oregon. The airport itself lies in Lane County. It is 135 miles south of Portland and approximately 20 miles south of Eugene,

¹ Cottage Grove State Airport does not have an official three-letter FAA identification code like larger airports--such as SEA for Seattle-Tacoma International--but rather has a code number (615). For simplicity, the airport will be referred to by the abbreviation CGSA from time to time in this report.

which is the County seat. CGSA is 19 statute miles southeasterly of Eugene's Mahlon Sweet Airport and 8 statute miles southwesterly from Creswell's Hobby Field.

The City of Cottage Grove is bisected by Interstate 5, the principal north-south arterial of Oregon. CGSA is located less than one mile from I-5. The airport is situated on 69 acres and is bounded on the east by the Row River. The foothills of the Cascade Mountain Range envelope the airport on the northeast, east, and southeast. Figure 1, Location Map, depicts the location of the airport in its regional setting. Figure 2, Vicinity Map, gives a graphic representation of the airport vicinity.

Cottage Grove State Airport is owned and operated by the State of Oregon, Aeronautics Division, and is classified as a General Aviation, Basic Utility airport in the National Plan of Integrated Airport Systems (NPIAS)² and the Oregon Aviation System Plan (OASP).³ In 1986, 32 single-engine aircraft were based at the airport, and an estimated 11,200 aircraft operations (takeoffs and landings) occurred.

AIRPORT DEVELOPMENT HISTORY

The City of Cottage Grove is actually a merging of two communities lying on the east and west banks of the Coast Fork of the Willamette River. The west side, Cottage Grove, incorporated in 1887, and in 1901 merged with "East Cottage Grove," which was also known as "Lemati," the Indian name for "peaceful valley."⁴

Southern Pacific Railroad began serving the area in 1871, promoting expansion. Gold was discovered in the Bohemia Mining District in 1905. Between 1890 and 1910 numerous sawmills were established, and the City's population grew from 750 to 1,800 during that twenty-year period.⁵

Cottage Grove State Airport is a relatively recent addition to the community. The airport was constructed in 1965 on a 40-acre site which was purchased with funds donated by the Woodard Foundation,⁶ a local charitable organization. Construction was funded by the Federal Aviation Administration and State of Oregon. From CGSA's inception, the State of Oregon Aeronautics Division has owned and operated this public-use airport.

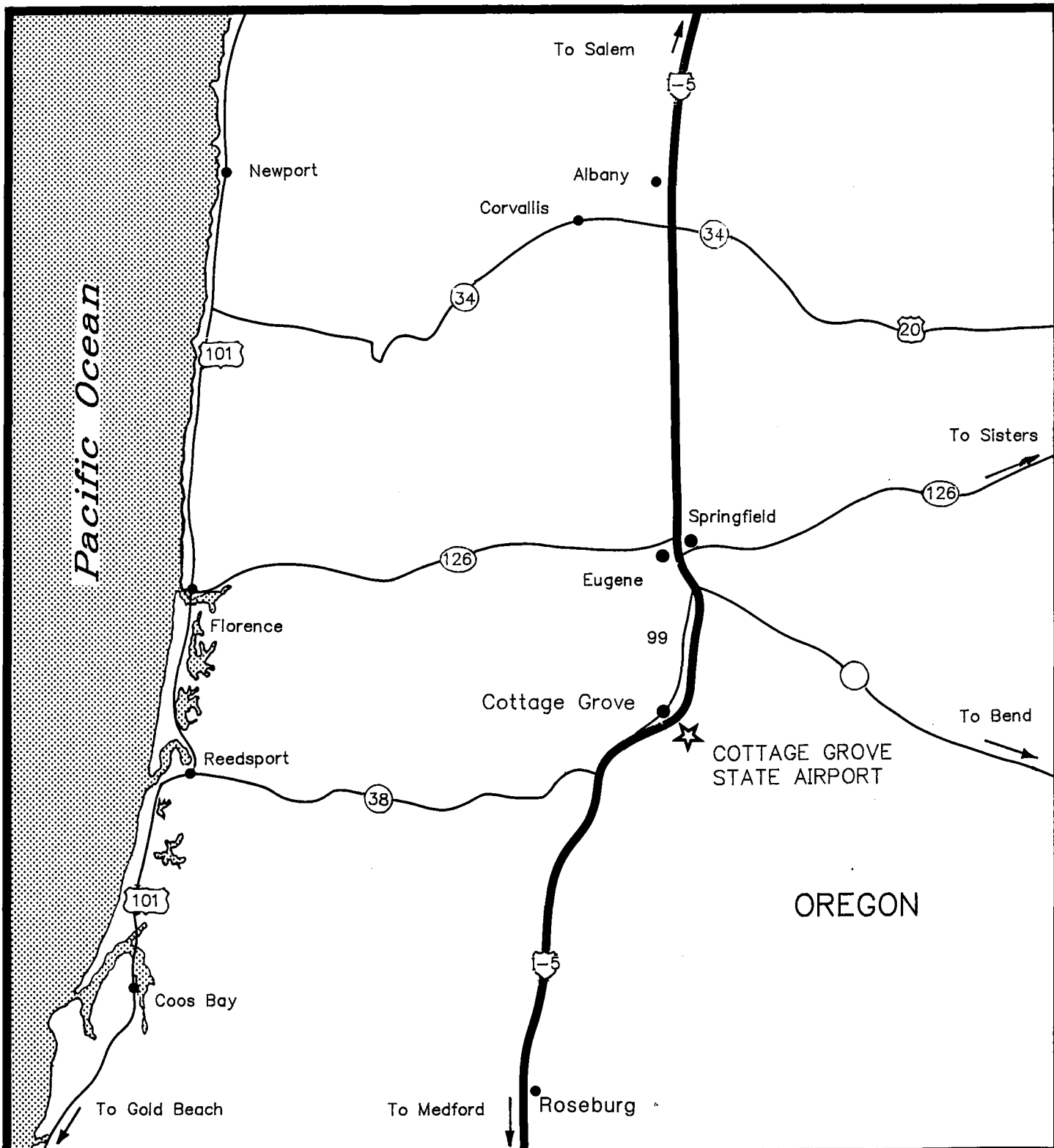
² National Plan of Integrated Airport Systems, FAA, August 1985.

³ Oregon Aviation System Plan, Aeronautics Division of Oregon Department of Transportation, 1983.

⁴ Golden Was The Past, 1850-1970, The Writers Discussion Group, 1970.

⁵ City of Cottage Grove Comprehensive Land Use Plan, October 27, 1980.

⁶ Roger Ritchey, Deputy Administrator, State of Oregon Aeronautics Division.



LOCATION MAP

COTTAGE GROVE STATE AIRPORT
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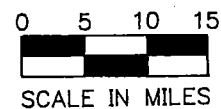
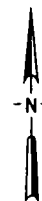
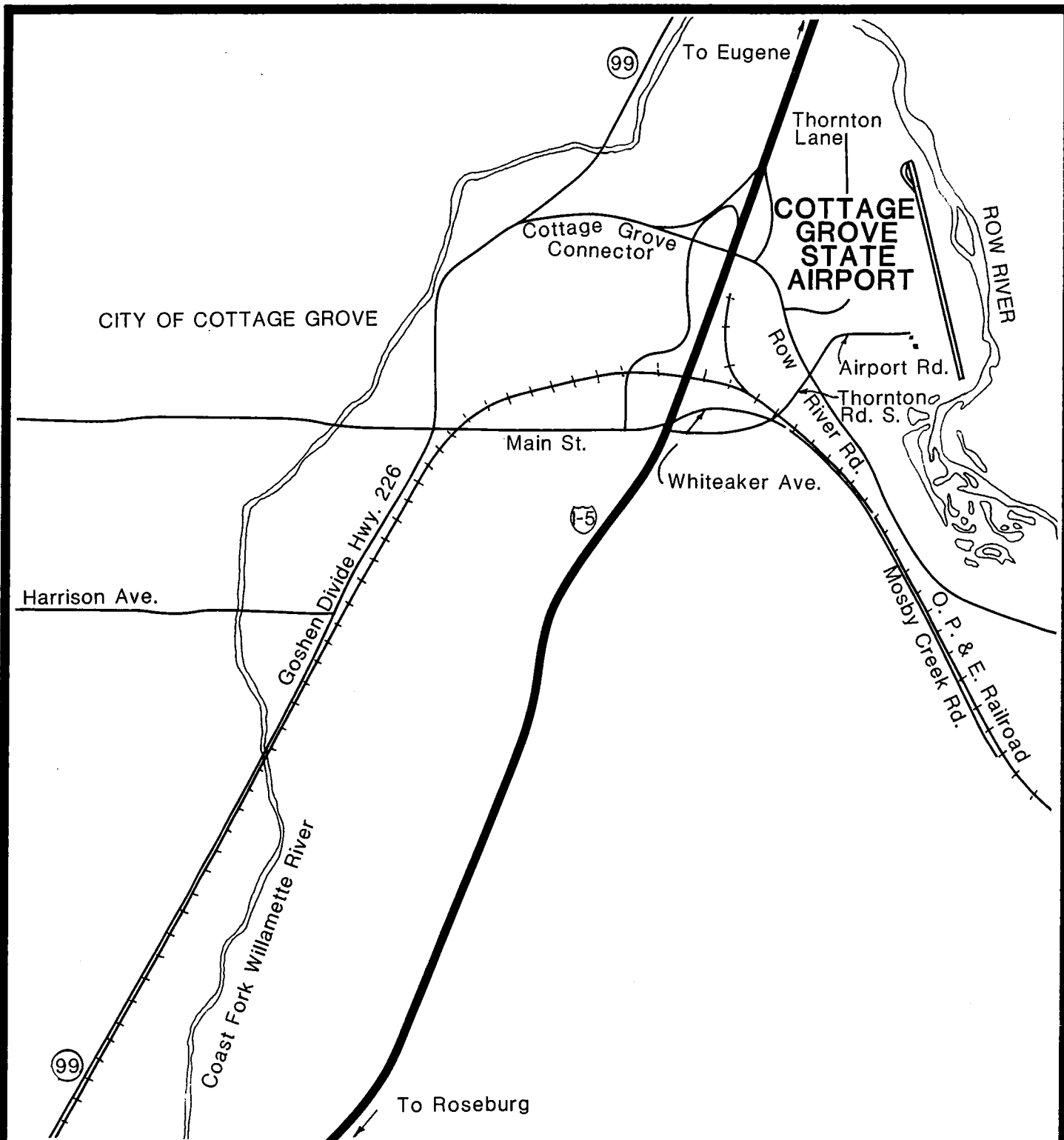
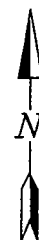


FIGURE 1



VICINITY MAP

COTTAGE GROVE STATE AIRPORT
AIRPORT MASTER PLAN UPDATE



0 1000 2000
SCALE IN FEET

FIGURE 2

In 1969, the Village Green donated a taxiway and tie-down area, and constructed a road from Bowling Green to intersect with Thornton Road South.

Community support was evident in 1977 when local citizens organized volunteer labor to upgrade the airport. Several businesses, as well as the City and State Aeronautics Division, donated equipment to accomplish needed improvements. During this effort, both sides and ends of the runway were cleared, the east-west taxiway was graded, the tie-down area was fenced, and the ground was treated for dust control.

A summary of airport improvements appears on Table 3A.

Table 3A
COTTAGE GROVE STATE AIRPORT IMPROVEMENTS

- 1969 Widening and lengthening of runway; Village Green donated a taxiway and tie-down area and constructed a road from Bowling Green to intersect with Thornton Lane.
- 1973 Pave tie-down area; construct east-west taxiway from terminal area to Village Green.
- 1974 Construct drainage system; additional tie-downs; NAVAIDS; LIRL; obstructional aids; pave east-west taxiway to service station.
- 1976 Repave and widen taxiway west of runway; pave aircraft apron; construct maintenance shop south of administration building (State contributed additional land for taxiway and parking area).
- 1977 Construct aircraft parking apron; construct connecting taxiway and vehicle parking area.

An Airport Layout Plan was accomplished in 1974 and has been updated consistently as changes have occurred on the airport. This Master Plan represents the latest in a series of planning efforts for CGSA.

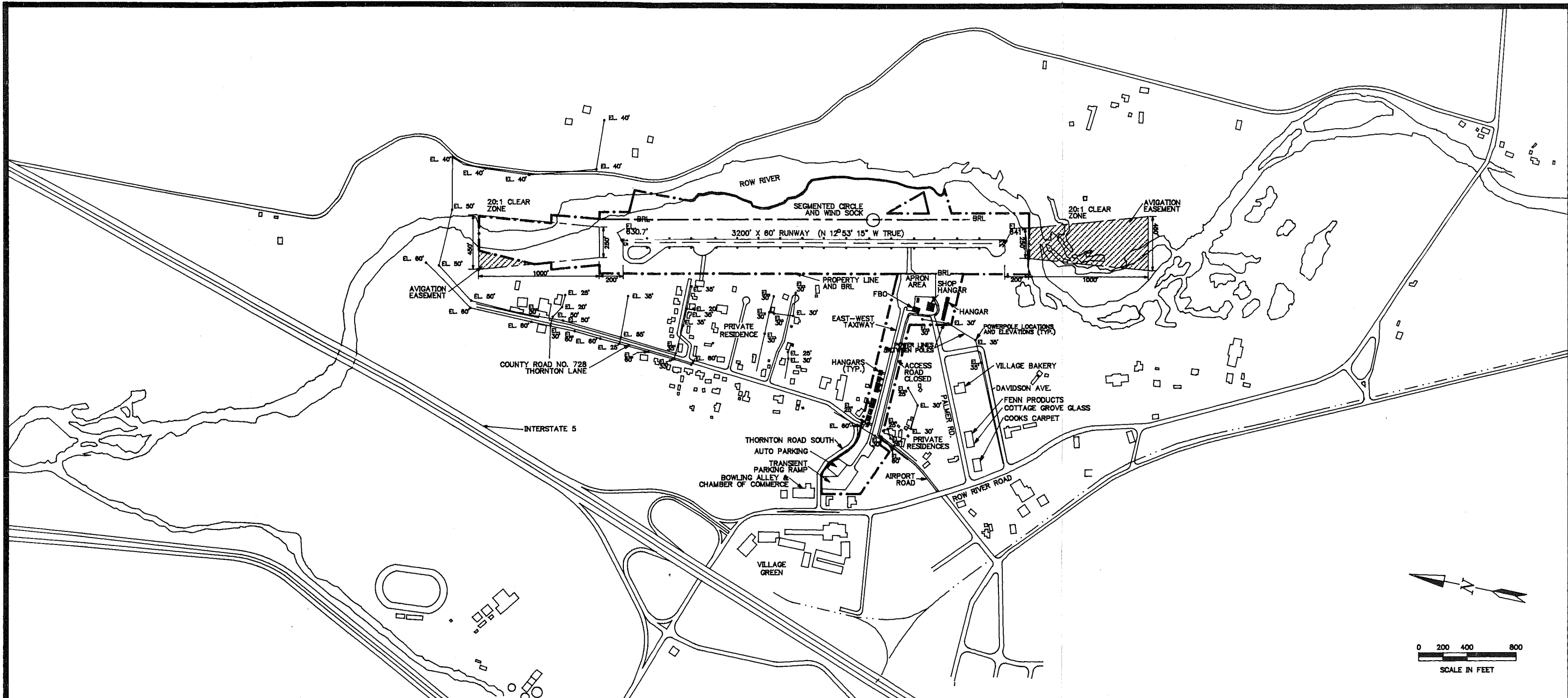
EXISTING AIRFIELD FACILITIES

Cottage Grove State Airport's airfield facilities consist of one active runway and the east-west taxiway, lighting, and navigational aids. Runway 15-33 is 3,200 feet long and 60 feet wide, and is constructed of asphaltic concrete with a strength rated at 12,500 pounds single-wheel (S) loading. The runway appears to be in good overall condition. The effective gradient on Runway 15-33 is 0.37 percent.

A 20-foot-wide terminal area taxiway provides access to hangar space west of the runway. Electrical power lines traverse the west end of this taxiway and are marked with two red spheres. Because the lines may serve as obstructions to helicopter operations, the final phase of this study will evaluate underground placement and overhead relocation. The taxiway was constructed in 1981 and also appears to be in good condition. Runway 15-33 is equipped with Low Intensity Runway Lighting (LIRL) and basic runway marking. A lighted 8' wind cone and segmented circle are located approximately 1,000' north of the Runway 33 threshold, east of the terminal area and Runway. A rotating beacon assists pilots in identifying the airport at night and in marginal weather. At present, no instrument approach to the airport is published by the FAA, and only visual approaches are approved. Figure 3, Existing Facilities, illustrates the layout of the runway and the taxiway. Airfield facilities are summarized in Table 3B, Airfield Facilities Data, and depicted in Figure 3.

Table 3B
AIRFIELD FACILITIES DATA
Cottage Grove State Airport Master Plan

	RUNWAY	
	15	33
Length (feet)	3,200	
Width (feet)	60	
Gradient	.37%	
Surface Material	Asphaltic/ Concrete	
Load-Bearing Capacity: Single-Wheel (pounds)	12,500	
Approach Slope Ratio:		
Existing	20:1	10:1
FAA Standards	20:1	20:1
Lighting	LIRL	
Marking	Basic	



COTTAGE GROVE STATE AIRPORT
COTTAGE GROVE, OREGON
MASTER PLAN UPDATE

EXISTING FACILITIES
(JULY 1987)

FIGURE 3

Existing Terminal Area Facilities

Figure 3 depicts facilities currently existing on the airport. The present terminal area includes the office of Cottage Grove Aviation, the fixed-base operator. The FBO operations and office area is housed in an approximately 1,220-square-foot building, which also includes a pilot's lounge. The FBO provides flight instruction and aircraft rental services. A UNICOM radio is located in the FBO office (Frequency 122.8).

Another primary structure at Cottage Grove State Airport is a hangar owned by Cottage Grove Aviation and used for aircraft maintenance, repair, and inspection service. The operation is housed in a 60' x 60' maintenance hangar to the southeast of the airport office. A seven-bay T-hangar is located approximately 75' south of the maintenance hangar.

Approximately 1,200 square feet of gravelled terminal area parking is located directly south and west of the FBO office. The airport access road feeds into the parking area from the west and is paved.

Six single-unit and 2 double-unit aircraft hangars are situated north of the east-west taxiway, which lies west of the terminal area. There are a total of 25 permanent tie-down locations, 14 of which are situated north of the FBO on the paved apron. Thirteen of these are located on airport property across from the Village Green, some of which are designated for transient aircraft. The remainder are grass tie-downs. This year, 38 additional temporary grass tie-downs were added to handle the increased demand which occurred when the Creswell Airport (Hobby Field) was closed for construction. These added tie-downs were not looked upon as meeting the future needs of the airport due to their temporary location.

The State of Oregon Aeronautics Division serves as the airport's general manager, handling lease management of the property upon which private improvements are constructed (all hangars and FBO facilities). The State also owns and leases the refueling facility directly north of the FBO building. An informal, cooperative arrangement exists whereby Cottage Grove Aviation personnel tend to some routine maintenance items, such as mowing and the replacement of runway light bulbs.

Auxiliary Facilities

The refueling facility is located adjacent to Cottage Grove Aviation on the terminal apron, and dispenses 80- and 100-octane aviation gasoline to aircraft, with one 10,000-gallon storage tank for each grade. Jet fuel is not presently available at CGSA.

FAA flight services are available from the McMinnville Automated Flight Service Station (FSS) which can be reached by radio or telephone using a toll free number. Weather information can be received from the McMinnville and Hobby Field FSS office or through the Eugene weather station located at Eugene's Mahlon Sweet Airport.

Ambulance and Fire Protection Service is provided to Cottage Grove State Airport by the City of Cottage Grove.

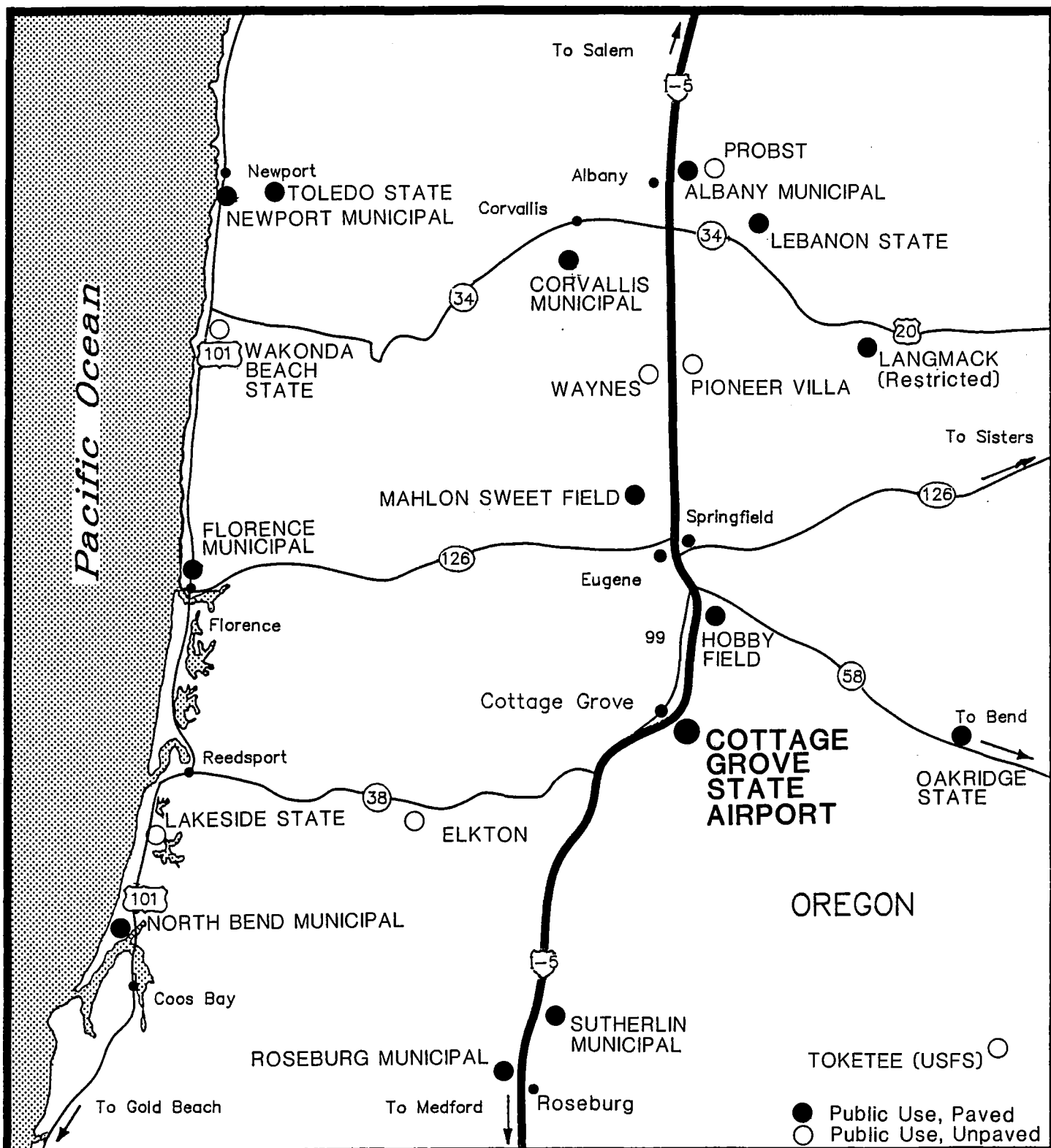
Hotel, motel, RV park, and restaurant facilities are available in Cottage Grove, with the Village Green Motor Hotel easily accessible from the satellite aircraft parking area.

AREA AIRPORTS

Cottage Grove State Airport is a General Aviation, Basic Utility airport. The FBO reports that there are currently approximately 32 based aircraft. The airport serves primarily single-engine piston aircraft. There is no scheduled airline service nor is there chartered service departing from CGSA. A June 1987 survey of airport users indicates that 39% of the aviation traffic is for business purposes, 53% is for private recreation, and 8% is for instructional purposes. The survey results are summarized in Appendix A.

The two area airports which impact Cottage Grove State Airport activity most significantly are Hobby Field, 8 statute miles northeast in Creswell, and Mahlon Sweet Airport, 19 statute miles to the northwest in Eugene (see Figure 4, Area Airports). Mahlon Sweet is an Air Carrier airport, as well as a full-service, corporate, general aviation facility with an ILS precision instrument approach and aircraft maintenance service. It will no doubt continue to be the airport of choice for larger corporate aircraft. Walker Field, a private airport located 3 miles south of Creswell, could have a slight impact on Cottage Grove State Airport if it were to close. This is not projected to happen in the near future.

Hobby Field has recently been developed to a standard similar to that of Cottage Grove State Airport, with a 60' x 3,100' runway. Based aircraft capacity is not being increased in conjunction with the current project. While the construction at Hobby Field has caused a temporary sharp increase in activity at CGSA, long-term competitive impacts pursuant to improvements at the airport at Creswell are not anticipated. Hobby Field draws activity primarily from South Eugene and the Creswell area. Cottage Grove appears to have the population, business, and attractions to stand on its own. While the NPIAS suggests that the Air Trade Area for general aviation facilities is 30 minutes' surface travel time, the Cottage Grove FBO reports an active market area that extends as far south as Roseburg and Sutherlin. This is supported by the Airport Users Survey (Appendix A). Businesses such as the Village Green and Weyerhaeuser have clientele who fly in for a specific business or leisure activity.



ROADS

Interstate 5, the principal north-south highway of Oregon and Washington, is less than a mile from Cottage Grove State Airport. Palmer Road, via Row River Road, serves as the primary access to CGSA. This is the primary surface route connecting CGSA to the City, to I-5, and beyond. Cottage Grove is served by seven motor carriers and by Greyhound Bus Lines.

RAIL

Cottage Grove is served by Southern Pacific, OP&E, and Burlington Northern railroads. There are currently no abandonment proceedings on any trackage in the immediate area.

SOCIOECONOMIC CHARACTERISTICS

A variety of historical and forecast socioeconomic information relating to Cottage Grove and Lane County has been collected for use in various elements of the Master Plan Study. Table 4-A, in the next chapter, "Aviation Demand Forecasts," shows that Cottage Grove grew by about 10% between 1970 and 1975. It continued to grow at a rate of nearly 7% in the next five-year period, and then lost population in the first part of the 1980s, as did Lane County as a whole. Community representatives report that this out-migration has ended with the stabilization and diversification of the County's historically timber-based economy, and growth is projected to increase by 30% over the twenty-year airport planning period.⁷

AIRPORT ENVIRONS

Climate

Weather conditions play an important role in the planning and development of an airport. Temperature is an important factor in determining runway length. Wind speed and direction determine optimum runway orientation, and the percent of time visibility is impaired due to cloud coverage is a major factor in determining the need for navigational aids and lighting.

The area around Cottage Grove State Airport has a predominantly modified marine climate with relatively cool summers and mild, wet winters. According to statistics from the Cottage Grove Planning Department, precipitation ranges from a mean low of 0.34

⁷ Portland State University Center for Population Research and Statistics.

inches in July to a mean high of 7.9 inches in November, with an average annual total of 47.96 inches. Moderate temperatures throughout the year provide only a 25-degree variance between the mean temperature in January and July.

Existing Land Use

All of the cleared land on the airport property is designated for and committed to aeronautical use. The predominant off-airport land uses in the vicinity of Cottage Grove State Airport to the north, east, and south are park land, agricultural, and forestry. With the exception of two small islands of land (see Figure 3, page 3-7, Existing Facilities), the Row River forms the easterly boundary of the airport. The bank following the river is wooded and interspersed with vegetation typical of streambanks, such as blackberry vines, grasses, and willow shoots.

A rural residential subdivision is located on the northwest side of the airport, with a few Industrial Commercial uses occupying land southwest of the runway.

The majority of the airport property is situated on the floodplain of the Row River as defined by the Federal Insurance Administration's Flood Insurance Study for Lane County, Oregon Unincorporated Areas and accompanying Flood Insurance Rate Maps.

Lane County operates a solid-waste transfer station approximately 1,100' east of the south end of the runway, across the Row River.

The potential for impacts relative to airport development upon residential uses proximate to the airport, solid waste, farmland, and floodplain issues will be examined in greater detail in the environmental element of the Master Plan Study. Depending upon recommended time frames for the phasing of improvements, an Environmental Assessment may be accomplished.

Regulatory Framework

As mentioned previously in this report, the State of Oregon Aeronautics Division owns and operates Cottage Grove State Airport. The airport property is located in Lane County and land-use regulation and development is within the County's jurisdiction. CGSA is zoned Airport Overlay and is governed by the Lane County Rural Comprehensive Plan, (1984).

The westerly-most portion of the airport, upon which are located hangars, tie-downs, and an access taxiway, is within the Urban Service Boundary of the City of Cottage Grove. The airport is contiguous to the City and is currently under consideration for inclusion in its entirety within the City's Urban Service Boundary. Because of the airport's proximity to the City, the City of Cottage Grove Comprehensive Land-Use Plan (1981) is also of importance in evaluating aviation planning issues. Compatible Land-Use issues will be analyzed in the environmental element of this Master Plan Study.

AIRSPACE AND INSTRUMENT APPROACH FEASIBILITY

A separate substudy was conducted concurrent with the inventory to determine if an Instrument Approach Procedure could be established that would serve the Cottage Grove State Airport. The following is based upon the text of the evaluation by Mr. Robert Schweitzer.

The evaluation was accomplished using current FAA criteria. When developing the procedure for publication, the FAA has great latitude in the use of judgment where a question of safety is involved. For this reason, there is a very good possibility that several hundred feet may be added to the approach minima due to the surrounding terrain at the missed-approach point.

The first step in the evaluation was to determine if the existing en route navigational aids would support the required transitions from the IFR enroute environment to the start of an approach procedure and the approach procedure itself.

The Eugene, Oregon, VOR will support a transition from the north to the Cottage Grove State Airport at the altitudes required to start an approach. However, the controlling terrain between the two points will limit the use of the VOR to the initial approach transition altitudes and restrict its use at the lower altitudes required for the remainder of the approach.

The Roseburg, Oregon, VOR will support transitions from the south but will require a higher transition altitude than the one from the north. This increase in altitude is due to the fact that the terrain between the VOR and the airport is several hundred feet higher than the terrain that controls the transition from the north. This same terrain would also limit the use of this VOR to the initial approach altitude.

There are no navigational aids in the Cottage Grove area that will support an instrument approach procedure. For that reason a study was made using the Non-Directional Beacon (NDB) criteria. This type of facility was chosen because it has the largest approach trapezoids and involves the lowest cost to buy, install, and maintain.

The second step was to determine the best approach direction that would yield the lowest approach minima.

Terrain that surrounds the airport on three sides from the north of the airport through the west will limit the approach course in the northwest quadrant and an approach to Runway 15.

The third step was to determine a facility location that would produce the lowest approach minima.

Due to the lower cost, ease of installation, and maintenance considerations, the first part of this evaluation was made with the NDB located on the airport. The airport locations were eliminated from consideration when it became apparent that close-in terrain surrounding the airport on three sides would produce approach minima several hundred feet higher than if the facility were located off the airport.

The next part of the evaluation was made with the NDB facility located at various points within an area 1 to 6 nautical miles from the Runway 15 threshold, and within 30 degrees of each side of the runway centerline extended.

The terrain that lies within this area has several high points with elevations within a few feet of each other. Their locations are such that, no matter where the facility is placed, at least one of these points will control the approach minima. These same high obstructions are so numerous that a smaller approach surface such as a VOR would lower the minima no more than 50 feet.

The lowest approach minima resulted with an NDB facility located 1 to 2 nautical miles from the runway threshold and on the extended runway centerline. This location resulted in an NDB procedure that had the lowest minima but would be limited to Approach Category A aircraft. This procedure would require a small 5-mile procedure turn and would produce circling minima of 1,660 feet HAA⁸ (2,300 feet MSL) and visibility minima of 1 1/4 miles.

To include Approach Category B aircraft, minima would require a standard size procedure turn. The larger area of a standard procedure turn will increase the approach minima to 1,780 feet HAA (2,420 feet MSL) and the visibility minima to 1 1/2 miles.

Approach Category C aircraft minima were not considered, due to the short runway length and the much higher approach minimum altitude that would be required by terrain in the missed approach segment.

⁸ Height Above Airport

A Loran C instrument approach would be possible with approach minima about the same as those for the NDB using the existing interim criteria. The approach would start over the Eugene VOR and could be aligned with Runway 15. Descent-rate limitations in the final segment would still limit the approach to circling only.

A preliminary evaluation was also made on the IFR departure procedures and the following minima appear possible:

Runway 15 - IFR departure from this runway may not be possible, due to the proximity of high terrain in the departure path.

Runway 33 - IFR departure from this runway is possible but will require a ceiling of 2,000 feet MSL and a visibility of 2 miles, unless a higher than standard climb gradient is imposed.

Recommendations

This element of the study is very basic, due to the limitations and the scope requested. However, the minima given will be within plus or minus 100 feet, provided that there are no man-made obstacles that do not show up on the charts used.

Because of the very high approach minima found in this study, it is recommended that a careful analysis of cost versus benefit be conducted before proceeding further on an instrument approach procedure for Cottage Grove State Airport.

The cost of the additional runway marking required for an approach procedure and the installation of a local altimeter setting source must also be considered.

A re-study of the Loran C procedure is recommended when the final criteria are adopted and the necessary monitor system is installed.

AVIATION DEMAND FORECAST

Chapter 4 AVIATION DEMAND FORECASTS

The purpose of this chapter is to provide forecasts of aviation demand for Cottage Grove State Airport at key intervals during the 20-year planning horizon. These will serve as the basis for planning future facilities to accommodate the area's aviation needs through the year 2007. Accurate forecasting in such a capital-intensive industry as aviation is essential. Nevertheless, forecasting any type of future activity remains more art than science. Regardless of methodology, assumptions must be made about how activities might change in the future. The objective of forecasting, therefore, is to provide a measure of those changes by which effects can be estimated and preparations made to accommodate the resulting conditions in a smooth and cost-effective manner. However, forecasts should be viewed only as guidelines.

FORECASTING CONSIDERATIONS

Airport activity forecasts are developed through a combination of mathematical, analytical, and judgmental approaches. Historical patterns are examined for trends and possible relationships between different phenomena. From these historical data, projections are made and then evaluated in comparison to forecasts from other sources. These may include other aviation demand forecasts or projections by governmental agencies for socioeconomic directions. Projections are produced through mathematical extrapolation of past growth trends or associations using standard statistical techniques, including:

- o **Trend Analysis:** Establishes growth curve lines to historical data points and extends a projected curve to indicate future levels.
- o **Correlation Analysis:** Socioeconomic and other aviation activity indicators (independent variables) are tested in a bivariate or multivariate regression analysis with a given aviation activity indicator (the dependent variable) to measure degree of association and therefore predictive reliability.
- o **Market Analysis:** Compares airport activity indicators with national or state levels and applies either a static or dynamic market share percentage for the airport to the forecast larger-market levels.

The analytical phase of the forecasting process extrapolates past trends or associations into future projections and evaluates the validity of mathematical indicators. For instance, several curvilinear statistical techniques are available to test the association between variables, and the highest correlation coefficient may not necessarily indicate the most appropriate

measure for realistic projection. By comparing several projections, a range or "envelope" of growth can be identified within which the future activity levels will probably occur.

Sound professional judgment is the final and most important part of forecasting. An experienced analyst examines the output of the foregoing steps and subjectively determines the preferred forecast. Among the intangibles to be considered are the uses for which the forecast is being developed, the character of the community, the status and potential changes in the business and economic environment, pertinent advances in aviation technology, the impact of new facilities, and the policies and resources of the airport's sponsor.

Even with the best analytical and judgmental techniques, however, a high level of confidence generally cannot be placed in forecasts extending beyond five years. Since airport development programs often require ten or more years to complete, it is therefore necessary to plan a capital improvements program with the flexibility to accommodate future activity levels which may vary from forecast levels. The development program must have provision for adjusting schedules to prevent overburdened facilities in the event of unanticipated growth, or avoid uneconomic, underutilized facilities should activity fall short of that forecast.

The following sections contain general aviation forecasts for Cottage Grove State Airport. These include aircraft operations, based aircraft, fleet mix, and peak demands (General Aviation passenger forecasts are presented in conjunction with related facility requirements in Chapter 5). The results of the forecasting process provide input for the demand/capacity and facility requirements analysis of the next chapter as well as all subsequent elements of this study.

POPULATION FORECASTS

Demographic and socioeconomic characteristics of a community are important factors in determining aviation demands. Historical and projected trends in the local and regional economy and the aviation market provide parameters within which airport activity can be gauged. Most important, however, are trends in the airport's service area population. Past growth trends in airport activity can be compared with historical population counts to identify correlations which can be extrapolated into the future and the extent to which population variables are reliable predictors of future aviation activity. This is an especially useful approach, as official population forecasts are normally available from government agencies and other sources. Table 4A depicts historical and forecast population for Lane County and Cottage Grove through the planning period. The planning period interval projections represent interpolations or extrapolations of official published population forecasts.

Table 4A
POPULATION FORECASTS
Cottage Grove State Airport

<u>Year</u>	<u>Lane County Population</u>	<u>Cottage Grove Population</u>
1970	215,401	6,004
1975	241,800	6,700
1980	269,500	7,190
1985	262,958	7,090
1986	261,650	6,906
1988	276,318	7,195
1992	296,419	7,772
1997	321,546	8,494
2002	346,674	9,216
2007	371,801	9,938

Sources: Portland State University, Center for Population Research and Statistics, and City of Cottage Grove, Comprehensive Plan. Interpolated and extrapolated by FORESITE Group, Inc.

GENERAL AVIATION FORECASTS

General aviation encompasses all facets of aviation except the commercial air carriers and military flying. Nationally, it accounts for over 80 percent of all civil aircraft operations, and includes all takeoffs and landings at Cottage Grove State Airport. Forecasts of general aviation activity will be used to plan for various airfield and landside facilities. Based aircraft, fleet mix, and aircraft operations are the primary forecast indicators for the types, sizes, and timing of facilities which will be needed to meet the needs of general aviation at the airport. The number of based aircraft is the most fundamental indicator of GA activity. After developing this forecast, other projected demands can be estimated in conjunction with community factors which influence the character and levels of activity at Cottage Grove State Airport.

Based Aircraft

Historical based aircraft counts for Cottage Grove State Airport were available from FAA Form 5010-1 Airport Master Records for the years 1970 through 1986. Given the apparent year-by-year trend, these counts appear to be reasonably reliable. There has been a slow but steady growth in the numbers of aircraft based at Cottage Grove during this period, even since 1980, while the national general aviation fleet and--most important--Lane County aircraft registrations have been declining.¹

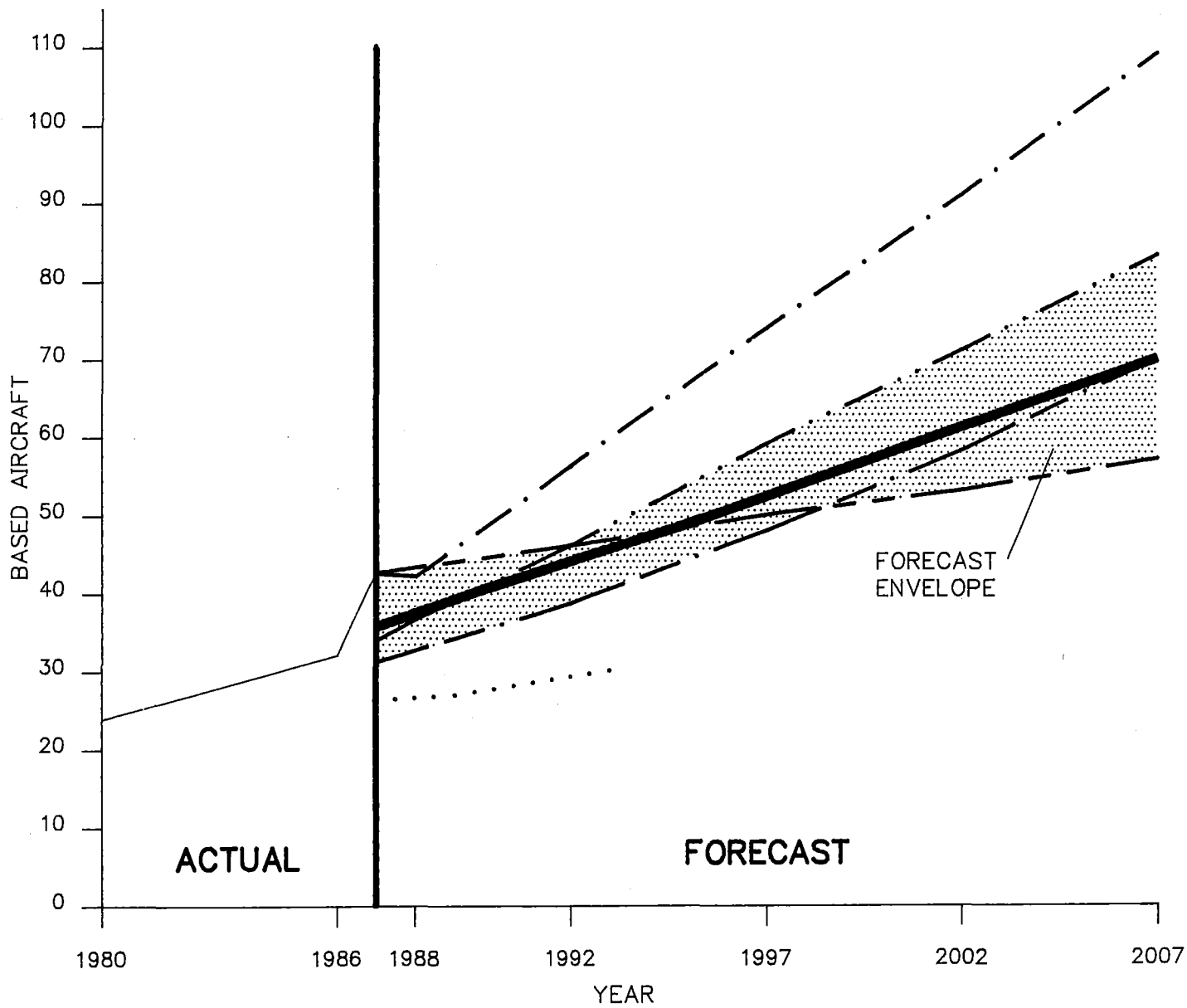
¹ Aircraft registrations are recorded by county of the registrant's address and do not necessarily represent county of aircraft basing.

Detailed historical information was available on registered aircraft in Lane County in the FAA Census of U.S. Civil Aircraft for calendar years 1977 through 1985. County registrations rose sharply in the 1970s, peaking at 610 in 1980, before declining to 494 in 1984. The increasing numbers and share of County aircraft based at Cottage Grove apparently represents a relocation of some users from other area airports, rather than a natural increase attributable to more localized socioeconomic or business growth in Cottage Grove. Because of the inverse relationship, however, Cottage Grove based aircraft cannot be projected in correlation with a continued decline of County registrations.

On the other hand, the rate of decline in Lane County has slowed and may soon stabilize, consistent with trends nationally and in the regional GA fleet. The FAA projects a slight growth in this fleet beginning in the early 1990s. This can be reflected in Lane County by projecting the current ratio of 14 registered aircraft per 10,000 population constant throughout the planning period. A "static share" of County registered aircraft based at Cottage Grove has been forecast for comparison by similarly holding constant the airport's 11 percent current share through the year 2007. The resulting static share forecast of county aircraft by county population is shown in Table 4B, along with other comparative forecasts, and these are depicted graphically in Figure 4A. This forecast, higher initially but with the lowest growth rate among the forecasts, forms the upper limit of the "forecast envelope" in the near term and the bottom end of the range in the latter years of the planning period.

County population and aircraft registrations also provided the independent variables in a dynamic, or trended, market share regression analysis. This technique yielded a high correlation coefficient ($r = .95$), but the cumulative growth factors of the independent variables produced a forecast of based aircraft for Cottage Grove that was unrealistically high, and it has been removed from further consideration.

Bivariate regression analyses were also conducted using Lane County population and registered aircraft separately. The rate of growth projected for each analysis was similar, with the population-based forecast indicating higher numbers of based aircraft in the long term. Both analyses produced correlation coefficients sufficient for predictive confidence but not extraordinary. As shown in Figure 5, these virtually parallel forecasts have been chosen to delineate much of the high and low parameters of the forecast envelope.



- . — . — TRENDEN SHARE, LANE CO. POP. & REG. AIRCRAFT
- . . — . . — COUNTY POP. REGRESSION SHARE
- — — — — OASP EXTRAPOLATED (PREFERRED)
- . — — — — TRENDEN SHARE, CO. REG. AIRCRAFT
- — — — — STATIC SHARE, CO. POP. & REG. AIRCRAFT
- FAA NPIAS

COTTAGE GROVE STATE AIRPORT
MASTER PLAN UPDATE
FORECAST OF BASED AIRCRAFT

FIGURE 5

Table 4B
BASED AIRCRAFT FORECASTS
Cottage Grove State Airport

YEAR	FAA	OASP	Static Share	Trended Share	Trended Share	County
	NPIAS		Lane Co. Pop. & Reg. Aircraft	Lane Co. Pop. & Reg. Aircraft	of County Reg. Aircraft	Population Regression Share
1983	25					
1984		30				
1986						
1987			43		31	34
1988				42		37
1989	27					
1990		41				
1992			46	56	39	46
1993	30					
1995		49				
1997			50	74	48	59
2002			53	91	58	71
2007			57	109	70	83

Upon evaluation of the foregoing analyses, it was noted that the forecast of based aircraft for Cottage Grove State Airport, as presented in the 1984 Oregon Airport System Plan (OASP), followed a median course among the other forecasts up to its endpoint in 1995. Extending this linear trend to the year 2007 continued its average center position within the forecast envelope. As any future levels within this range are considered possible, the reasonable approach is to select the mean. Therefore, the OASP forecast, extrapolated to 2007, is the preferred forecast of based aircraft at Cottage Grove State Airport.

Because of terrain restrictions to significant expansion and the availability of transport category accommodations at Mahlon Sweet Field in Eugene, Cottage Grove State Airport is expected to remain a Basic Utility airport. Its role will continue to be that of serving primarily small, piston-powered, general aviation aircraft. At the time of this study, only one twin-engine aircraft is based at the airport, the remainder being single-engine planes. The based aircraft fleet mix should remain more than ninety percent singles with a few twins. The possibility exists for one or more turboprop aircraft among these. Although the airport will experience limited operations by the smallest business jets, none are anticipated to be based there. The based aircraft forecast for Cottage Grove State Airport throughout the planning period is presented in Table 4C.

Table 4C
BASED AIRCRAFT FORECAST
 Cottage Grove State Airport

	<u>1988</u>	<u>1992</u>	<u>1997</u>	<u>2002</u>	<u>2007</u>
Single-Engine Piston	36	42	49	57	64
Multi-Engine Piston	<u>1</u>	<u>2</u>	<u>4</u>	<u>5</u>	<u>6</u>
TOTAL BASED AIRCRAFT	37	44	53	62	70

General Aviation Operations

Aircraft operations at an airport are categorized as either local or itinerant. A local operation is a takeoff or landing by an aircraft in the traffic pattern or within sight of the air traffic control tower or executing practice instrument approaches at the airport. Itinerant operations are all operations other than local operations.

Estimates of general aviation aircraft operations over time at Cottage Grove State Airport were available only on the 5010-1 forms. However, the estimates changed only three times in sixteen years, with periodic estimates being carried over through intervening years. The March 1986 figure of 19,200 operations was carried over since January 1980, for instance. On-site activity sample counting between April 1985 and March 1986 placed the level at a more credible estimate of 11,158, indicating that the trend of activity at Cottage Grove has paralleled the decline experienced nationally since 1980. As the "historical" counts from the 5010-1 forms were not considered adequately reliable from which to extrapolate trends or correlate with other variables, no regression analyses were conducted to project future aircraft operations.

Typically, general aviation operations are related to the changing numbers of aircraft based at an airport over the years. However, operations appear to have declined at Cottage Grove State Airport even as the number of based aircraft has grown. This may be explained in part by the relocation of aircraft from other airports. Some of these are used primarily for itinerant flights away from Cottage Grove as indicated by a recent estimate which indicates a comparatively high percentage (62.5%) of itinerant operations for an airport of this type. To extrapolate the inverse relationship between based aircraft and operations would project continued decline into the future--an invalid technique and improbable result.

Therefore, a static ratio of 350 operations per based aircraft, as indicated by the recent activity counts, has been selected as the preferred forecast of aircraft operations at Cottage Grove State Airport. Moreover, the current estimated dichotomy of

itinerant and local operations is projected unchanged, as shown in Table 4D. The ratio of operations to based aircraft is somewhat lower than the national average, indicating that economic recovery in the region is lagging compared to other areas. However, the relatively high percentage of itinerant operations indicates significant business aviation use of the airport, primarily businessperson-pilots flying their own single-engine aircraft. This is confirmed in a recent survey conducted by FORESITE Group in June of 1987 which indicates 40% of the activity at CGSA is for business purposes (Appendix A). The operational fleet mix is forecast to remain more than ninety percent single-engine aircraft.

Table 4D
AIRCRAFT OPERATIONS FORECAST
Cottage Grove State Airport

<u>Year</u>	<u>Based Aircraft</u>	<u>Local Operations</u>	<u>Itinerant Operations</u>	<u>Total Operations</u>
1986 (est.)	32	4,184	6,974	11,158
1988	37	4,850	8,100	12,950
1992	44	5,775	9,625	15,400
1997	53	6,950	11,600	18,550
2002	62	8,000	13,350	21,350
2007	70	9,200	15,300	24,500

PEAKING CHARACTERISTICS

Like many other public facilities, various components of airports must be designed to meet demands during periods of high activity levels. Typical measures of peak activity include peak month, busy-day, and design hour. Peaking characteristics of aircraft operations are used to evaluate and project airfield facility requirements and certain requirements for the terminal area. The following peak measures have been used in this analysis:

- o Peak Month - the month in a calendar year with the highest total for a given activity (operations, etc.).
- o Busy-Day - the busy day in a typical week of the peak month (used particularly for planning itinerant GA apron space).
- o Design Day - the average day within the peak month (the month's total activity count divided by the number of days in the month).
- o Design Hour - the peak hour within the design day (for airfield and landside demand/capacity analyses).

The peaking characteristics used for analysis are planning standards only and do not necessarily represent absolute peak demands. Only the peak month is an absolute maximum within any calendar year. The other planning "peaks" may be exceeded from time to time.

Design Hour Operations

In the absence of detailed monthly data, a ratio of peak-month operations to total annual operations typical of similar airports was chosen, and this can be expected to remain relatively constant throughout the planning period. The design day is the average day of the peak month, with 3.2 percent of peak-month operations. Specific facility requirements are determined based upon the design-hour within the average day of the peak month. Although no hourly data were available for Cottage Grove State Airport, a design-hour peaking factor (from observations at comparable airports) was used to derive an operational figure representing the typical busy hour in the peak month. Design-hour aircraft operations are forecast for the planning period in Table 4E

Table 4E
DESIGN HOUR OPERATIONS
Cottage Grove State Airport

	<u>1988</u>	<u>1992</u>	<u>1997</u>	<u>2002</u>	<u>2007</u>
Annual Operations	12,950	15,400	18,550	21,350	24,500
Peak Month (.115)	1,489	1,771	2,133	2,455	2,818
Busy-Day (.032)	48	57	68	79	90
Design Hour (.20)	10	12	14	16	18

SUMMARY

The forecasts provided in this chapter will form the basis for subsequent tasks in this Master Planning Study, including demand/capacity, facility requirements, and financial analyses. The following chapter will determine the airport's existing component capacities and the facility improvements which will be needed to meet the future demand that have been forecast in this chapter. Table 4F has been provided as a convenient summary of the major forecasts.

TABLE 4F
SUMMARY OF AVIATION FORECASTS
Cottage Grove State Airport

	<u>1988</u>	<u>1992</u>	<u>1997</u>	<u>2002</u>	<u>2007</u>
Based Aircraft:					
Single Engine Piston	36	42	49	57	64
Multi-Engine Piston	<u>1</u>	<u>2</u>	<u>4</u>	<u>5</u>	<u>6</u>
TOTAL BASED AIRCRAFT	37	44	53	62	70
Annual Operations:					
Itinerant	8,100	9,625	11,600	13,350	15,300
Local	<u>4,850</u>	<u>5,775</u>	<u>6,950</u>	<u>8,000</u>	<u>9,200</u>
TOTAL OPERATIONS	12,950	15,400	18,550	21,350	24,500
DESIGN-HOUR OPERATIONS	10	12	14	16	18

**AIRPORT
FACILITY REQUIREMENTS**

Chapter 5 AIRPORT FACILITY REQUIREMENTS

In this chapter, forecasted aviation demand has been related to airfield and terminal area facility requirements. Projected requirements have been compared to the existing facilities and any deficiencies noted. From this analysis, a sequential schedule of demand will be derived for the various airport services and facilities.

AIRFIELD AREA PLANNING

A determination of airfield requirements necessitates technical analyses of eight important factors:

1. Meteorological conditions
2. Landing aid analysis
3. Airspace analysis
4. Airfield demand/capacity analysis
5. Runway length
6. Taxiways
7. Pavement strength
8. Airfield lighting

Wind/weather characteristics at an airport are important because, to a great extent, they determine the principal direction of aircraft operations, the need for a crosswind runway, and the number and type of terminal navigational aids that may be required.

Obstructions are a part of the airspace environment of an airport and affect operational minima and therefore the desirability of various landing aid programs. The smooth flow of air traffic is dependent upon approach and departure control procedures. The procedures in effect for aircraft operations to and from an airport warrant investigation to determine the degree of constraint, if any, on airspace and runway capacity.

Airfield capacity is dependent upon aircraft mix and runway configuration, usage, and length. The capacity is identified as hourly capacity and annual service volume (ASV) during VFR and IFR weather conditions.

Runway length determines the maximum stage length (non-stop haul distance) that a particular aircraft can travel without reducing payload--passengers, cargo, and/or baggage. The runway length requirements depend upon the type of aircraft and the desired maximum stage length.

The pavement strength of runways, taxiways, and apron areas must be adequate to sustain loads imposed by the weight of critical aircraft anticipated to operate on the particular surfaces.

The results of these analyses are interdependent in that the recommended improvement program must satisfy the respective requirements of each of these airfield requirements.

METEOROLOGICAL CONDITIONS

It is essential to analyze weather conditions at an airport, because these parameters determine when and on which runway aircraft operations are conducted. An analysis of weather conditions is necessary to determine the number and direction of required runways, the most efficient use of airfield facilities, and the requirements for certain navigational aids (NAVAIDs).

Ceiling and visibility greatly affects the flow of air traffic. When visibility is equal to or greater than three statute miles and the ceiling is equal to or greater than 1,000 feet, aircraft may operate under visual flight rules (VFR). If either the ceiling or the visibility falls below these specified minima, aircraft using the airport must operate under Instrument Flight Rules (IFR). These rules require that approaches to the airport be cleared of obstructions above a specified imaginary plan surface and that landing aircraft and runways be equipped with certain navigational aids (NAVAIDs). At present there are no instrument operations at Cottage Grove State Airport. The possibility of a future will be discussed in more detail later in this chapter.

Wind also affects the operation of an airport, in that pilots usually prefer to operate into the wind in order to reduce aircraft ground speed. During landing, it is especially desirable that the crosswind component remain at a low velocity, thus requiring only a minimal correction for wind during final approach and landing procedures.

Analyses of ceiling, visibility, and wind data are facilitated through the use of a wind rose, which is a graphic representation of wind velocity and direction. No measured wind speed and direction data is available at the Cottage Grove State Airport. A 10-year period of general and pilot observations concluded that the relationship of the runway alignment and wind speed direction is satisfactory. A recent Master Plan study at Creswell Hobby Field¹ determined the wind coverage at that airport to be in excess of 95%. The close proximity of the two airports and the same runway alignment support the on-site observations over a 10-year period as stated above.

¹Hobby Field Airport Master Plan, 1983, DEVCO Aviation Consultants.

Crosswind limitations experienced by a pilot making a landing approach are a function of the aircraft's stall speed, pilot proficiency, and other factors. For general planning purposes, a crosswind component criterion of 12 mph for general aviation aircraft has been established by the FAA.² The larger basic transport aircraft can operate safely with higher crosswinds, due to their increased weight and power; therefore, the FAA has established 15 mph as the funding criterion for basic transport crosswind runways. FAA criteria recommends use of a crosswind runway if the primary runway does not have 95 percent wind coverage. On the basis of the wind analysis, Runway 15-33 meets and exceeds the 95 percent wind coverage; thus, no crosswind runway is required at the Cottage Grove State Airport.

LANDING AID ANALYSIS

During IFR conditions, aircraft desiring to land at an airport are required to adhere to FAA-approved instrument approach procedures. There are generally two basic types of IFR approach procedures--nonprecision and precision. Minima for each of these procedures are specified for various aircraft speed and weight combinations.

In the performance of a precision approach, an aircraft is guided along both vertical and lateral paths. A precision approach to a runway is typically provided by an instrument landing system (ILS), which consists of a localizer and glide slope facility. The localizer aligns the aircraft with the runway centerline and the glide slope provides descent information. In addition to the localizer and glide slope facilities, the runway is typically equipped with high intensity runway lighting (HIRL) and medium intensity approach lighting systems (MIRL) with runway alignment indicator lights (MALSR).

During performance of a nonprecision approach, an aircraft is guided to the vicinity of the airport by one of several NAVAIDs. The aircraft may then descend to a specified altitude, the minimum descent altitude (MDS). Provided visual contact is made with the airport, the aircraft may proceed with the approach and landing--either straight-in or circling unless specifically designated--on the specified runway as per published approach procedures (NOTE: Instrument approaches are developed and approved by the FAA and are published by the U.S. Department of Commerce, as well as by several commercial concerns). If the airport is not sighted, the aircraft must terminate the approach at the MDA and either initiate a second approach or proceed to another airport.

²FAA Advisory Circular 150/5300-4B, Utility Airports--Air Access to national Transportation, June 24, 1975, as revised.

Under the provisions of the Facilities and Equipment (F&E) element of the Airport and Airways Improvement Act of 1982, certain navigational aids qualify for 100 percent FAA installation and maintenance funding. Qualifications for this funding are based on the number of annual instrument approaches (AIAs) at the airport.

Additionally, FAA funding assistance can be obtained for navigational aids under the provisions of the Airport Improvement Program (AIP) of the 1982 Act. Under the provisions of the AIP, navigational aids would be eligible for approximately 90 percent federal funding assistance. Maintenance would be provided by the airport. It should be noted that, due to the limited availability of the F&E program funds and the high demand for AIP funds, very few electronic facilities are installed under AIP.

There are certain landing aids designed specifically to aid the pilot during visual (VFR) flying weather. The visual approach slope indicator (VASI) or precision approach path indicator (PAPI) systems are designed to provide visual approach slope (descent) guidance information for both day and night operations. The VASI/PAPI systems do not have any measurable effect on the capacity of an airport, but significantly aid pilots in executing safe landings. The FAA's minimum requirement for approval of a PAPI installation is 6,000 landings annually on the subject runway end. Runway End Identifier Lights (REIL) consist of two flashing lights located at the end of the runway and provide positive runway end identification.

A study of the potential of instrument flight procedures was accomplished, as discussed in Chapter 3. The conclusion of that study is that a Non-Directional Radio Beacon (NDB) would lower the operating minima into the airport, but the study questioned the benefit of such an improvement due to the high cost of the installation. It does appear that there is substantial evidence to conclude that many of the airport's users could benefit by the installation of an NDB facility.

AIRSPACE ANALYSIS

Cottage Grove State Airport has two airspace environments--one present during VFR conditions and one present during IFR conditions which require landing aids, as discussed earlier. The VFR airspace is typically defined in relation to an airport control zone, extending from the surface to 3,000 feet above the ground and within a radius of 5 statute miles, plus any extensions to accommodate approach and departure routes. Airport advisory service is provided by the airport through the use of Unicom radio.

Airport and runway navigational aid requirements are based upon FAA guidelines contained in DOT/FAA Handbook 7031.2C, Airway Planning Standard Number One, and FAA Advisory Circular 150/5300-2D, Airport Design Standards, Site Requirements for Terminal Navigational Facilities. Candidacy for various NAVAIDs and air traffic control facilities is established by computing a ratio of annual instrument approaches (AIAs) versus the FAA's qualifying AIAs. Cottage Grove State Airport does qualify at this time for a nonprecision instrument approach.

Additional facilities which could increase the hourly capacity of Cottage Grove State Airport during instrument weather conditions are airport surveillance radar (ASR) and an air traffic control tower (ATCT). Forecasts indicate that operations will remain substantially below existing capacity in the absence of such facilities. Therefore, airspace and NAVAID factors are not expected to be a constraint on the airport's capacity to accommodate anticipated (or even somewhat higher) demands.

AIRFIELD DEMAND/CAPACITY ANALYSIS

The term "capacity" implies that a given aircraft movement rate produces specified delays. Once a certain level of delay has been chosen as representing a maximum acceptable level, the capacity is reached when the movement rate reaches that level of delay. At general aviation airports, runways may generally be considered to have reached their hourly capacity when delays to departures average two minutes during the peak hours of the week. The hourly capacity can be exceeded, but the average delays during these periods will be increased beyond what is considered to be acceptable. The annual service volume (ASV) of an airport's runway or runway system is reached when the delay to operations over one-a-year period reaches a predetermined level, as defined and integrated within the FAA's capacity model, AC 150/5060-5, Airport Capacity and Delay, dated September 23, 1983, as revised.³

Aircraft mix (population) affects the capacity of the airfield. Aircraft using Cottage Grove State Airport are predominately single-engine piston aircraft. This is significant, because a mix of predominantly light aircraft will produce a much higher capacity for a given runway configuration than a mix which includes a substantial number of heavier aircraft, such as business jets or commercial transport aircraft. Through the forecast period, utility category aircraft are anticipated to account for more than 95 percent of all operations at the airport.

³This methodology, developed for the FAA in 1976 and found in Report No. FAA-RD-74-124, Techniques for Determining Airport Airside Capacity and Delay, provides more realistic measures of airfield capacity than the previous methodologies.

Based upon statements made by airport tenants and users and supported by Oregon Aeronautics Division staff, the most important improvement needed at the airport is a parallel taxiway. Safety and delays experienced during heavy-use periods were cited as the primary reasons for the need of this facility.

At the present time and throughout the planning period, the airport, according to FAA criteria, will not have an annual capacity problem. Based upon observations and concerns of the owners and users of CGSA, a parallel taxiway is vital to the expansion of the aircraft activity. The addition of the parallel taxiway will enhance the airport's operational capacity, as well as positively impact the safety of its operation. From the users' survey (See Appendix A), safety was the most important concern to the pilots and could have a major impact on the airport's overall competitiveness with other area airports. Spring and summer aircraft fly-ins also tax the airport capacity; thus, users point to these fly-ins as added reasons that a parallel taxiway is important.

RUNWAY LENGTH

Runway length requirements depend upon critical aircraft, the nonstop stage length to be flown, aircraft useful load (includes passengers, cargo, and fuel), and the aircraft performance characteristics. The takeoff performance characteristics of an aircraft, in turn, are affected by the environment at an airport. The more important environmental factors are:

- o Elevation of the airport
- o Ambient temperature
- o Surface wind
- o Runway gradient

The airport elevation is 641 feet MSL. For planning purposes, the ambient temperature is defined as the normal maximum temperature during the hottest month (82°F).⁴ A zero-wind situation poses the critical wind condition from the standpoint of runway length calculations. The existing percent-effective gradient of Runway 15-33 is 0.32 percent.

Table 5A outlines the runway length requirements for basic utility and general utility category aircraft. The highest use of the airport is by Basic Utility II category aircraft, but the airport is used occasionally by General Utility category aircraft. The operators on the Airport and the Community feel that the airport should be planned to accommodate the larger twin-engine turboprop and jet engine aircraft. To accommodate larger corporate aircraft on a regular basis would require a minimum runway length of 3,700 feet.

⁴FAA Advisory Circular 150/5300-4B, Utility Airports--Air Access to National Transportation, through Change 8, date 7-3-85.

The number of jet and turboprop aircraft presently utilizing the airport is low and, without a longer runway, is not projected to grow during the planning period. It is believed by many that a longer runway would stimulate this segment of the general aviation market and that a lengthening of Runway 15-33 is justified. Following a review of the potential market for larger aircraft utilizing the Cottage Grove State Airport, the Consultant recommends examining the feasibility of extending the runway to 3,700 feet in the analysis of airport development alternatives.

Table 5A
RUNWAY LENGTH REQUIREMENTS
Cottage Grove State Airport

	<u>Design Group</u>	<u>Approach Speed (kts.)</u>	<u>Aircraft Wingspan (ft.)</u>	<u>Maximum Design Weight(lbs.)</u>	<u>Required Runway Length(ft.)</u>
BASIC					
UTILITY					
Stage I	I	<91	<49	12,500	2,800
Stage II	I	<91	<49	12,500	3,100
GENERAL					
UTILITY	I	<121	<49	12,500	3,700

TAXIWAYS

Taxiways are constructed to facilitate aircraft movement to and from the runway. Taxiways parallel to the runway provide enhanced and increased capacity of the airfield system by reducing the runway occupancy time. In addition, exit taxiways aid in quickly channeling aircraft off the runway, thus increasing runway utilization. The spacing of exit taxiways relates directly to the type and size of aircraft for which the taxiway is designed. The smaller aircraft land and take off in a shorter distance; thus, taxiways designed for these aircraft are spaced more closely together.

The location of taxiways has a significant effect on the total capacity of the facility. The airport capacity issue discussed previously assumed construction of a parallel taxiway 150 feet from the Runway 15-33 centerline. In addition to the connecting taxiways at both ends of the runway, an additional mid-field taxiway will be required.

PAVEMENT STRENGTH

The pavement strength analysis of the runway, taxiway, and apron areas was based on the present and future critical aircraft operating on each surface to be paved. The critical aircraft

(in terms of pavement design) is defined as that which, because of its weight and landing gear configuration, produces the greatest pavement loading. Occasional operations conducted by aircraft with higher wheel loading--such as an aircraft diverted from its destination and required to land at Cottage Grove State Airport--should not be enough to adversely affect the conditions of the paved areas it traverses.

Facility requirements analysis indicates that the existing 12,500-pound weight-bearing capacity is adequate through the planning period. This strength would accommodate all General Utility category general aviation aircraft expected to use this airport on a regular basis. As discussed in the inventory, the condition of Runway 15-33 is good and does not need an immediate overlay for either strength or maintenance.

AIRFIELD LIGHTING

Runway 15-33 is currently lighted by low intensity runway lighting (LIRL). The lights on the runways are considered in fair condition and will require replacement within the planning period. Upgrading the runway lights to medium intensity will be required if a nonprecision instrument approach is approved for CGSA.

Taxiway lighting is often difficult to justify on a cost-benefit basis. Taxiway reflectors are quite adequate for most aviation operations. Ultimate lighting (medium intensity) of the primary taxiways is recommended, as funds become available.

The airport beacon is presently shielded by trees, making it difficult for pilots to see. The beacon is obsolete; and a new one, placed on an Occupation Safety and Health Administration (OSHA) approved tower, will be recommended.

TERMINAL AREA PLANNING

Future Aviation Museum

Oregon Aviation Museum, Inc., was organized in 1983 as a non-profit corporation with the following stated purposes:

- o To collect, preserve, restore, and exhibit Oregon aviation artifacts, including flight exhibitions.
- o To establish facilities for public viewing and exhibition of Oregon aviation artifacts and related objects.
- o To establish a resource center for Oregon aviation historical research.

- o To establish facilities for the restoration and preservation of museum artifacts.

After considering numerous locational criteria including accessibility to an airport, central location within the state of Oregon, accessibility and visibility from I-5, and nearby tourist facilities, the group selected Cottage Grove State Airport as a viable location for a future aviation and rail museum. A potential site has been identified and is shown on the Airport Layout Plan (Drawing 1, page 7-2). Representatives of this organization have contacted the property owners and are currently negotiating to secure the parcel.

Terminal area planning for the Cottage Grove State Airport was carried out to facilitate plans for this cultural resource and to ensure compatible visual aesthetics with the proposed museum.

BUILDINGS AND HANGARS

General Aviation Terminal Space

Public area for the general aviation passengers and pilots is presently provided by the fixed based operator. Future expansion of these facilities will be controlled by the factors of competition. If the terminal area becomes congested, there will be economic incentives for the local operators to expand their public areas.

Hangar Space

The space required for hangars and associated hangar apron at Cottage Grove State Airport is determined by the numbers and types of aircraft based at the airport. At present, there are 17 individual and T-hangar units, all of which are occupied. A few aircraft are stored in the one conventional hangar, which is used primarily for aircraft maintenance and short-term shelter. The remaining based aircraft are parked on the apron or grass tie-down area. State Aeronautics personnel report that there is a waiting list for hangar space.

Based upon (a) the existing pattern of demand at Cottage Grove State Airport, (b) forecasts for its future based aircraft fleet mix, and (c) trends at other airports, several assumptions have been made to forecast future hangar demand. In April 1987, 63 percent of all based aircraft were in hangars. Combining apparent local demand with common planning factors, it was assumed that the owners of 65 percent of the forecast aircraft would desire covered storage.

From these numbers, a determination was made for the percentage of aircraft which would utilize conventional-type open-space hangars as opposed to the individual T-hangar units. On this basis, it was assumed that 14 percent of the aircraft would be housed in conventional hangars. An area equal to 10 percent of

all the hangar space on the airport should be allocated to maintenance shop facilities, to be housed in conventional hangar space (i.e., FBOs). The remaining aircraft requiring storage will be in T-hangars. Aircraft hangar storage demand and capacity are shown in Table 5B.

Table 5B
HANGAR AND HANGAR APRON DEMAND/CAPACITY
Cottage Grove State Airport

	Existing Capacity	Forecast Demand			
		1998	1992	1997	2007
Number of Aircraft to be Hangared	22	24	29	34	45
Conventional Hangar Positions	5	5	6	7	10
Conventional Hangar Area (SF)	8,000	8,000	9,000	11,000	15,000
T-Hangar Positions	17	19	22	26	35
T-Hangar Area (SF)	27,000	30,000	35,000	41,000	55,000
Hangar Apron Area (SF)	34,000	38,000	45,000	52,000	71,000

Total hangar space needs were calculated after determining the number of positions. FAA planning standards were used, including 1,575 square feet per aircraft in T-hangars and 1,500 SF for aircraft in conventional hangars. However, it is recognized that smaller single-engine aircraft do not require this much hangar space. In addition, hangar apron areas should equal hangar area at a minimum. Hangar shop area and conventional apron area requirements are depicted in Table 5B.

There is an immediate demand for additional hangar storage capacity. Single-hangar units could be constructed in the near term, with a 9- to 10-unit T-hangar in the latter portion of the planning period. The recommended airport development plan will identify an area where additional T-hangar buildings can be constructed beyond the planning period, or if demand exceeds expectations. It is recommended that private development provide these non-eligible but needed airport improvements on leased land.

An important demand, as stated by airport users at Cottage Grove State Airport, will be for conventional hangar space. A portion of the long-term demand, in terms of gross area requirements, is projected for such facilities. It is anticipated that a combination of fixed-base-operator-type facilities and corporate

private hangars will be required through the planning period. The development of such facilities represents an opportunity for private investment in needed airport improvements at the airport (and in the community) which the State has encouraged. Areas for FBO and corporate general aviation development will be identified on the recommended airport development program plans.

Aircraft Parking Apron

Paved parking apron should be provided for at least the number of locally-based aircraft not stored in hangars, plus the number of transient aircraft parked on the "busy day." The amount of space required was converted from the number of positions forecast to be needed, including the planning standards of 300 square yards (SY) per based aircraft tie-down and 360 SY for itinerant tie-downs. FAA AC 150/5300-4B provides the methodology to determine itinerant parking requirements from an analysis of busy-day itinerant operations. Table 5C summarizes the aircraft parking apron capacity and demand.

The FBO apron area is already beyond capacity, with based aircraft tied down in spaces normally designated for transient aircraft. Some of the based aircraft in this area are tied down in grass. This area has tie-down spaces for approximately 18 single-engine aircraft. The itinerant parking apron adjacent to Row River Road is open and can provide space for almost 12 aircraft. This area appears to be more than adequate for transient aircraft operators using the Village Green, but few based aircraft owners desire to permanently park there. Security for stored aircraft is inadequate due to the lack of fencing and poor lighting around the apron area. Proper fencing would be a good addition to the area, but would not eliminate the potential for theft or increase the demand for based aircraft parking. Police patrolling and new lighting would probably have the most positive impact. The area requiring additional tie-down apron is in the FBO area.

With the 12 aircraft parking spaces for transients adjacent to the highway, only 5 or 6 additional transient aircraft tie-down locations need to be added in the FBO area. This information on transient tie-down requirements is the best estimate of the FBO. By the year 2007, a total of 17,000 SY of tie-down apron will be needed. Pavement strength should be for 12,500 pounds single-wheel load factor for all transient and based aircraft tie-down aprons.

Table 5C
LOCAL AND ITINERANT APRON DEMAND/CAPACITY
Cottage Grove State Airport

	<u>Existing Capacity</u>	<u>Forecast Demand</u>			
		<u>1988</u>	<u>1992</u>	<u>1997</u>	<u>2007</u>
Local Ramp					
Nonhangared Based					
Aircraft	13	14	16	20	25
Apron Area (SY)	3,900	4,100	4,800	6,000	7,500
Itinerant Apron					
Itinerant Parking Pos.	12	18	22	24	26
Apron Area (SY)	4,400	6,500	8,000	8,600	9,500

General Aviation Passenger Facilities

A general aviation "terminal" facility has several functions, including administration, pilot's lounge/flight planning area, food service area, waiting area, restroom, storage, and other needs. The space is not necessarily limited to a single building, but normally includes that provided by fixed based operators. Adequate public parking must be provided for these activities (at each location, if not contiguous).

General Aviation Traffic

General aviation-related automobile trip characteristics differ from itinerant and local operations. Local operations, which are principally training or instructional flights, are usually conducted in a series of touch-and-go operations. The aircraft generally carries only the pilot and, occasionally, an instructor (1.5 passengers per average flight). As such, a single local flight can account for approximately eight local operations (average of four touch-and-go's per flight). Itinerant flights, on the other hand, typically carry an average of 2.9 occupants. Other studies have indicated that the average automobile occupancy is approximately 1.2 itinerant aircraft passengers per one-way automobile trip and 1.0 for local passengers. Thus, each local aircraft operation generates approximately 0.1875 one-way vehicular trips and each itinerant operation generates 1.21 one-way trips. For planning purposes for general aviation activity, it is projected that the above trip-generation characteristics will remain fairly stable throughout the forecast period.

Application of the trip-generation characteristics for general aviation discussed above resulted in the following forecast volumes for general aviation vehicular traffic entering or leaving the airport:

Table 5D
GENERAL AVIATION
VEHICULAR ACTIVITY FORECAST, ONE WAY
Cottage Grove State Airport

<u>Year</u>	<u>Annual Vehicular Traffic</u>	<u>AADT^a</u>	<u>DHV^b</u>
1987 estimated	10,000	27	14
1992	12,600	35	17
1997	15,300	42	20
2007	20,000	55	25

a Average annual daily traffic

b Design-hour vehicles

General Aviation Passengers

By refining the above calculations, the number of general aviation enplaned passengers can be established. It should be noted that these values include pilots and instructors. Projections of general aviation enplaned passengers were developed and are tabulated in Table 5E.

Table 5E
FORECASTED GENERAL AVIATION ENPLANED PASSENGERS
Cottage Grove State Airport

<u>Year</u>	<u>Total Annual General Aviation Operations</u>	<u>Total Annual General Aviation Enplaned Passengers</u>
1987 (est. actual)	12,200	13,700
1992	15,400	17,300
1997	18,500	20,800
2007	24,500	27,600

Aircraft Fuel Storage Facility

The present fuel storage facility has two tanks, each having a 10,000-gallon capacity. One tank stores 80 octane and the other one stores 100 octane AVGAS. According to the FBO who dispenses the fuel, the storage capacity is adequate to handle today's demand and will be adequate to handle the future needs as projected in the study. Total fuel sales amounted to 62,000 gallons from May 1986 to April 1987. The average fuel sales per month is between 5,000 and 6,000 gallons. Higher sales are experienced during summer months when there is more aircraft activity. After a trial period of providing Jet A fuel on the airport, it was determined that the airport could not support the addition of Jet A fuel service, based upon projected demand now or in the future.

Circulation

In the foreseeable future, access to the airport and operations area is anticipated to continue to be via Palmer Road from Row River Road. Palmer Road provides single-lane access to and from the existing FBO parking area. The Highway Capacity Manual Special Report 87, National Research Council, 1965, that the per-lane, peak-hour capacity (assuming a 50-50 directional split) is 700 vehicles. The anticipated peak-hour, one-way vehicular traffic generated by all airport activity in 2007 is well below that capacity.

Automobile Parking

Parking area requirements in the vicinity of the general aviation terminal/FBO area include space requirements for passengers, employees, and visitors. The number of general aviation automobile parking spaces required is a function of the number of general aviation aircraft operations occurring over the busiest 3-hour period per week and the average aircraft occupancy. A 3-hour period was chosen, since FAA statistics indicate that the "average" general aviation flight lasts approximately 2.5 hours.

The busiest 3-hour period was determined from FAA data as reported in the General Aviation Activity Survey, prepared by the Department of Transportation, Federal Aviation Administration, and from the Consultant's experience with similar airports. As discussed previously in this chapter, local operations generate 0.1875 one-way trips, and itinerant operations 1.2 one-way trips.

The nature of general aviation flying is such that the average flight occurs within a single day and has a duration between 2 and 3 hours. Therefore, automobiles are usually left at the airport. Estimated future parking requirements are indicated on Table 5F.

Table 5F
GENERAL AVIATION AREA PARKING FORECASTS
Cottage Grove State Airport

<u>Year</u>	<u>Total General Aviation Operations</u>	<u>Automobile Parking Spaces^a</u>	<u>Area Requirements (acres)^b</u>
1987 (estimated)	12,200	42	0.32
1992	15,400	51	0.40
1997	18,500	60	0.46
2007	24,500	75	0.58

a Includes employee parking

b Assumes approximately 130 spaces per acre

SUMMARY

The intent of this analysis has been to ascertain the minimum development necessary to provide for unconstrained growth of Cottage Grove State Airport.

The projected terminal area requirements are summarized in Table 5G by planning period. It should again be cautioned that the decision to provide terminal area facilities is subject to many variables, not the least of which is economic considerations. Should additional State funds be available, it may be economically feasible to provide additional parking and hangar space to attract potential revenue-producing concessions. Additionally, it is possible that increase demand can be realized if increase capacity is provided in advance.

Table 5G
SUMMARY OF TERMINAL AREA REQUIREMENTS
Cottage Grove State Airport

<u>Facility</u>	<u>Existing</u>	<u>1987-92</u>	<u>1992-97</u>	<u>1997-2007</u>
Conventional Hangar Space	5	6	7	10
T-Hangar Spaces	17	22	26	35
Tie-Downs, Based	13	16	20	25
Tie-Downs, Transient	12	22	24	26
Auto Parking	42	51	60	75

**AIRPORT
DEVELOPMENT ALTERNATIVES**

Chapter 6

AIRPORT DEVELOPMENT ALTERNATIVES

This chapter is organized to identify alternatives for the future management, operation, and development of Cottage Grove State Airport. This analysis deals with on-airport development only. In subsequent analysis, the data detailed in this effort will be integrated with off-airport non-aviation-related development and economic factors to define the best way to develop and manage the Cottage Grove State Airport.

Previous chapters have been organized to provide the basis for this analysis. The Inventory was accomplished to set a sound foundation for the study effort. Future aviation demands have been calculated and compared with existing capacity in order to determine what facilities might be required now and in the future. This effort will be presented for public input and subsequently modified to serve as a foundation for a solid, usable plan. From this analysis, we have identified some items which are required to meet minimum safety considerations, certain items which would improve airport utilization and/or enhance operational capacity, and other items which might simply be desirable. In this chapter, all of these factors are brought together as a first step in the analysis of alternatives for the development of Cottage Grove State Airport.

FAA Airport Design Group Designations

Consideration of aircraft physical characteristics is the most important element in the design of airport facilities. Airplane wingspan is the principal physical characteristic affecting airport design. The FAA has developed the following Airplane Design Groups, as a factor of airplane wingspan. The following are airplane design groups as established by the FAA:

<u>Airplane Design Group</u>	<u>Wingspan</u>
I ^a	Up to but not including 49 feet (small airplanes)
I	Up to but not including 49 feet
II	49 feet up to but not including 79 feet
III	79 feet up to but not including 118 feet
IV	118 feet up to but not including 171 feet
V	171 feet up to but not including 197 feet
VI	197 feet up to but not including 262 feet

Source: FAA Advisory Circular 150/5300-12, dated
2-28-83.

DEVELOPMENT CONSIDERATIONS

The development standards listed below have been extracted from FAA guidelines based upon the following airport role characteristics:¹

Design Aircraft: Basic Utility Stage II or General Utility Stage I - single-engine and small, twin-engine business and air-taxi-type airplanes with a maximum gross landing weight of 12,500 pounds.

Design Group I and I^a - aircraft with wingspans up to but not including 49 feet (15 meters).

To facilitate readers' understanding, the 12,500-pound maximum gross weight relates to aircraft design specifications of the aircraft manufacturer. This means that larger aircraft without passengers, etc., could use the airport for practice. The 12,500-pound aircraft category includes the Beech King Air and Cessna Citation twin-engine business types. The Design Group I aircraft designation relates to aircraft with a wingspan of less than 49 feet. This would relate to all of the single-engine as well as the smaller, twin-engine business type aircraft.

INTRODUCTION TO DEVELOPMENT ALTERNATIVES

The existing Cottage Grove State Airport has been developed around the Basic Utility Airport standards, due to the majority of aircraft that use the facility. Normally, the standard planning process dictates that planning for future facility needs only takes into account the anticipated aircraft that will use the airport. In the case of the Cottage Grove State Airport, there is the possibility that improvements to facilities and runway length will draw larger types of turboprop and pure jet aircraft.

As one of the alternatives, the Consultant looked at the expansion of the airfield to meet General Utility type airport standards. It was determined early in the Master Planning process that topographical and environmental constraints might be the greatest deterrents to extending the runway to the General Utility design standards. However, it was determined that the Master Plan should look at the feasibility of extending the runway and should evaluate the cost/benefit of such an alternative.

¹ FAA Advisory Circular 150/5300-4B, Utility Airports - Air Access to National Transportation, through Change 8, dated July 3, 1985.

The standards have been extracted from FAA Advisory Circular (AC) 150/5300-4B as revised, AC 150/5300-12, and Federal Aviation Regulation (FAR) Part 77.

Table 6A
AIRPORT TYPES

	<u>Basic Utility Stage I</u>	<u>Basic Utility Stage II</u>	<u>General Utility</u>
RUNWAY LENGTH (feet)	2,600	3,100	3,700
<u>AIRPLANE DESIGN GROUP</u>			
	<u>I^a</u>	<u>I</u>	<u>I</u>
RUNWAY 15-33:			
Width (feet)	60	60	60
Strength (pounds)	12,500	12,500	12,500
RUNWAY SAFETY AREA			
Length (feet)	240 (beyond runway end)	240 (beyond runway end)	240 (beyond runway end)
Width (feet)	120	120	120
MARKINGS	Basic	Basic	Basic
TAXIWAYS			
Width (feet)	25	25	25
Markings	Basic with guidance signs	Basic with guidance signs	Basic with guidance signs
SEPARATIONS:			
Parallel Taxiway (ft)	150	225	225
Building			
Restriction (ft)	125	200	200
Taxiway to Aircraft			
Parking (ft)	44	44	44

^a Airports served only by small (12,500 lb.) aircraft.

Source: FAA Advisory Circular 150/5300-4B, as revised through Change 8

Other than the longer runway length alternatives, the other alternatives will center around taxiway and terminal area development. The forecasts of aircraft activity show a requirement for a number of short-term as well as long-term improvements. The process of alternative development deals with the facility requirements and the different methods for accomplishment.

No-Build Alternative

The Do-Nothing Alternative, commonly called No-Build, is always a consideration. The decision not to provide additional airfield or general aviation facilities would likely have an impact on the airport's capacity to meet future demand. This alternative does not imply letting existing facilities deteriorate or become unsafe. Therefore, ongoing maintenance and operating costs would continue to be a public burden.

This alternative would essentially allow the State to operate CGSA as a Basic Utility type airport with little expansion of the general aviation facilities. The airport, at the present time, is not a self-sustaining operation, requiring State contributions for capital improvements. This alternative would continue the deficit operation unless the total burden for sustaining the airport was passed on to existing tenants and aircraft owners, who would be unlikely to accept this.

Existing runway lengths and weight-bearing capacities of all facilities are adequate to handle the types and weight of a majority of aircraft presently using the facilities. The aircraft storage facilities would offer a significant constraint to expanding based aircraft. The present taxiway situation is considered less than desirable. Without a parallel taxiway, CGSA will experience decreased capacity and delays, as well as the safety factors inherent with dual use of the runway as a taxiway. The condition of most other facilities is rated as very good to excellent and will not require immediate maintenance attention.

Clearly, the airport will be inadequate to satisfy all demands projected in this report. It is important to point out, however, that the runway is not anticipated to be at capacity through the planning period. And since, by law, the State will not be able to restrict aircraft from using the runway unless they would create damage (too heavy for example), it is likely that the operational activity will continue to increase.

Service From Another Airport Alternative

An analysis of area airports within a reasonable distance of the Cottage Grove State Airport was conducted. Three airports were found within a reasonable distance from Cottage Grove--Creswell's Hobby Field, Eugene's Mahlon Sweet Airport, and Walker Field, a private airport.

Mahlon Sweet Airport, approximately 30 minutes to the north, is the air carrier airport serving the area. It has runways long enough for large corporate and transport aircraft and a precision Instrument Landing System (ILS). Mahlon Sweet will continue to be the airport of choice for the larger, more sophisticated corporate jet aircraft. Creswell's Hobby Field, located eight miles to the north of Cottage Grove, is a facility similar to that at the Cottage Grove State Airport. Creswell now has a 3,100-foot-long runway, though this recent project does not increase its based aircraft capacity. Walker Field is a private airport and is not capable of expanding or serving the existing demand at Cottage Grove State Airport.

It has been questioned whether both (Hobby Field and Cottage Grove) airports are required, due to their close proximity and the fact that they serve similar sizes of aircraft. It is the Consultant's opinion that both airports serve a market specific to their location. Hobby Field draws business mostly from South Eugene and the Creswell area, while Cottage Grove State Airport serves its immediate area and as far south as Roseburg. Cottage Grove State Airport also has businesses like the Village Green, which have customers who use the airport as access to their specific activity. Without the airport, much of this business would be lost. The Cottage Grove State Airport also serves a much larger and more industrial commercial community than Hobby Field. A survey (Appendix A) conducted in June of 1987 has confirmed that a significant amount of activity at CGSA is business-related.

Alternative 1 - New Runway

This alternative assumes that Cottage Grove State Airport changes from a Basic Utility to a General Utility type of airport. This alternative requires a 500-foot extension to the runway or, as shown, a realignment of the runway. The realignment and reconstruction of the entire runway would be required due to the location of the Row River at both ends of the runway. At the north end of the runway, the edge of the river is approximately 520 feet from the existing pavement surface. At the south end, the runway is only 180 feet from the edge of the river.

With the requirement for a 240-foot safety area at each end of the runway, it is apparent that the runway could not be extended to the south. Maintaining the safety area to the north of the existing runway end would only allow for an extension of 200 feet, or 300 feet short of the desired runway length of 3,700 feet. Therefore, achieving the desired runway length would require runway relocation, with the north end positioned to the east approximately 100 feet from the existing runway end. Even though the relocated runway would allow for the additional 500 feet of pavement, it would position the runway for ultimate extension to approximately 4,700 feet, if demand were to dictate the need.

This alternative would require the purchase of approximately 14.8 acres of property, including about 7.9 acres of farmland, 5.5 acres of residential property, and 1.2 acres of industrial land. The estimated cost to acquire these properties is \$70,000. Estimated construction costs for the relocated runway and parallel taxiway total \$574,000, bringing the construction and land acquisition total estimate for this development alternative to \$644,000.

Further, this alternative would require an additional approximately 9.68 acres in the proposed northerly clear zone and 1.45 acres in the proposed southerly clear zone be controlled under aviation easements.

There are several environmental factors which would have to be analyzed in detail before selecting this alternative for development. Of the alternatives under consideration, Alternative 1 would involve the greatest amount of construction within the designated 100-year floodplain of the Row River. It would also require the greatest amount of land acquisition. Alternative 1 would require the acquisition and relocation of residences. It would also require acquisition and conversion of property classified as prime agricultural land. Were this development pursued, some clearing of trees would likely be required northwest of the existing runway.

Adjustments to the southerly aviation easements would be needed while, to the north, significant new land should be included in aviation easements. It appears that Alternative 1 has the potential for negative noise impacts upon existing residential land uses to the north and west, although this will be verified or disproven when noise contours are generated at a later point in the study.

The additional runway length would allow the airport to move up to the General Utility Airport type, thus allowing the use of the facility by larger and more sophisticated pure jet, corporate aircraft, in addition to the Cessna Citation which already uses the airport.

Alternative 2 - Runway Extension and Parallel Taxiway

It was determined early in the study process that one of the alternatives should be the determination of the maximum length to which the existing runway could be extended. The existing runway exceeds the design standards as established by the FAA for a Basic Utility Airport. The runway length required for a Basic Utility Stage II runway is 3,100 feet. The existing runway is 3,200 feet long.

We have already determined in Alternative 1 that the runway cannot be extended to the south, while still maintaining the 240-foot safety area. The safety area is now approximately 180 feet, or 60 feet too short. By maintaining a 240-foot safety area to the north, the runway could be extended approximately 190 feet.

No additional land would be required for the runway extension or a parallel taxiway constructed with a 150-foot separation (see taxiway discussion under Alternative 3).

Estimated costs for construction of the 200-foot runway extension and parallel taxiway would be approximately \$201,000. Land purchase would not be required with this development alternative, although obtaining an avigation easement over an additional 2.8 acres in the proposed northerly clear zone is recommended.

A significant amount (approximately half) of the development proposed in Alternative 2 is within the floodplain. The runway extension and a small portion of the connecting taxiway are situated in the floodway, which is considered to be the most sensitive area in a floodplain. Lane County requires that fill and construction within the floodway be designed by an Oregon-licensed engineer and meet all provisions of the Floodplain Combining Overlay Zone.

Under this development alternative, a small amount of agricultural land in the northerly clear zone would be recommended to be controlled through an avigation easement. Some trees and riparian vegetation may be required to be removed or reduced in the proposed Clear Zone.

The longer runway length would give an added margin of safety and could enhance aircraft operating performance, but it would not be adequate to allow use by larger aircraft permitted at General Utility type airports. It has been determined that the additional 190-feet of runway, although beneficial, would not expand the use of the airport to larger types of aircraft than presently exist.

Alternative 3 - Parallel Taxiway

This alternative assumes that no new runway will be constructed and the present runway will not be extended. A parallel taxiway has been proposed as the only significant improvement. The taxiway will run the full length of Runway 15-33 and be 25 feet wide. The FAA Design Standard for the facility is Design Group I. This alternative shows the runway-taxiway separation at 150 feet. This separation would relegate the airport to single-engine and small, twin-engine aircraft but would eliminate the need for acquiring residential, industrial, and agricultural properties. The taxiway would maximize the airport's capacity by reducing the time each arriving or departing aircraft spends on the runway. Adequate access taxiways would have to be constructed with the primary taxiway to insure smooth traffic flow. The addition of a parallel taxiway would substantially increase the airport's safety and capacity.

Construction costs for the taxiway are estimated to be \$160,000. No land acquisition or additional avigation easements are required under this development scenario.

By placing the taxiway in the configuration proposed in Alternatives 2 and 3, adequate land area to the west of the taxiway would remain for clearances established by the FAA.

The 150-foot runway/taxiway separation would also leave approximately 90 feet of land to the west of the taxiway that would be available for grass overflow tie-downs. The land necessary for this option is owned and controlled by the state.

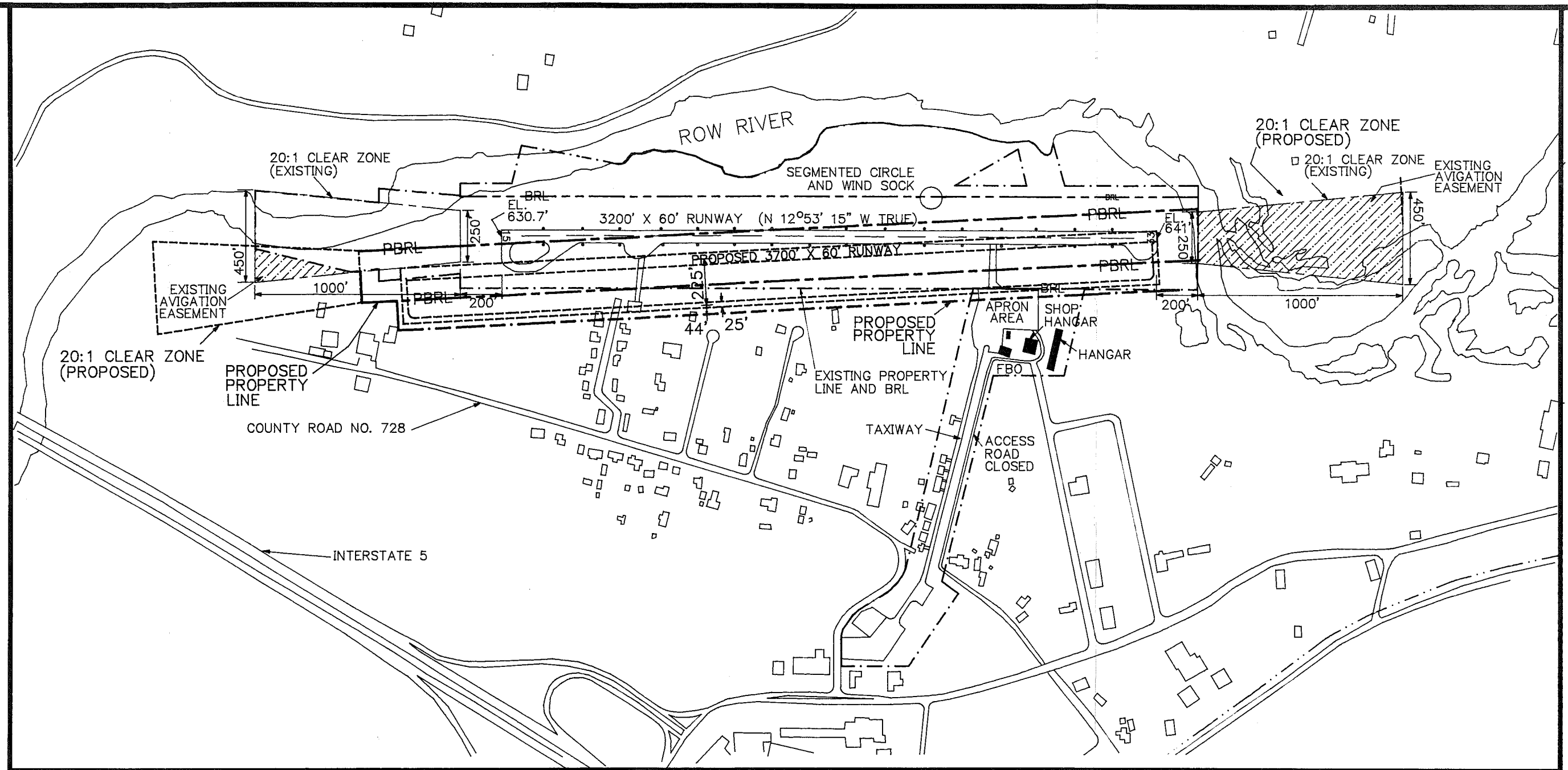
Development proposed under Alternative 2 also involves construction and fill within the floodplain, thus would have to comply with requirements of the County Floodplain Combining Overlay Zone. It appears that there would be limited construction and fill in the floodway associated with the connecting taxiway.

PRELIMINARY CONCLUSIONS

Each alternative was analyzed relative to cost factors and compared for environmental concerns in this portion of the study, although cost estimates and environmental issues developed were limited to the significant improvements. In this Report, general aviation development alternatives are depicted in conceptual terms and drawings. Based on information available at this juncture of the study, the Consultant's preliminary conclusion was that Alternative 2 would best serve Cottage Grove State Airport.

Though a preliminary selection has been made, the alternatives for airfield and terminal area development will continue to be evaluated throughout remainder of the planning process. Input received will be utilized in the determination of the preferred alternative, which will be reflected in the final Plan.

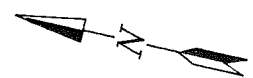
Chapter 7, Airport Plans, of this Report will document the final Plans, which will be developed once the preferred alternative has been selected. More specific cost estimates will be addressed in Chapter 8, Implementation Plan.



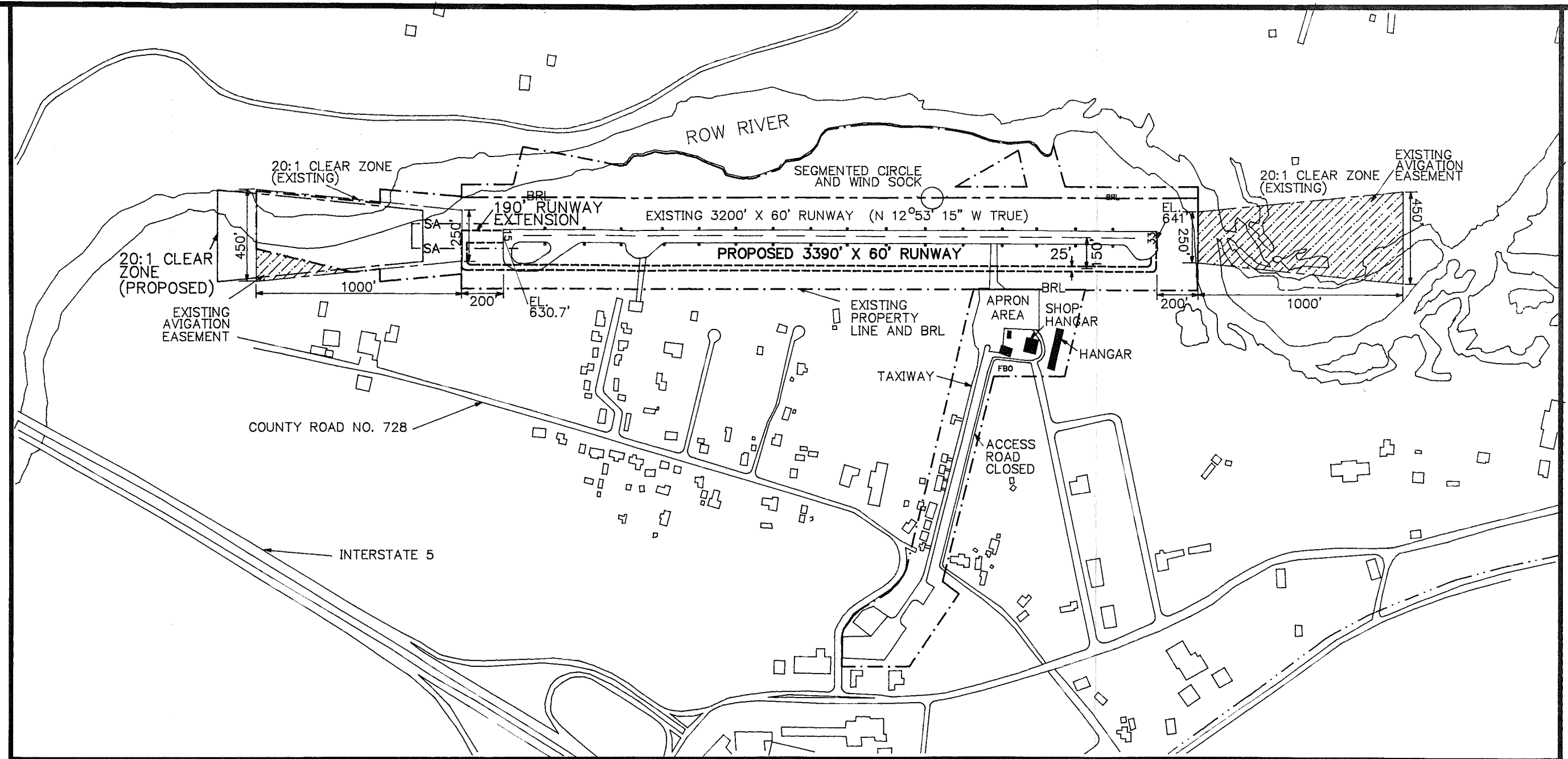
COTTAGE GROVE STATE AIRPORT
MASTER PLAN UPDATE

DEVELOPMENT ALTERNATIVE 1 NEW RUNWAY

FIGURE 6



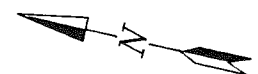
0 600 1200
SCALE IN FEET



COTTAGE GROVE STATE AIRPORT
MASTER PLAN UPDATE

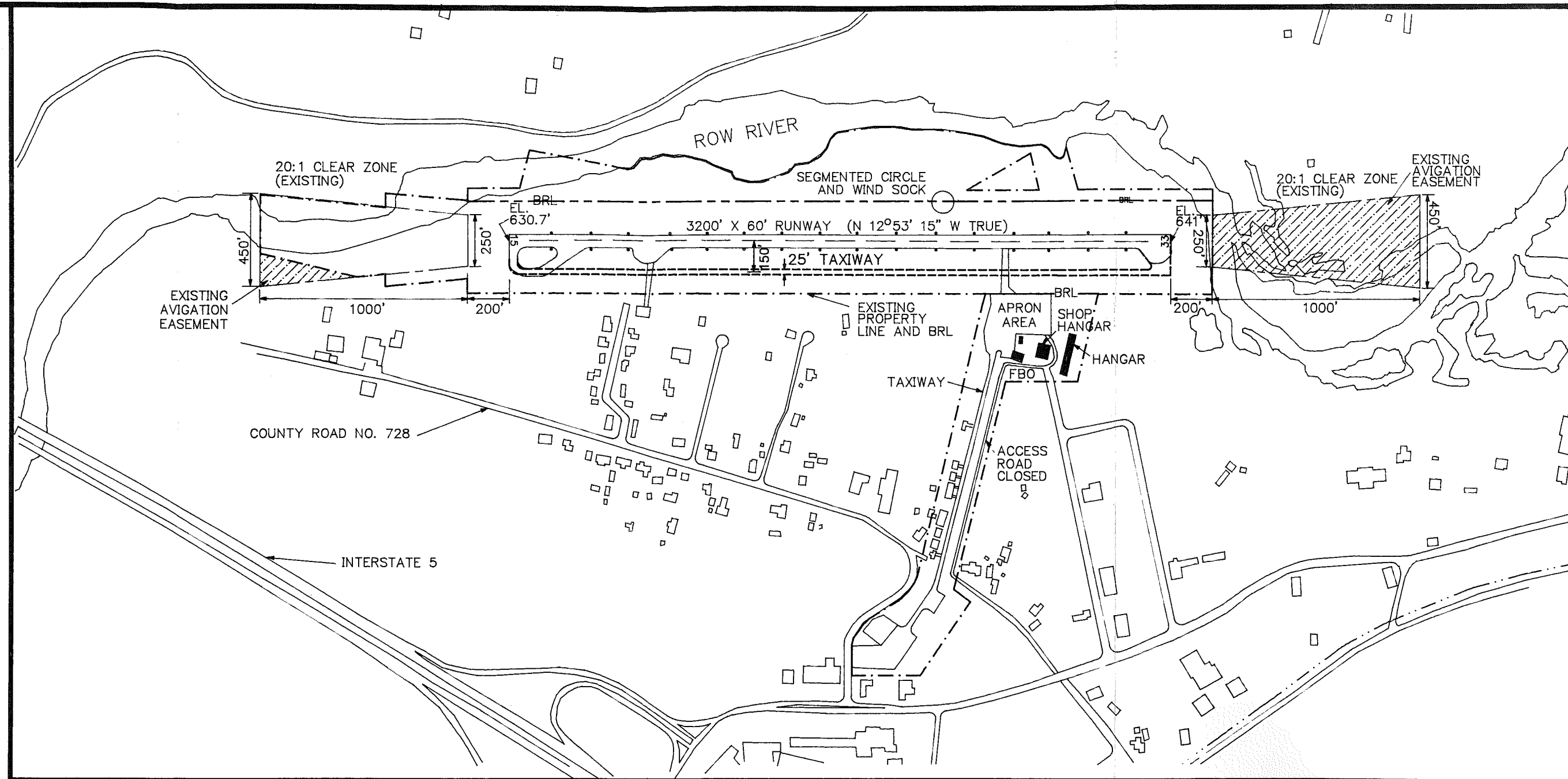
DEVELOPMENT ALTERNATIVE 2 RUNWAY EXTENSION (PREFERRED ALTERNATIVE)

FIGURE 7



0 600 1200
SCALE IN FEET

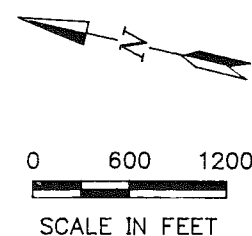
FORESITE GROUP



COTTAGE GROVE STATE AIRPORT
MASTER PLAN UPDATE

DEVELOPMENT ALTERNATIVE 3 PARALLEL TAXIWAY

FIGURE 8



AIRPORT PLANS

Chapter 7 AIRPORT PLANS

INTRODUCTION

This chapter discusses the physical airport plan, as established in this study process. This plan has evolved through meetings with airport tenants and the public. The data developed in Chapters 3 through 6 has been assessed, discussed, and, in many cases, revised. The plan has been reviewed by the Federal Aviation Administration (FAA) Seattle Airports District Office and reflects their comments.

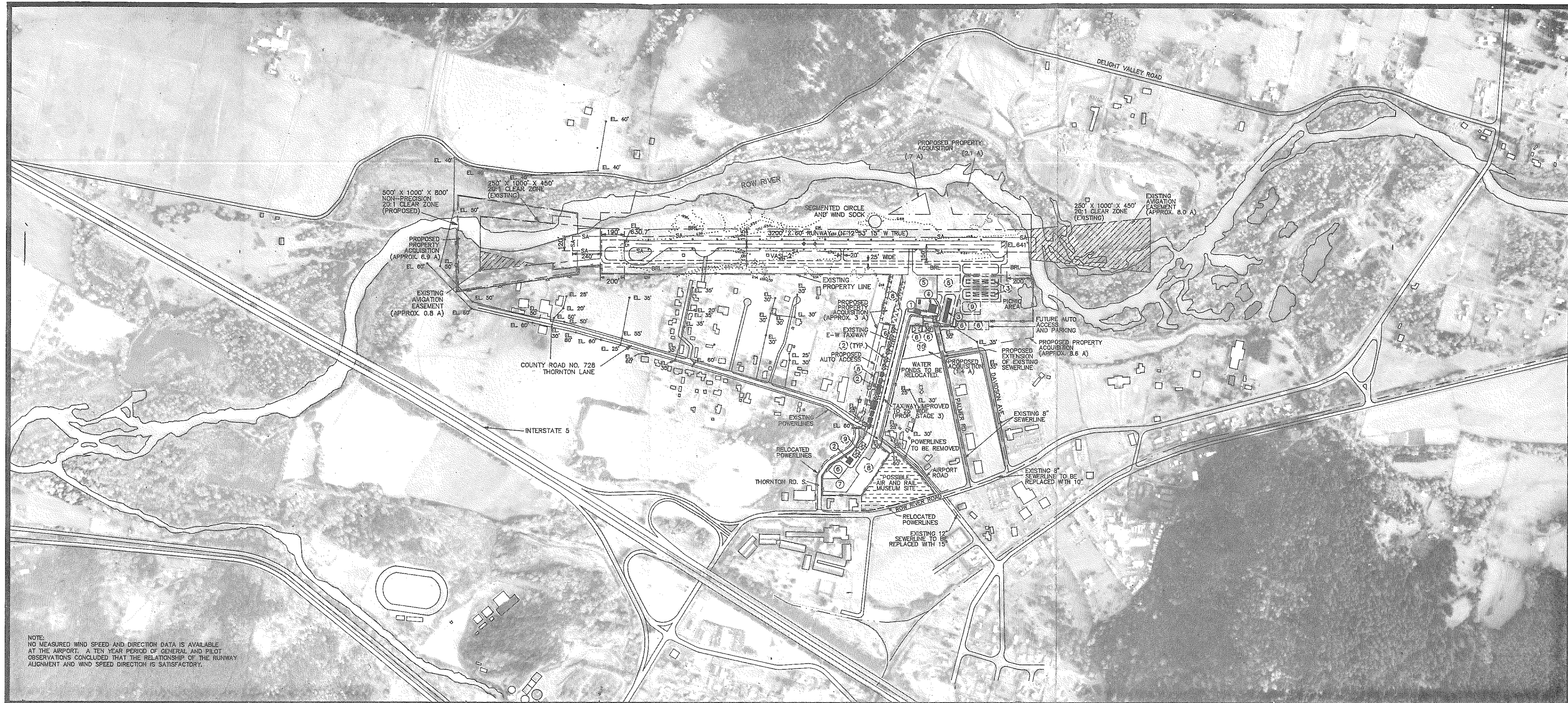
The Oregon State Aeronautics Division, considering public input and comments by various interest groups has approved the following plan and has directed the preparation of the final implementation elements as documented in this and subsequent chapters of this report.

AIRPORT LAYOUT PLAN

The Airport Layout Plan for Cottage Grove State Airport is shown on Drawing 1. The Plan is designed to meet the requirements for Airport Master Plans, as established by the FAA in Advisory Circular (AC) 150/5070-6A, and was developed using current FAA standards applicable to the types of activity and aircraft anticipated at the airport throughout the forecast period. Major recommended airport improvements are as follows:

- o Land Acquisition
- o Runway 15-33 Extension
- o Parallel and Access Taxiway
- o Aircraft Parking Apron Expansion
- o Staged Expansion of General Aviation Facilities
- o NDB/Loran C Instrument Approach
- o Obstruction Removal

Land acquisition is always a very important element in maintaining the viability of an airport. The State of Oregon owns or controls most of the land required for the existing operation and activity level. Additional land is required if the State is to provide adequate facilities to meet the projected growth in activity. The land proposed for acquisition is located primarily in the terminal/hangar area. This land would be used for the development of new aircraft storage facilities, an aircraft parking apron, and expansion and improvement of automobile parking. An additional small parcel of land will allow greater utilization of the east-west taxiway by larger aircraft taxiing to the westernmost transient parking area.

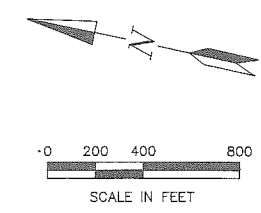


BASIC DATA TABLE AIRPORT DATA		
	EXISTING	PROPOSED
AIRPORT ELEVATION (MSL)	641'	SAME
AIRPORT REFERENCE POINT (ARP)	44°48' 00" N LAT. 123°01' 40" W LONG.	44°48' 00" N LAT. 123°01' 41" W LONG. (APPROX.)
TERMINAL NAVIGATIONAL AIDS	NONE	NDB
NORMAL MAXIMUM TEMPERATURE IN HOTTEST MONTH	82° F	SAME
AIRPORT SYSTEM CLASSIFICATION	GENERAL AVIATION	SAME
TAXIWAYS		
MARKING	BASIC	SAME
LIGHTING	MIRL	SAME
WIDTH	AS SHOWN	SAME
SURFACE	ASPHALTIC CONCRETE	SAME

BASIC DATA TABLE RUNWAY DATA		
	EXISTING	PROPOSED
RUNWAY LENGTH (FEET)	3200'	3390'
RUNWAY WIDTH (FEET)	60'	SAME
EFFECTIVE GRADIENT (%)	.37%	SAME
PERCENT WIND COVERAGE	SEE NOTE	SAME
RUNWAY INSTRUMENTATION	VISUAL	NONPRECISION
PAVEMENT STRENGTH (LBS.)	12,500 SW	SAME
APPROACH SURFACE (FAR PART 77 MINIMUM)	R/W 15 - 20:1 R/W 33 - 10:1	R/W 15 - 20:1 R/W 33 - 20:1
OPERATIONAL ROLE (FAR PART 77)	BASIC UTILITY	SAME
LIGHTING	MIRL	SAME
MARKING	BASIC	SAME
LANDING AIDS	VASI-2	SAME
SURFACE MATERIAL	ASPHALTIC CONCRETE	SAME

BUILDINGS/FACILITIES		
	EXISTING	PROPOSED
FBO	(1)	(1)
HANGAR	(2)	(2)
T-HANGAR	(3)	(3)
SHOP/HANGAR	(4)	(4)
APRON AREA	(5)	(5)
AUTO PARKING	(6)	(6)
TRANSIENT RAMP	(7)	(7)
TIE DOWN AREA	(8)	(8)
CORPORATE HANGAR AREA	(9)	(9)
EXPANSION AREA	(10)	(10)

LEGEND		
	EXISTING	PROPOSED
FACILITIES		
BUILDINGS		
AIRPORT PROPERTY LINE		
SECURITY FENCE		
BUILDING RESTRICTION LINE (BRL)		N/A
SAFETY AREA (SA)		N/A
AIRPORT REFERENCE POINT (ARP)		



FEDERAL AVIATION ADMINISTRATION	STATE OF OREGON DEPARTMENT OF TRANSPORTATION DIVISION OF AERONAUTICS	COTTAGE GROVE STATE AIRPORT COTTAGE GROVE, OREGON AIRPORT LAYOUT PLAN DRAWING NO. 1 OF 3
APPROVAL DATE: _____ SEE APPROVAL LETTER: _____	APPROVAL DATE: _____ SEE APPROVAL LETTER: _____	DATE: DECEMBER 1987 DRAWN BY: L.D. HOWARD APPROVED BY: J.R. HAWLEY
SIGNATURE _____	SIGNATURE _____	
 Corvallis, OR Irvine, CA Seattle, WA		

The preparation of this document was financed in part by a planning grant from the FAA as provided under Section 505 of the Airport and Airway Improvement Act of 1982. This document does not necessarily reflect the views of the FAA until reviewed and signed.

Another two parcels recommended for acquisition are in the expanded clear zone for the 190-foot extension to the north end of Runway 15-33. This area includes one parcel that is now controlled by an avigation easement. We are recommending that the State acquire the property rather than purchase a new and more restricted easement. Ownership provides full control over development and obstruction removal. Continual costs to lower tree heights and other obstructions can be eliminated with ownership. Obstructions can then be removed rather than cut back, thus providing a safer unobstructed area for an aircraft to use in case of an emergency. There are two small land parcels directly east of the runway that the State does not own. We are recommending that these also be included in the total program.

The most significant improvement to enhance the airport operation is the parallel taxiway and access taxiways. The taxiway will greatly improve the flow and safety of aircraft operating at the airport. Aircraft presently using the airport must either complete a 180-degree turn and taxi down the runway following a landing, or taxi down the runway in preparation for a takeoff. Operating in this fashion offers the potential for an accident. A parallel taxiway would allow aircraft to exit and enter a runway at several points, thus eliminating time-consuming taxiing and an unsafe condition. The projected airport activity alone cannot support the addition of a parallel taxiway, but the improved safety of the airport's operation can easily justify its development.

A runway extension of 190 feet is proposed for the north end of Runway 15-33. The added length was established by using FAA AC 150/5300-4B criteria for Design Group I^a airports, which requires that a 240-foot safety area exist off of each end of the runway. The Row River winds around the north end of the runway, thereby limiting the ultimate length of the facility in its existing location.

A thorough evaluation of the future critical aircraft produced no justification for assuming the existing mix of aircraft would change. Although an occasional pure jet lands at the airport, the Consultant could not support pure jet traffic on a frequent basis. It was determined that larger corporate type aircraft, which are unable to land and take off from Cottage Grove State Airport, would prefer to use Eugene's Mahlon Sweet Airport because it has longer runways and a precision instrument approach.

A General Utility runway was considered as a development alternative in Chapter 6. To achieve the full runway length for that FAA category airport, it was determined that construction of a new 3,700-foot runway would be required. The new facility was designed to remain clear of the Row River to allow development of the facility's full projected length, with additional length possible if later demand warrants. This alternative was also the most costly, due to the need for construction of a totally new runway and the sizable amount of land to be acquired.

TERMINAL AREA PLAN

The need for additional tie-down apron and storage hangars to meet the projected demand was the next important improvement. Based and transient aircraft parking tie-downs are required in the terminal/hangar area. As previously discussed, land must be acquired to facilitate the development of new apron areas and to construct additional T-hangars. An area to the north of the existing aircraft parking area will be paved in stages for future based aircraft tie-downs. A small unpaved area to the east and adjacent to the existing terminal apron will be paved to accommodate based as well as transient aircraft.

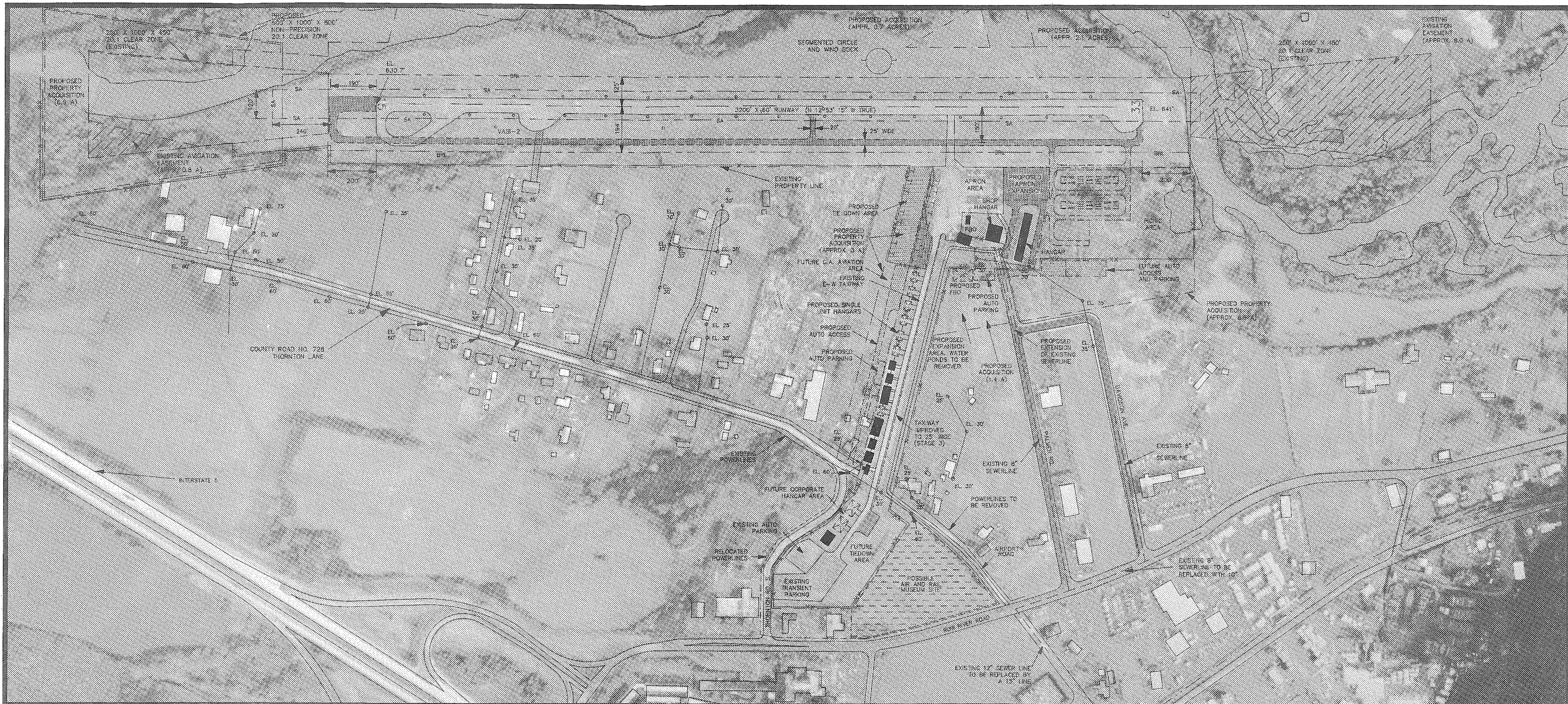
An area directly south of the present T-hangar will be acquired to allow construction of two, possibly three, ten-unit T-hangars. T-hangars are normally developed by private companies and individuals, not by the airport owners. This allows the owners to invest their limited dollars in airfield improvements. The area is large enough to allow replacing one T-hangar unit with three corporate hangars if there is a demand for this type of facility. Based upon the projected needs for hangar space, two T-hangar units will be more than adequate to meet the airport's projected demand.

Several locations are shown on the ALP and Terminal Area Plan (Drawing 2) for automobile parking. Each area requires separate access, and they serve entirely different use areas. One area is an expansion of the existing terminal/hangar auto parking areas, while another is to the north of the east-west taxiway. The latter auto parking area would require that a new roadway be developed adjacent to the row of single-unit hangars which now have access via the east-west taxiway.

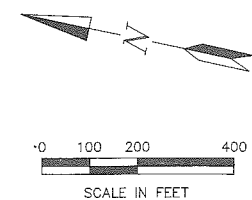
There is adequate space along the east-west taxiway for an additional seven single-unit hangars. An additional six units could be built in the open area to the northeast of the transient tie-down apron off Thornton Road. Open land, across the transient apron area and owned by the State, is also available for tie-down apron, if future demand warrants. This area has been designated for future aviation development beyond the planning period.

A parcel of privately owned property, north of the intersection of Row River Road and Airport Road, has been designated as a potential site for a future aviation and rail museum. Representatives of Oregon Aviation Museum, Inc., a non-profit corporation which was formed in 1983 for the purpose of establishing a facility to house aviation historical artifacts, have contacted the owners of this parcel and are negotiating toward securing the property as a future museum site.

Single-unit hangars that are privately owned are increasing in popularity and are well established on CGSA. It appears that this trend will continue through the planning period. It is the Consultants' recommendation that only the area adjacent to the



LEGEND		STAGED DEVELOPMENT	
	EXISTING	PROPOSED	
FACILITIES	—	—	STAGE I (1988 - 1992)
BUILDINGS	■	▨	STAGE II (1992 - 1997)
AIRPORT PROPERTY LINE	—	—	STAGE III (1997 - 2008)
SECURITY FENCE	—X—X—	—XX—XX—	
BUILDING RESTRICTION LINE (BRL)	—BRL—	N/A	
SAFETY AREA (SA)	—SA—	N/A	
AIRPORT REFERENCE POINT (ARP)	⊕	⊕	



FEDERAL AVIATION ADMINISTRATION APPROVAL DATE: _____ SEE APPROVAL LETTER: _____ SIGNATURE _____		STATE OF OREGON DEPARTMENT OF TRANSPORTATION DIVISION OF AERONAUTICS APPROVAL DATE: _____ SEE APPROVAL LETTER: _____ SIGNATURE _____		COTTAGE GROVE STATE AIRPORT COTTAGE GROVE, OREGON TERMINAL AREA PLAN DRAWING NO. 2 OF 3 DATE: DECEMBER 1987 DRAWN BY: L.D. HOWARD APPROVED BY: J.R. HAWLEY	
FORESITE GROUP Corvallis, OR Irvine, CA Seattle, WA					

east-west taxiway be used for single-unit hangar development. The proposed hangar area to the south of the terminal/hangar area should remain for either multiple-unit T-hangars or corporate hangars.

At least one conventional hangar is projected to be required by the year 2008, and an area just west of the airport FBO Office has been reserved for this purpose. The present locations of the airport's FBO office, lounge, and maintenance hangar restrict the location for the development of a second conventional hangar. Conventional FBO hangars need adequate airfield exposure to be competitive with other airport service companies. This proposed location uses the land available with reasonable exposure to the flying public. Adequate parking requirements have also been determined and provided for in the plan, as mentioned previously.

Throughout the planning period, the terminal area will provide for approximately 45 based aircraft. New hangars will provide space for about 15 additional aircraft, with an additional 25 aircraft tie-down spaces provided for the increasing demand for based and transient aircraft.

Access

The existing access to the airport is Palmer Road, via Row River Road. This is a fairly recent improvement, changing the airport entrance. Palmer Road was designed and constructed for the surrounding industrial park. The original entrance road which ran parallel to the east-west taxiway has been closed. The Airport Master Plan shows no changes to the primary airport entrance throughout the planning period.

A new access road is recommended to be constructed to the north of the hangars that open up to the east-west taxiway. This new roadway would provide access to all the existing hangars as well as to the proposed tie-down area to the north of the terminal area. The new roadway would provide vehicle access to the row of hangars, eliminating the need to drive down or across the east-west taxiway.

Access to the airport, as discussed earlier, is now very good, due to the new roadway and limited industrial development. Increased construction of new facilities in the industrial park will tend to reduce the airport's exposure to Row River Road. At the present time, airport directional signage is adequate to help people find the airport entrance. In the future, however, the sign program should be expanded to include the marketing of airport and aviation services.

Automobile parking is presently adequate, since most aircraft owners and users drive directly to their aircraft or hangars. It is unsafe to mix aircraft and automobile operations, and the Master Plan provides a means to accomplish separation of these activities. New auto parking areas have been designed, along with the expansion and improvement of the existing terminal area parking lot. The proposed hangar area to the south of the terminal will have adequate parking for aircraft owners renting hangar space.

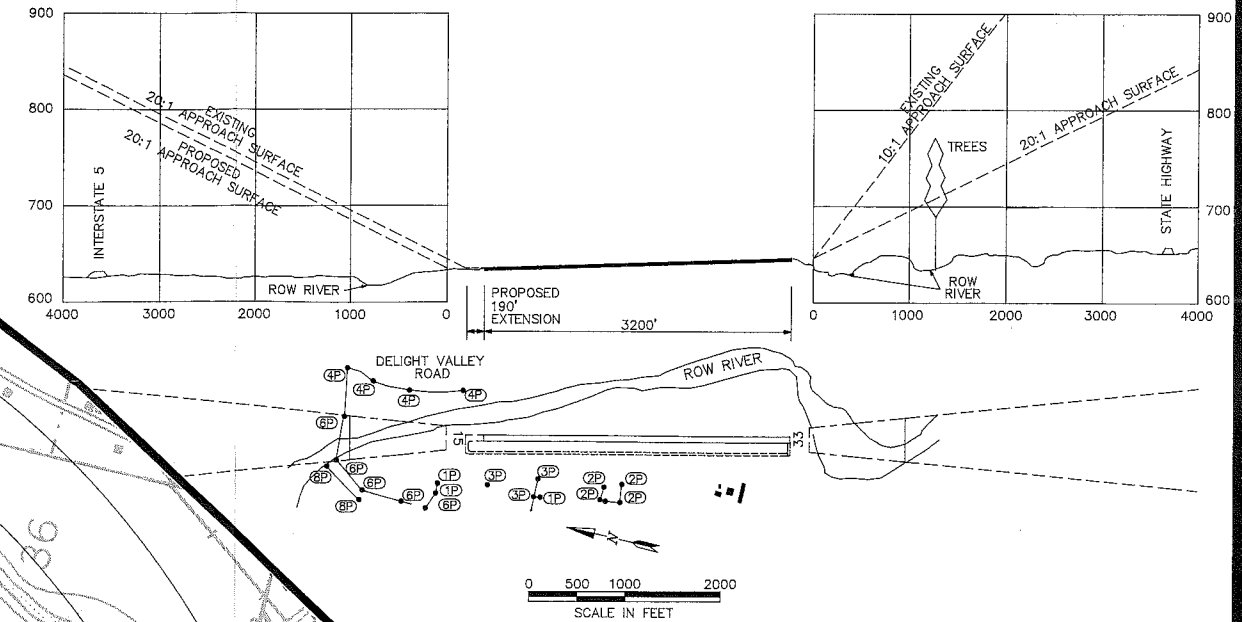
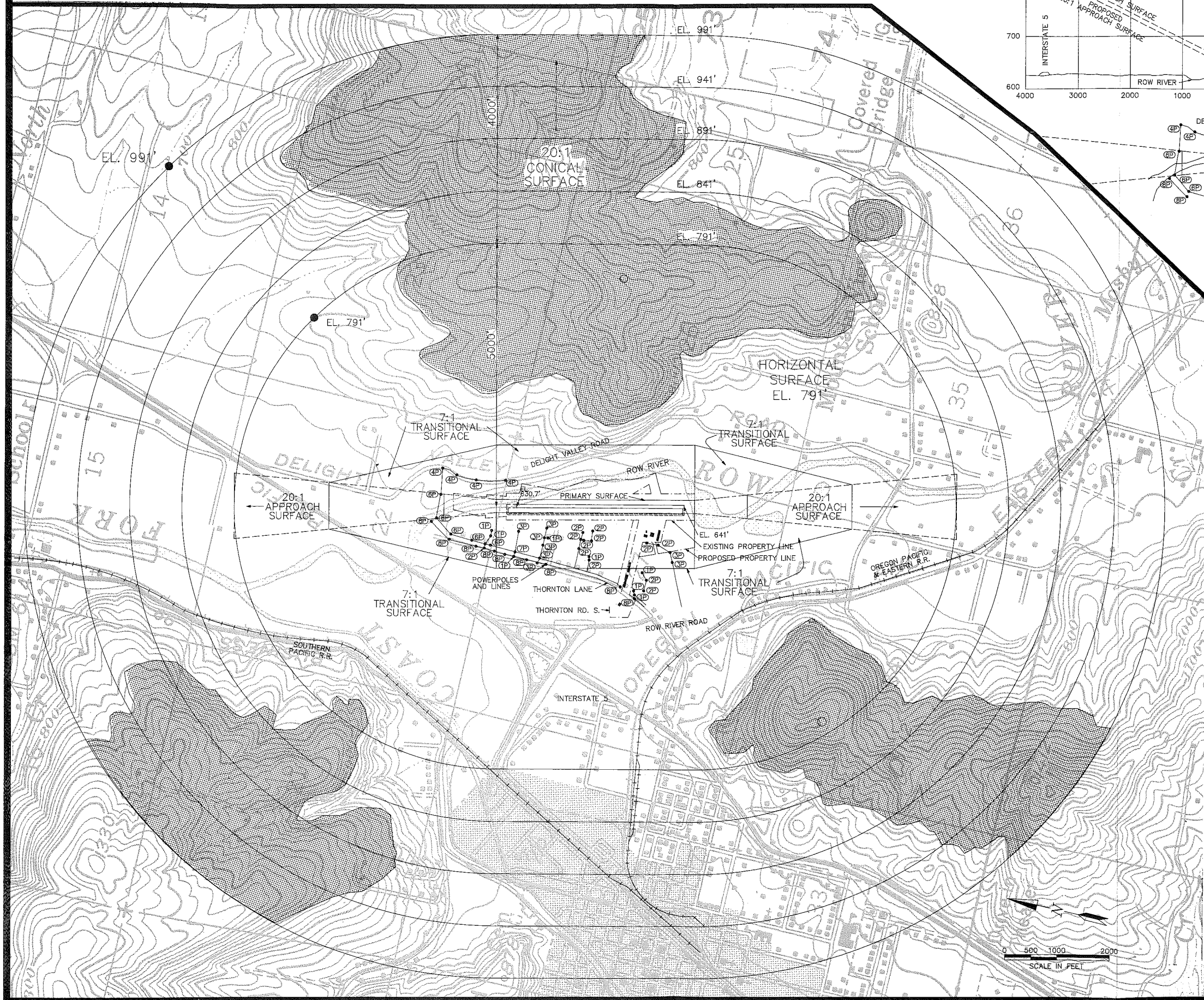
Approach and Clear Zone Plan

To safeguard the operation of aircraft into, out of, and around an airport, the Federal Aviation Administration has adopted, as part of the Federal Aviation Regulations (FAR), Part 77: **Objects Affecting Navigable Airspace**. Subpart C of Part 77 establishes standards for determining obstructions to air navigation by means of imaginary surfaces which provide vertical height limits for man-made and natural objects within the airspace above designated areas on and around the airport. Utilizing Part 77 criteria, Drawing 3, Approach and Clear Zone Plan, depicts the imaginary surfaces developed for Cottage Grove State Airport for the operating conditions anticipated through the forecast period.

FAR Part 77 also establishes a federal procedure whereby the height of any proposed man-made structure or natural growth can be monitored to determine what effect it may have upon aircraft operations. It is recommended that the existing regulations be reviewed and that any changes required be adopted by the City of Cottage Grove and Lane County.

As shown on Drawing 3, there are obstructions which penetrate the existing and ultimate FAR Part 77 imaginary surfaces. As depicted in FAR Part 77, a penalty of 15 feet must be added to highways to compensate for vehicle movement. None of the existing area roadways open to the public represent obstructions to the present facilities at the airport. The obstructions which exist are:

- o Terrain obstructions to the Horizontal and Conical Surfaces to the east, northwest, and southwest of the airport. It is recommended that obstruction lights be placed on the terrain at the highest elevation within the Horizontal Surface.



PLAN AND PROFILE
RUNWAY 15-33

LEGEND

EXISTING FACILITIES	—
PROPOSED FACILITIES	- - -
RAILROAD	+ + +
APPROACH SURFACE	- - -
TRANSITIONAL SURFACE	- - -
PRIMARY SURFACE	- - -
OBSTRUCTION TO SPECIFIED SURFACE	■
EXISTING POWERLINES AND POLES	•
POWERPOLE ELEVATION CODE (SEE NOTES)	GP
PROPOSED LOCATION FOR NEW OBSTRUCTION LIGHTS	○
TREES	⊕

NOTES:
POWERPOLE ELEVATIONS ARE CODED WITH THE FOLLOWING SYMBOLS:

1P - EL. 25'	5P - EL. 45'
2P - EL. 30'	6P - EL. 50'
3P - EL. 35'	7P - EL. 55'
4P - EL. 40'	8P - EL. 60'

FEDERAL AVIATION ADMINISTRATION	STATE OF OREGON DEPARTMENT OF TRANSPORTATION DIVISION OF AERONAUTICS	COTTAGE GROVE STATE AIRPORT COTTAGE GROVE, OREGON
APPROVAL DATE: _____ SEE APPROVAL LETTER: _____	APPROVAL DATE: _____ SEE APPROVAL LETTER: _____	APPROACH AND CLEAR ZONE PLAN
SIGNATURE _____	SIGNATURE _____	DRAWING NO. 3 OF 3
FORESITE GROUP Corvallis, OR Irvine, CA Seattle, WA		DATE: DECEMBER 1987 DRAWN BY: J.D. HOWARD APPROVED BY: J.R. HAWLEY

Table 7A
STAGED DEVELOPMENT PROGRAM
Cottage Grove State Airport

PHASE I (1988-1992)

- 1988--
 - No Proposed Development
- 1989--
 - o Land Acquisition
 - o New Airport Beacon
- 1990--
 - o Land Acquisition
 - o Precision Approach Path Indicator (PAPI)
- 1991--
 - o New Parallel Taxiway
 - o Runway Extension
 - o Access Taxiway
- 1992--
 - o Apron Expansion Phase I - 2 areas
 - o Existing T-hangar Taxiways
 - o Power Line Relocation

PHASE II (1993-1997)

- o NDB Installation
- o Medium Intensity Runway Lights (MIRL)
- o New T-hangar of 10+ Units
- o Apron Expansion 200' x 100'
- o East-West Hangar Roadway
- o East-West Hangar Area and Apron Auto Parking Lot

PHASE III (1998-2008)

- o FBO Hangar
- o FBO Area Auto Parking
- o Widen East-West Taxiway
- o New T-hangar (10+ Units)
- o Apron Overlay
- o Runway Overlay

IMPLEMENTATION PLAN

Chapter 8 IMPLEMENTATION PLAN

FINANCIAL AND DEVELOPMENT PROGRAM

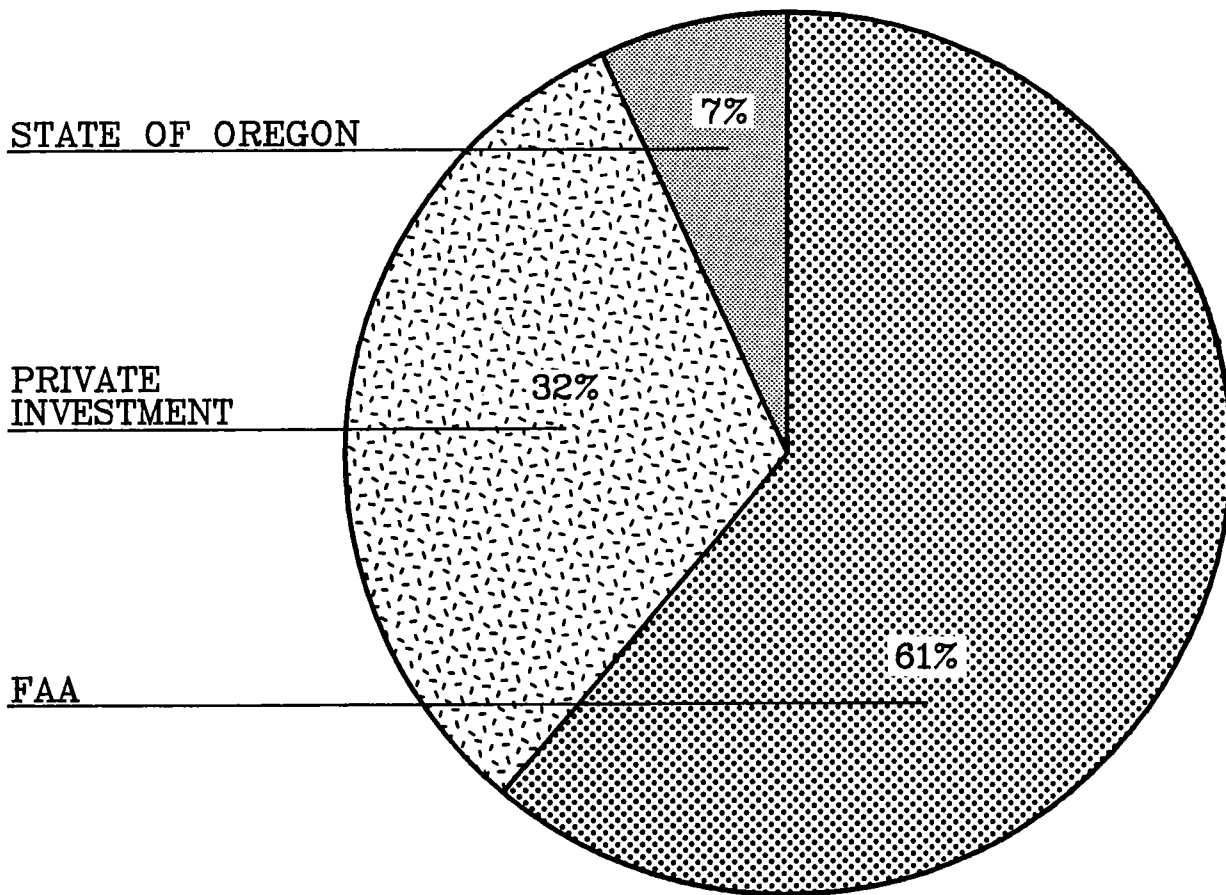
The foregoing chapters in the Master Plan have established the facility needs and recommended plan for Cottage Grove State Airport throughout the planning period. The purpose of this chapter is to provide an implementation program through which the recommendations can be realized in an effective and economical manner. The following sections present schedules and cost summaries for the development items; and the airport's projected revenue, expense, and financing options for capital improvements.

The cost of providing needed facilities at the airport will not be borne entirely by the State of Oregon. Several sources for development funding exist, including the Federal Aviation Administration and private enterprise (see Figure 9). As proposed, the State would be responsible for approximately seven percent of the total development cost during the planning period.

The primary source for airport development funding is aviation users, both locally and nationally. FAA grants fund 90 percent of eligible projects at airports such as Cottage Grove. Money in this program is derived from user fees deposited in the National Aviation Trust Fund. The Federal Airport Improvement Program (AIP) is the current legislation authorizing the dissemination and use of these funds, which have been collected through excise taxes on airline tickets, aviation fuel, accessories, aircraft registrations, and other aviation uses. The AIP expired in late 1987; however it is expected that a similar program will be enacted by Congress to carry on this process, which has been in effect for forty years.

Funds for general aviation airport development projects come from the FAA's discretionary part of the Trust Fund annual budgeted amount. These grants are distributed on a priority basis, with priorities established by FAA regional offices on the basis of number and dollar amount of applications received. Consequently, the State will be competing with other airport sponsors in the Northwest Mountain Region and throughout the nation for this type of assistance.

The State, as the owner of 36 airports, must also establish priorities for its available but limited funding. The available state or local funds appear to be more limited than federal and thus determine whether a project is funded or not. State funds are used to assist smaller airports with projects that do not receive federal aid and can also be used to maintain and develop State-owned airports. State funding has not been adequate to allow participation in federally funded projects similar to a few



COTTAGE GROVE STATE AIRPORT
COTTAGE GROVE, OREGON

**DEVELOPMENT PROGRAM
FUNDING SOURCES**

other states. For planning purposes, it's fair to assume that the State will be responsible for funding 10 percent of the cost on FAA-eligible projects for State-owned airports.

Activity at the airport is also a significant source for development funding. Use of the airport's facilities generates a variety of revenues, including land and facility rentals, tie-down leases, fuel flowage fees, and other use charges. The development of additional airport facilities and services for aviation uses can contribute greatly toward maximizing potential airport income.

The private sector is the final source for airport development funding. Private investment can contribute greatly toward the provision of needed aviation facilities not eligible for FAA funding, such as FBOs and T-hangers. This Master Plan identifies the improvements which will be needed and can best be provided by private development. The leasing of surplus airport property for farming or non-aviation activities is, of course, also a source of income.

AIRPORT DEVELOPMENT SCHEDULES AND COST SUMMARIES

Prior to formulation of the development schedules, the cost of each proposed improvement has been estimated. The figures used for all development items throughout the planning period are expressed in 1987 dollars and (except where noted) include 25 percent overhead for administration, engineering, and contingencies. For future implementation of this plan, management can convert the 1987-based figures by adjusting for subsequent inflation. The interim change in the United States Consumer Price Index (USCPI) provides a mechanism through the following formula to yield a multiplier ratio:

$$\frac{X}{337.7} = Y$$

Where:

X = USCPI in any given future year
337.7 = USCPI in 1987
Y = Conversion factor

Dividing the future CPI by the 1987 CPI provides a conversion factor (Y) which, in turn, is multiplied by the 1987-based cost estimates to provide appropriate amounts in any future re-evaluation. Only national CPI data should be used, as local or regional indices may vary. Consumer price index information may be obtained from the U.S. Bureau of Labor Statistics and the economic research units of some banks and councils of governments.

The airport development schedules have been established in consideration of their economic feasibility for implementation, i.e., the expected financial capability of the sponsor, as well as in response to anticipated demand for aviation facilities. Accordingly, the prioritized schedules have been divided into three stages covering the entire planning period. Table 8A shows that the total cost of developing Cottage Grove State Airport to meet forecasted needs will total nearly \$2.2 million by the year 2008.

The following sections outline the recommended development program and detailed funding distribution assumptions. The scheduling has been prepared in accordance with facility requirement determinations, as well as funding availability. It should always be remembered that, in addition to funding considerations, airport development should take place in response to demand, or actual activity, rather than merely be scheduled according to the time frame corresponding to earlier forecasts of future activity. Therefore, should significant variations from forecast trends occur, facility development scheduling may have to be adjusted in the future.

STAGE I DEVELOPMENT

Stage I projects comprise the initial five years of the planning period, and have been further sub-divided into single-year phases for the years 1988 through 1992. No new projects are programmed for the year 1988. The State budget has already been set for 1988, and it is too late to enter a new project at this time.

The most significant thrust of Stage I will be the acquisition of land which will allow the Cottage Grove State Airport to expand to meet the projected planning period demand and beyond. Without the additional land, much of the development program proposed in this Master Plan will not be implemented. The property to be acquired falls into two distinct categories. Approximately four acres of land proposed for acquisition lies within the extended clear zone for the additional length of runway scheduled for 1991. The other significant area of land acquisition is located in the terminal area. The proposed land acquisition of approximately 13 acres is in several locations within the designated terminal area. The primary acres to be acquired are for the expansion of aircraft parking and new T-hangars. The acquisition of land will take up the second and third year of the development program. It would be likely that federal funding of the land acquisition would be under only a one-year program.

Additional projects falling within the 1991 time frame would be the construction of the parallel taxiway and exit taxiway to serve Runway 15-33. Along with development of the taxiways will be the proposed runway extension of 190 feet. There are economies of scale by completing those projects simultaneously.

A much-needed addition to the airport to handle the anticipated demand will be the expansion of the existing Terminal area aircraft parking apron and the development of a new parking area to the north of the terminal area. Stage I improvements, as outlined in Table 8A and graphically depicted on the Terminal Area Plan (Drawing 2, Page 7-3) will cost an estimated 0.783 million dollars.

STAGE II DEVELOPMENT

Stage II includes the period 1993 through 1997 and consists of airport enhancement type projects, as well as a sizable private development project. Major expenditures during this stage of the plan will expand the airport's capability to house and park additional aircraft. Improvements in this stage include an expansion of the apron area to the north of the east-west taxiway. The area will consist of approximately 4,200 square yards and will allow parking for an additional 35 to 40 aircraft. A new access roadway paralleling the east-west taxiway will be designed to serve the proposed single-unit hangars as well as the new tie-down apron. An auto parking lot will be developed at the north end of the new road to service the existing hangar area and new tie-down apron. One new T-Hangar of 9 to 10 units south of the Terminal Area is also proposed during this phase of the development program. New runway lighting and taxiway signs will also be improvements that will enhance the airport operations. The existing Low Intensity Lighting would be replaced with Medium Intensity Runway Lights. The upgrading of the runway lighting system will also be a requirement once a nonprecision instrument approach is established for the airport. During this stage in airport development, it is proposed that a nondirectional radial beacon be installed to provide the navigation radio signal to achieve nonprecision instrument approach capability. Extension of utility service will be necessary to existing and new development areas.

This second five-year portion of the planning period will cost the State approximately \$29,000, or one-quarter of its total development costs for the planning period, but--together with federal and private financial participation-- will result in new facilities worth nearly \$1.9 million. Many of these improvements will result in increased revenue-generating potential. Stage II projects are listed in Table 8A and depicted on Drawing 2.

Table 8A
PROJECTED DEVELOPMENT COSTS, SCHEDULE, AND FUNDING
Cottage Grove State Airport

Stage I 1988-1992

	<u>Sponsor</u>	<u>FAA</u>	<u>Other</u>	<u>Total</u>
<u>1988</u>				
No Development				
<u>1989</u>				
1. Land Acquisition	\$10,500	\$ 94,500	-0-	\$105,000
2. New Beacon	150	1,350	-0-	1,500
TOTAL 1989	\$10,650	\$ 95,850	-0-	\$106,500
<u>1990</u>				
3. Land Acquisition	\$ 9,800	\$ 88,200	-0-	\$ 98,000
4. Precision Approach Path Indicator (PAPI)	1,500	13,500	-0-	15,000
TOTAL 1990	\$11,300	\$101,700	-0-	\$113,000
<u>1991</u>				
5. Parallel Taxiway	\$19,970	\$179,730	-0-	\$199,700
6. Runway Extension	3,780	34,020	-0-	37,800
7. Access Taxiway	600	5,400	-0-	6,000
TOTAL 1991	\$24,350	\$219,150	-0-	\$243,500
<u>1992</u>				
8. Apron Expansion	\$16,600	\$149,400	-0-	\$166,000
9. T-Hangar Taxiway	1,340	12,060	-0-	13,400
10. Powerline Relocation	14,000	126,000	-0-	140,000
TOTAL 1992	\$31,940	\$287,460	-0-	\$319,400
 TOTAL STAGE I	 \$78,240	 \$704,160	 -0-	 \$782,400

Table 8A
PROJECTED DEVELOPMENT COSTS, SCHEDULE, AND FUNDING
Cottage Grove State Airport

STAGE II - 1993-1997

	<u>Sponsor</u>	<u>FAA</u>	<u>Other</u>	<u>Total</u>
11. NDB	\$ 6,550	\$ 58,950	\$ -0-	\$ 65,500
12. MIRL	9,380	84,420	-0-	93,800
13. Taxiway Signs	1,250	11,250	-0-	12,500
14. T-Hangar	-0-	-0-	140,000	140,000
15. Apron Expansion	4,460	40,140	-0-	44,600
16. E-W Road & Parking	4,370	39,330	-0-	43,700
17. Sewerline Installation	3,000	27,000	-0-	30,000
TOTAL STAGE II	\$29,010	\$261,090	\$140,000	\$430,100

STAGE III - 1998-2008

	<u>Sponsor</u>	<u>FAA</u>	<u>Other</u>	<u>Total</u>
18. T-Hangar	\$ -0-	\$ -0-	\$140,000	\$140,000
19. Conventional Hangar	-0-	-0-	320,000	320,000
20. Widen E-W Taxiway	4,200	37,800	-0-	42,000
21. Overlay E-W Taxiway	2,900	26,100	-0-	29,000
22. FBO Auto Park	1,800	16,200	-0-	18,000
23. Runway Overlay	25,800	232,200	-0-	258,000
24. Apron Overlay	16,800	151,200	-0-	168,000
TOTAL STAGE III	\$51,500	\$463,500	\$460,000	\$975,000

TOTAL ALL STAGES	\$158,750	\$1,428,750	\$600,000	\$2,187,500
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STAGE III DEVELOPMENT

Stage III (1998-2008) projects will complete the airport's development program for the planning period. Most of these projects will consist of expanded aircraft parking and storage facilities, as well as pavement improvement and rehabilitation for surfaces not constructed or overlaid since the beginning of the planning period. Among the new improvements will be a new general aviation conventional hangar for expansion of existing FBO operations. An additional 10-unit T-hangar will round out the aircraft storage facilities during this planning period. The east-west taxiway will be widened to provide improved aircraft taxiing to the western transient apron parking area and single unit hangars developed along the east-west taxiway. An improved auto parking area is located adjacent to the existing and new FBO facilities. It is anticipated that the runway and existing aircraft apron area will need rehabilitation with an asphalt overlay during this time frame. These projects, summarized in Table 8A and displayed on Drawing 2, will total almost one million dollars. The State's share of that expenditure would be approximately \$51,500, with private funding accounting for over \$460,000.

FINANCIAL PROJECTIONS

The financial management and reporting system for Cottage Grove State Airport is based on the accrual method of accounting, with this airport treated as only one of many segments within the State of Oregon general fund. The specific account falls within the Aeronautics Division Fund. Each airport within the Aeronautics Fund is only treated as a cost center, it is the total division account which must balance out. The feasibility for the development program is not as dependent upon the revenues and expenses of the Cottage Grove State Airport as it is of the overall Aeronautics Division Fund. However, it remains necessary to evaluate the operating income and expenses to determine its contribution to the total program.

Historical revenues and expenditures have been analyzed in conjunction with aviation activity and other indicators for the same period to ascertain possible relationships and trends. Where appropriate, these relationships have been projected into the future. Historical data was limited and was based on such State accounting records as were available. Of the information obtained, several years' worth provided little historical trend evaluation. The State has a uniform rates and charges schedule applicable to all State-owned airports. Again, the flexibility to unilaterally adjust the rates and charges of one airport could not be accomplished and will not be recommended. This section of the study, then, has been limited to projecting the income and expenses at the Cottage Grove State Airport throughout the

planning period. These projections show the airport's contribution to the overall Aeronautics Fund and will also be a good indicator of the economic support of its programmed capital improvements.

PROJECTED REVENUES

Operating revenues are generated on the Cottage Grove State Airport by user fees and lease agreements. Between the fiscal years 1982 and 1985, operating expenses have exceeded income by an average of 22 percent, with the prior biennium 1979/81 showing a loss in excess of \$8,100. Since the biennium 1985/1987, the airport has been producing a surplus of funds. For this period, the revenues exceeded expenses by 24 percent.

As shown on Table 8B, the new surplus is projected to remain fairly small for future biennium accounting periods. Only after substantial new facilities to house additional aircraft have been constructed on the airport will revenues start to outgain expenses by a substantial margin. Throughout the planning period, accumulated surpluses alone will not be adequate to fund the projected local share of capital improvement projects. Unfortunately, once the surpluses start amounting to something substantial and are adequate for funding sizable projects, most of the important work will already have been completed.

The airport presently has a limited variety of sources for income, and this is not anticipated to change throughout the planning period. The largest source of revenue comes from land leases, which provide almost 50 percent of the airport's total revenue; this margin is expected to remain the same, with a possibility of slight growth. Fuel tank rentals and the fuel flowage fee together account for 45 percent of today's income, with tie-down fees providing the balance of the revenue. The existing contract limits the fuel facility rental fees to no more than \$2,000 in any one contract year, while the fuel flowage income will increase in accordance with the rise in aircraft activity and/or a raise in the flowage fee itself. It is anticipated that the need for increased revenues in the later years would require a bump in the fuel flowage fee from the present \$.03 per gallon. A \$.01 cent increase has been proposed for the 1992/1994 biennium, with another increase prior to the 2006/2008 biennium.

Land-lease income was projected at the top side of the State's approved property lease rates. All Cottage Grove State Airport leased land can be considered high quality and improved property to support the higher fee per square foot. Any adjustment to State policy will have a substantial impact on the total airport revenues. The sizable increases in rental income occur during the 1996/1998 biennium period and again during the years surrounding the 2006/2008 biennium. Tie-down fees showed increases in accordance with the amount of new paved apron area. The total income would increase, if only because of the change

TABLE 8B
PROJECTED REVENUES AND EXPENDITURES
Cottage Grove State Airport

	<u>1985/86</u>	<u>1986/87</u>	<u>1985/87</u>	<u>1987/89</u>	<u>1992/94</u>	<u>1996/98</u>	<u>2001/03</u>	<u>2006/08</u>
REVENUE:								
Land Lease	\$ -0-	\$1,435.81	\$1,435.81	\$6,600.00	\$7,100.00	\$11,100.00	\$11,600.00	\$13,400.00
Hangar Rental	3,581.96	1,530.87	5,112.83	-0-	-0-	-0-	-0-	-0-
Tie-Down Fees	434.34	500.00	934.34	1,000.00	1,400.00	2,000.00	3,000.00	4,000.00
Flowage Fees	907.98	1,951.56	2,859.54	2,860.00	3,000.00	4,400.00	4,800.00	6,400.00
Fuel Tank Rental	<u>1,210.66</u>	<u>2,129.47</u>	<u>3,340.13</u>	<u>3,360.00</u>	<u>2,000.00</u>	<u>2,000.00</u>	<u>2,000.00</u>	<u>2,000.00</u>
TOTAL REVENUE	\$6,134.94	\$7,547.71	\$13,682.65	\$13,820.00	\$13,500.00	\$19,500.00	\$21,400.00	\$25,800.00

	<u>1985/86</u>	<u>1986/87</u>	<u>1985/87</u>	<u>1987/89</u>	<u>1992/94</u>	<u>1996/98</u>	<u>2001/03</u>	<u>2006/08</u>
EXPENSES:								
Aircraft Rental	\$ 239.70	\$ -0-	\$ 239.70	\$ 300.00	\$ 300.00	\$ 300.00	\$ 350.00	\$ 400.00
Meals & Lodging	310.71	568.26	878.97	1,250.00	1,450.00	1,500.00	1,600.00	1,700.00
Travel	6.76	13.19	19.95	30.00	40.00	50.00	50.00	60.00
Freight	6.05	-0-	6.05	20.00	25.00	30.00	30.00	30.00
Communications	551.06	668.19	1,219.25	1,420.00	1,500.00	1,600.00	1,700.00	1,800.00
Printing & Photo	276.96	91.20	368.16	300.00	300.00	300.00	300.00	300.00
Maintenance	174.00	577.73	751.73	1,300.00	1,700.00	1,900.00	2,100.00	2,300.00
Equipment Repair	2.85	158.00	160.85	300.00	300.00	300.00	300.00	400.00
Electricity	314.95	282.04	596.99	650.00	850.00	900.00	900.00	900.00
Grounds Maintenance	-0-	680.84	680.84	1,000.00	1,000.00	1,200.00	1,500.00	1,800.00
Supplies	99.31	15.96	115.27	150.00	200.00	200.00	200.00	200.00
Salaries & Wages	1,135.31	3,294.23	4,429.54	6,800.00	7,000.00	8,000.00	8,500.00	9,000.00
In-Kind Match	<u>-0-</u>	<u>1,563.90</u>	<u>1,563.90</u>	<u>-0-</u>	<u>-0-</u>	<u>-0-</u>	<u>-0-</u>	<u>-0-</u>
TOTAL EXPENSES	\$3,117.66	\$7,913.54	\$11,031.20	\$13,520.00	\$14,665.00	\$16,280.00	\$17,530.00	\$18,890.00

from grass to paved tie-down facilities. The increase in paved apron areas for permanent tie-downs should also result in added revenues. The projected increase in tie-down income will be realized if the based aircraft projections hold fairly firm. An increase or decrease in airport activity will impact the total revenue from the tie-down area.

PROJECTED EXPENDITURES

Airport operating expenses for the past several years at Cottage Grove State Airport have been examined, and it is apparent that this airport has been operated and maintained in a cost-effective manner for a facility with its activity. The cross-utilization of management and maintenance personnel for all State-owned airports provides a high level of control and support at a minimum cost. With the limited revenue-generating capability of the smaller general aviation airports, full-time management cannot be justified and, in this specific instance, is not required. Even without on-site management at this airport, as well as the other State facilities, this is one of the most well-maintained and managed airports in the State.

Cost centers corresponding generally to the budget account groupings have been evaluated for historical trends and relationships. Where applicable, relationships have been extrapolated into the future with forecast activity indicators. It was necessary to break down the biennium recorded expenses into individual years to establish annual patterns versus one-time expenses. The following discussion of expense centers will conform to the established accounts. Only significant categories of expenses will be discussed, with the others being combined with like purposes.

Aircraft rental, meals, lodging, and travel time are expenses associated with absentee-type management. These expenses, although not substantial, are extremely reasonable for the amount of travel involved in the maintenance of this facility. Aircraft rental has not been an annual expense, but it is expected to become one as activity and development increase at the Cottage Grove State Airport. Use of an aircraft to attend meetings or inspect the airport is a very cost-effective method for accomplishing those tasks. The continued use of an aircraft for this purpose is easily supported, and we project that this activity will increase in future years. Meals and lodging are also associated with maintenance personnel traveling to the site for the purpose of maintaining the airfield. As the activity grows at the airport, so will the requirement for expanded maintenance. As discussed above, even as a group of expenditures, the cost of maintaining this airport is extremely low.

Communications is one of the higher expenses and, with absentee management, is quite normal and expected. This cost is anticipated to increase in relationship to the rise in activity and development projects. The long-term increase in communications expense should be maintained with the 5 to 7 percent range. As with most businesses, the use of the telephone can have a substantial impact on maintaining a reasonable travel expense. Printing and photography costs will continue to remain at approximately the same level. The majority of this expense is for new airport photos taken by the State Highway Division, and this should remain fairly stable throughout the planning period.

The airport maintenance category consists primarily of expenditures for contractors hired to repair electrical equipment or paved facilities. Considering the age of the airport, there is no reason to believe that these expenditures will not go up. New facilities, as well as reasonable wear and tear, will continue to require increased contract maintenance work. Equipment repair is a relatively small expense and has varied substantially from year to year. It is anticipated that the costs will not go down, and with expanded use of existing equipment, a small initial increase with a leveling off over the planning period can be expected.

Electrical expenses include electricity for runway lights, security lighting, and the lighting of State-owned and -controlled facilities. All privately owned buildings have their own power sources which are metered separately from the state facilities. The expense is projected to increase with the development of the runway extension. The new lighting will include the additional runway lights. The taxiway is not anticipated to be lighted. The capital improvement program shows new medium intensity lighting being installed in the Stage II period. This will have an effect on electrical usage in the 1992/1994 biennium. New apron security lighting, also planned with apron expansion projects, will boost electricity usage, and this is reflected in the projected expenses in future years.

Another grouping of expenses is that of supplies. This category includes grounds maintenance, electrical, and miscellaneous supplies. All show increases over the years, due to expanded facilities or a higher usage of the airport facilities. Grounds maintenance supplies continue to be projected as the most significant cost of the groups, with electrical supplies being next. Electrical supplies will grow as new facilities are added, but it should remain a small part of maintaining the airport.

Age of equipment may play a larger role in further annual electrical costs. It can be anticipated that, once the runway lighting has been switched to MIRLS, the cost of light bulbs will more than double.

The last category, largest of the airport expense categories, is for Salaries and Wages. The cost has varied substantially year by year, and the fluctuation will continue on into the future. This category includes maintenance personnel as well as a portion of management costs. Labor expenditures are projected to increase each year, due to the aging of airport facilities as well as the expansion and development of new facilities. Annual increases should remain low--approximately 1.5 to 2.0 percent. It should be pointed out that cross-utilizing personnel for maintenance of the State's airports greatly reduces the individual airport's cost and still maintains a high standard of performance.

CONCLUSION

The foregoing has established that Cottage Grove State Airport is capable of meeting the aviation needs of the Cottage Grove area well into the future. Every indicator points to the airport's being a worthwhile investment, and the new improvements will enhance its importance. Although substantial public investment will be required to provide the recommended improvements, the resultant facilities will further expand this already significant economic asset to the community. Continued sound professional management with enlightened policy guidelines, combined with the comprehensive recommendations of this Master Plan, should help keep the Cottage Grove State Airport on a progressive course. Implementation of this program will enable the State to provide an airport that will not only serve the needs of the city and the area, but also be a source of pride for the community.

ENVIRONMENTAL REVIEW

Chapter 9 ENVIRONMENTAL REVIEW

INTRODUCTION

This review is designed to identify potential adverse environmental impacts which might occur should the recommended alternative be pursued by the State of Oregon Aeronautics Division at the Cottage Grove State Airport. In accordance with FAA Guidelines, FAA Order 5050.4A, Airport Environmental Handbook¹, this environmental review considers twenty Specific Impact Categories. Each category has been addressed for the proposed development alternative (Preferred Alternative).

The checklist, Table 9A, in this summary is adapted from "Environmental Consequences - Specific Impact Categories" as given in FAA Order 5050.4A. The table lists the twenty categories where environmental impacts are possible, should the airport development master plan be implemented. The word "investigate" opposite a specific impact category indicates a potential impact in that category. When a potential impact is identified in Table 9A, it does not mean that the impact will be negative or that it is even significant. The potential impacts identified in Table 9A are evaluated in this summary to determine their significance and estimate their level of importance.

NOISE

The impact of aircraft noise is potentially the most critical of all environmental effects associated with airport development and aircraft operations, thus, "noise" will receive the most detailed treatment in this environmental summary. Aircraft noise exposure is more likely to have a negative behavioral and subjective effect on people, rather than to cause physical injury. Behavioral effects involve interference with activities such as speech, learning, and sleeping. Subjective effects are described by terms like "annoyance" and "nuisance." The magnitude of the problem depends on the volume, frequency, and time of day of aircraft operations, the types of aircraft, and the character of land use in the area exposed.

¹ Airport Environmental Handbook, Order 5050.4A, Department of Transportation, Federal Aviation Administration, October 1985.

Table 9A

CHECKLIST OF SOURCES OF POTENTIAL ENVIRONMENTAL IMPACTS
Cottage Grove State Airport

<u>Sources of Potential Environmental Impact</u>	<u>Preferred Alternative</u>
Noise	Investigate
Compatible Land Use	Investigate
Social Impacts	Investigate
Induced Socioeconomic Impacts	Investigate
Air Quality	Investigate
Water Quality	Investigate
Special Land Uses, DOT Act, Section 4(f)	Investigate
Historic, Architectural, Archaeological, and Cultural Resources	Investigate
Biotic Communities	Investigate
Endangered and Threatened Species, Flora, and Fauna	Investigate
Wetlands	No
Floodplain	Investigate
Shoreline Management	No
Coastal Barriers	No
Wild and Scenic Rivers	No
Farmland	Investigate
Energy Supply and Natural Resources	Investigate
Light Emissions	Investigate
Solid Waste Impacts	Investigate
Construction Impacts	Investigate

Notes: Investigate = Possible impact might result; assessed
in more detail in this Summary.

No = No impact anticipated; source by alternative
has been eliminated from further consideration.

Background

Noise is most often defined as unwanted sound. However, sound is measurable, whereas noise is subjective. The relationship between measurable sound and human irritation is the key to understanding aircraft noise impact. A rating scale has been devised to relate sound to the sensitivity of the human ear. The A-weighted decibel scale (dBA) is calibrated to the faintest sound audible to the average young male ear. The human ear often judges an increase of 10 decibels as a doubling of sound. The relative loudness of several common sounds is related to the dBA, as shown in Figure 10, Common Noise Levels on dBA Scale.

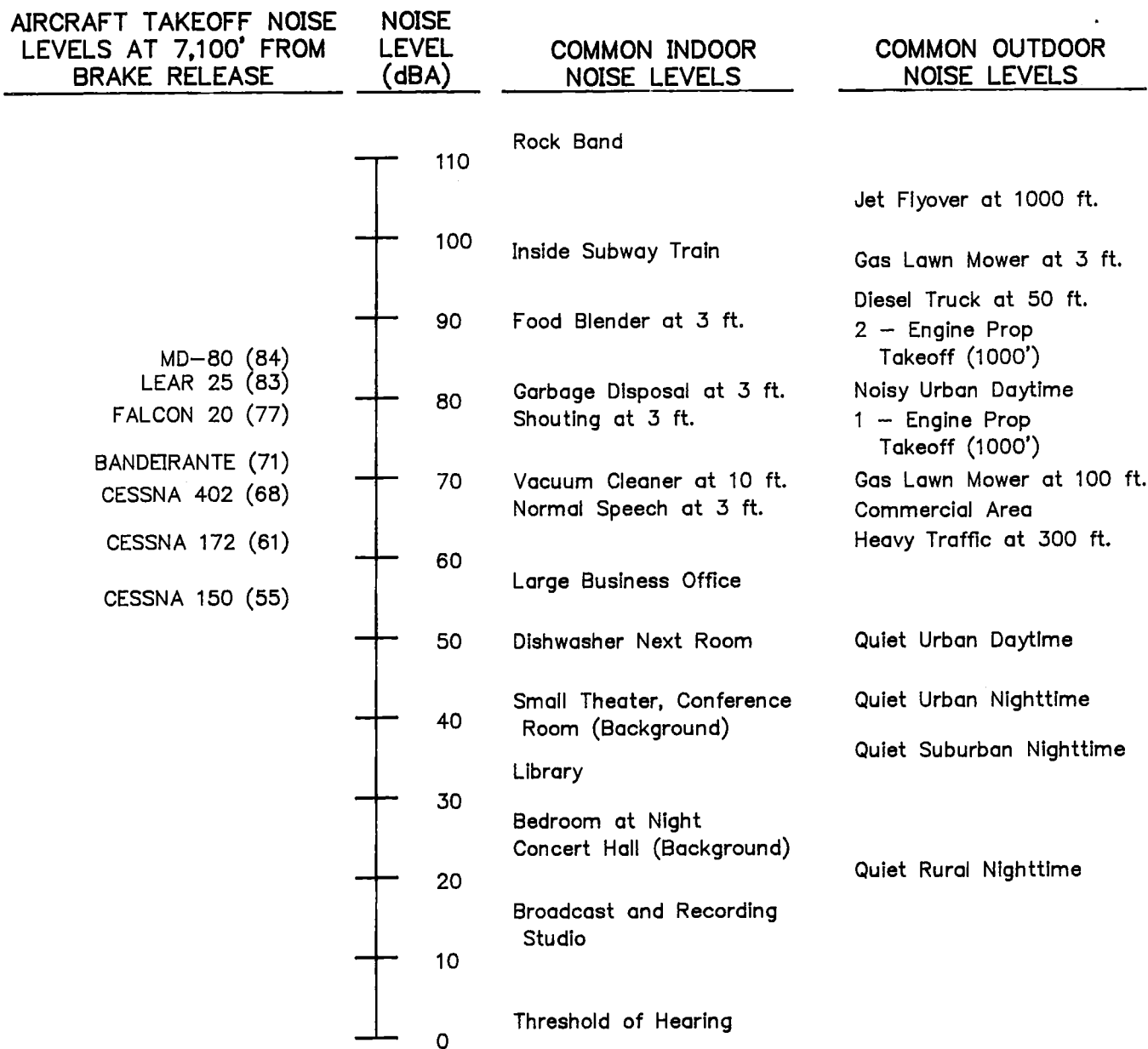
The difficulty lies in determining what amount and what kind of sound constitutes noise. The vast majority of people exposed to aircraft noise are not in danger of direct physical harm. However, much research on the effects of noise has led to several generally accepted conclusions:

- o The effects of sound are cumulative, and, therefore, the duration of exposure must be included in any evaluation of noise.
- o Noise can interfere with outdoor activities, speech, and other communication.
- o Noise can disturb sleep, TV/radio reception, and relaxation.
- o When community noise levels have reached sufficient intensity, community action can occur, including organizations pressing for new regulations or filing civil nuisance or inverse condemnation suits.

Research has also shown that individual responses to noise are difficult to predict. Some people are annoyed by each perceptible noise event, while others show little concern over the most disruptive of events. However, it is possible to predict the responses of groups of people. Consequently, community response, not individual response, has emerged as the prime index of aircraft noise measurement.

Ldn Methodology

On the basis of the findings described above, a methodology has been devised to relate measurable sound from a variety of sources to community response. Termed "Day-Night Average Sound Level" (Ldn), this metric has been adopted by the U.S. Environmental Protection Agency, Department of Housing and Urban Development, Oregon Department of Environmental Quality (DEQ), and the Federal Aviation Administration to use in evaluating noise impacts.



Source: Caltrans Noise Manual, California State Department of Transportation,
March 1980, pg. 1-1-4.

COMMON NOISE LEVELS ON dBA SCALE

Cottage Grove State Airport
Airport Master Plan

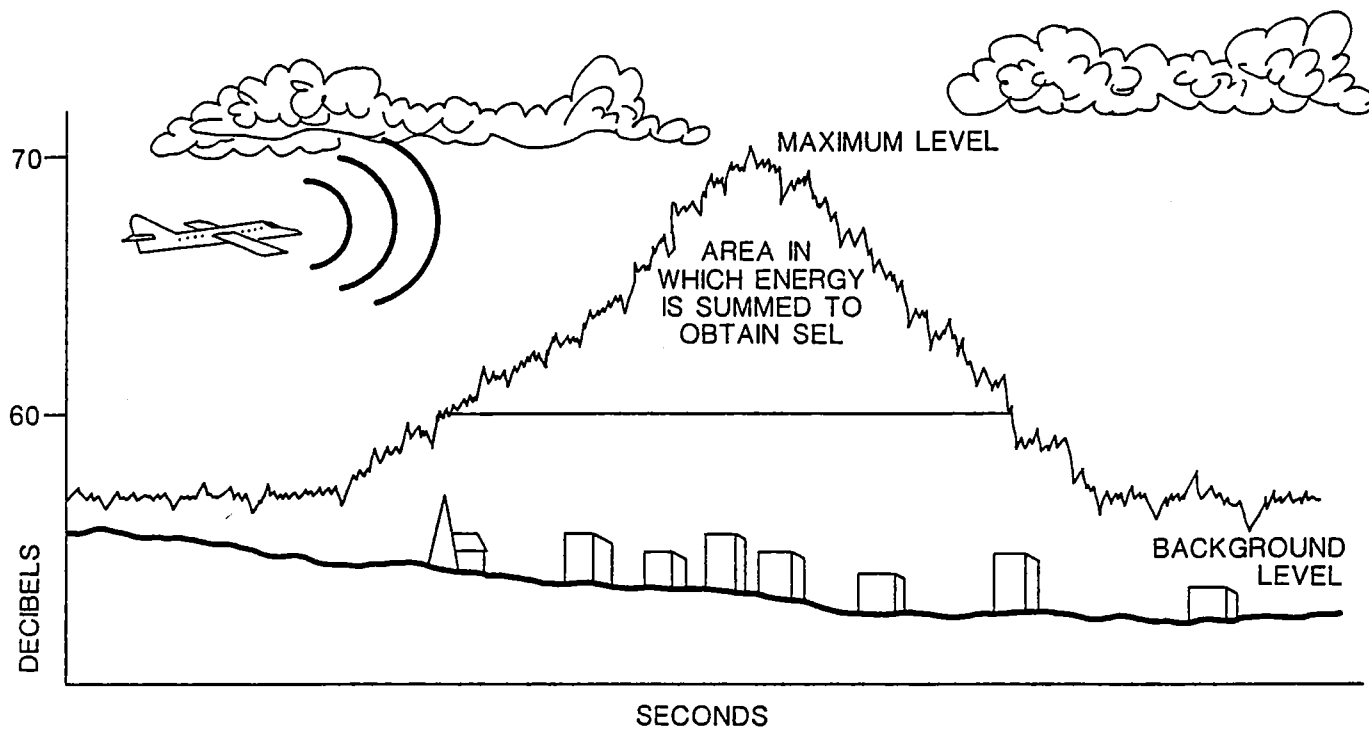
The basic unit in the computation of Ldn is the sound exposure level (SEL). An SEL is computed by adding the dBA level for each second of a noise event above a certain threshold. For example, a noise monitor located in a residential area with a background noise level of 45 dBA receives the sound impulses of an approaching aircraft and records the dBA reading for each second of the event as the aircraft approaches and departs the site. Each of these one-second readings is then added logarithmically to compute the SEL. Because of the logarithmic calculation, noise levels below 10 dBA of the maximum level are insignificant in terms of Ldn value. The description of an SEL is presented graphically in Figure 11, Description of Single Event Level.

The computation of an airport Ldn involves the addition, weighting, and averaging of each SEL to achieve an Ldn level at a particular location (Figure 12, Airport Ldn Noise Contour). The SEL of each noise event occurring between the hours of 10:00 p.m. and 7:00 a.m. is automatically weighted by adding 10 dBA to the SEL to account for the assumed additional irritation perceived during that period. All SELs are then averaged over a given time period (day, week, year) to achieve a level characteristic of the total noise environment.

Stated simply, an Ldn level is approximately equal to the average dBA level during an entire time period, with a weighting for nighttime noise events. Thus, a 65 Ldn level describes an area as having a time-averaged constant noise level of 65 dBA during the daytime and 55 dBA during the nighttime, even though the area would experience noise events higher and lower than 65 dBA.

The main advantage of Ldn is that it provides a common measure for a variety of different noise environments. The same Ldn level can describe both an area with very few high-noise events and an area with many low level events.

Ldn is commonly applied in order to predict community response. Survey research relating Ldn levels to community reaction is shown in Figure 13, Community Reaction to Noise, and summarized below. Generally, at a 65 Ldn level, 33 percent of those exposed will be highly annoyed and 5 percent will actually complain. As noise exposure drops to 60 Ldn, 24 percent are projected to be highly annoyed and 2 percent to be complaining. On the basis of such community reaction research, several government agencies have devised standards for acceptable land use within areas impacted by aircraft noise. The Environmental Protection Agency, Department of Housing and Urban Development, and the Federal Aviation Administration have adopted standards and guidelines recognizing that most common urban uses are compatible at levels below 65 Ldn. Acceptability standards and guidelines will be discussed in greater detail in this Summary under the section on "Compatible Land Use."



DESCRIPTION OF SINGLE EVENT LEVEL

Cottage Grove State Airport
Airport Master Plan

AIRPORT Ldn NOISE CONTOUR

Cottage Grove State Airport
AIRPORT MASTER PLAN

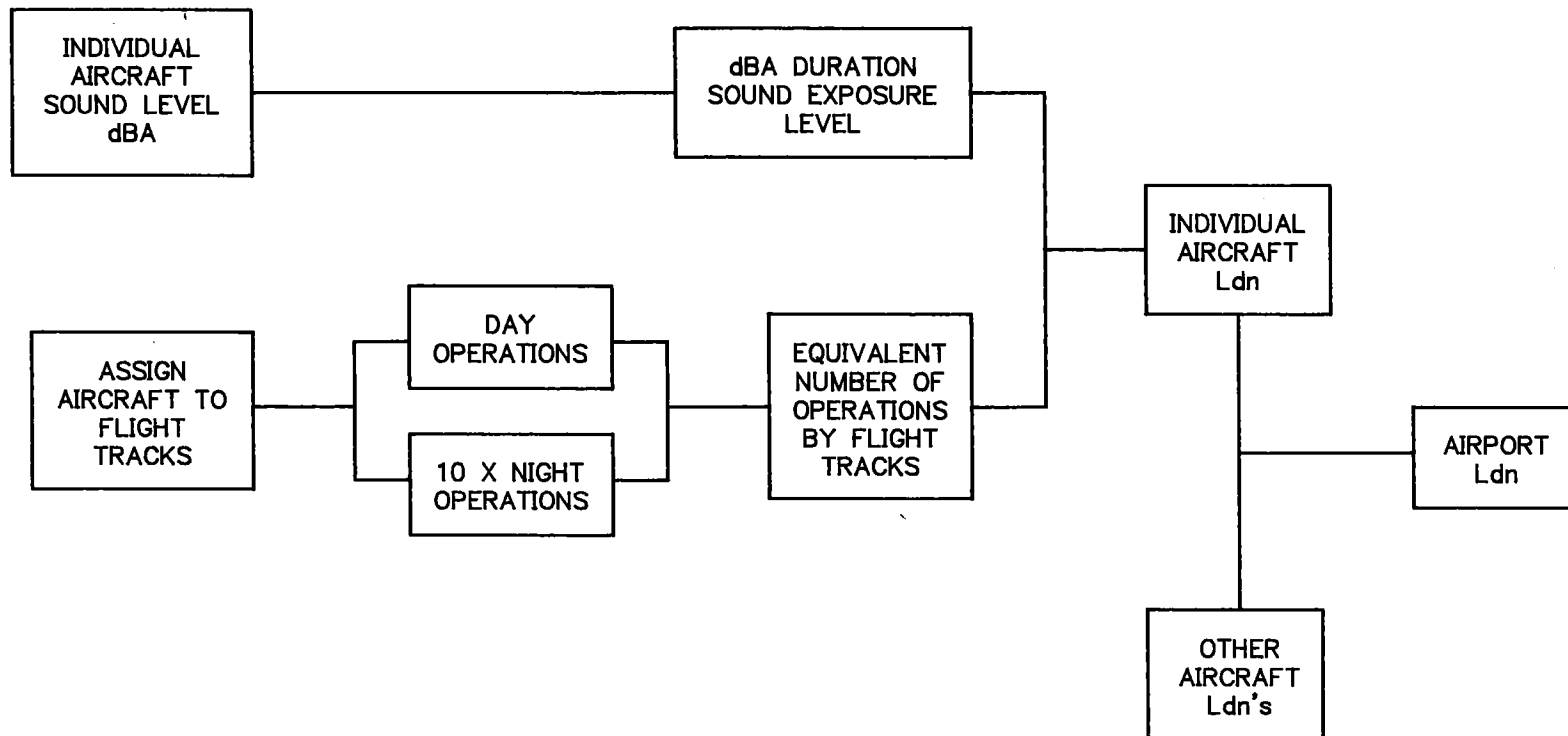
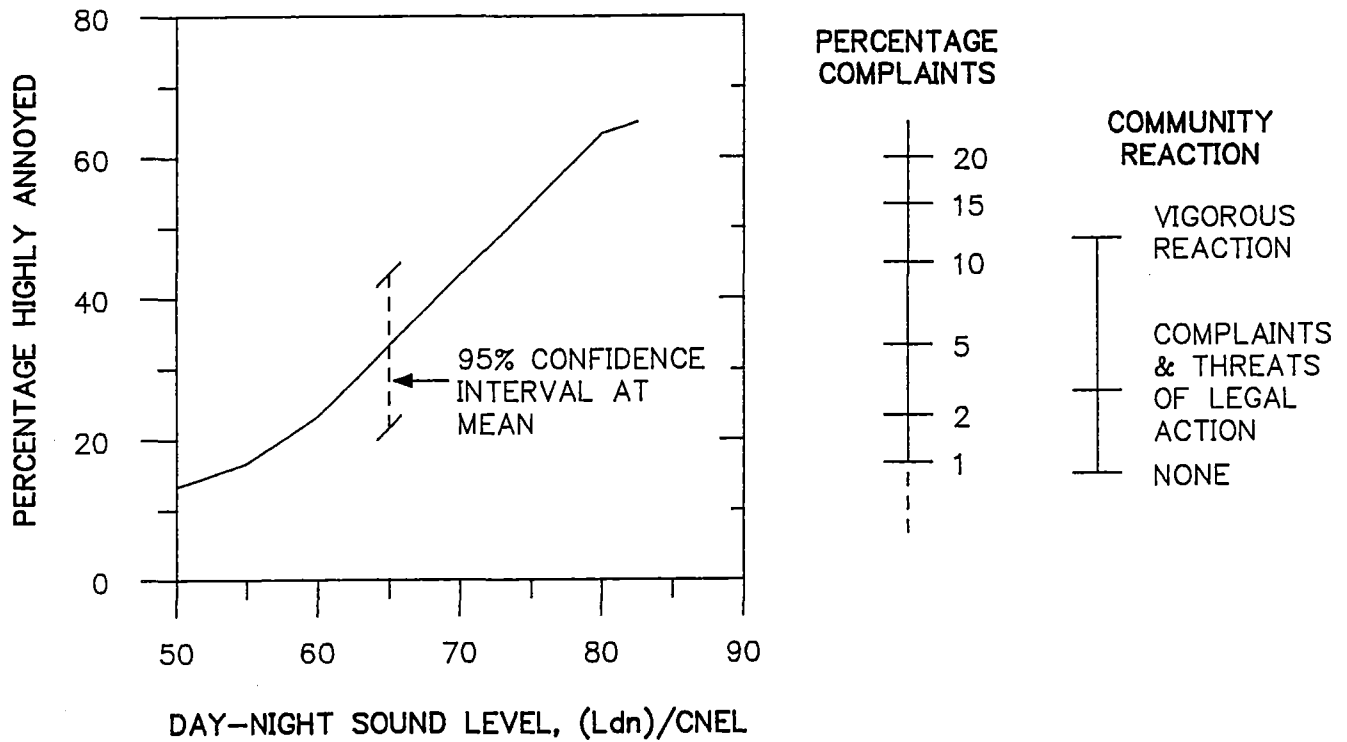


FIGURE 12



SOURCE: "IMPACT OF NOISE ON PEOPLE", FAA, OFFICE OF ENVIRONMENTAL QUALITY, MAY 1977.

COMMUNITY REACTION TO NOISE

Cottage Grove State Airport
Airport Master Plan

Ldn levels are typically depicted as contours. Contours are an interpolation of noise levels drawn to connect all points of a similar level. Contours appear similar to topographic contours and form concentric "footprints" about a noise source. It is these footprints of Ldn contours drawn about an airport which are used to predict community response to the noise from aircraft using that airport.

Airport Noise Contours

The FAA's Integrated Noise Model (INM), Version 3.8, was used to generate Ldn noise contours for Cottage Grove State Airport. Contours were developed in compliance with DEQ Noise Regulations (OAR Chapter 34, Division 35). The computer program is given data describing the size and location of the airport runway, the location of aircraft flight tracks, and noise characteristics of aircraft using the runway and flight tracks. Additional detailed information concerning glide slopes, aircraft weight, thrust settings, etc., is also given to the program. The program then computes the noise levels at points around the airport and plots the contours.

Aircraft operations are infinitely varied. Individual aircraft operations differ substantially in weight, engine type, fuel load, takeoff angle, flap settings, and a wide variety of other characteristics. Air traffic control procedures, winds, visibility, and pilot techniques also influence the operations and flight tracks of individual aircraft.

Computer modeling is limited in its response to these variations of flight and noise. Only so many flight tracks can be included in the computer data, and those tracks must be precisely identified, rather than defined as corridors or wide paths. The weight, thrust, flight angles, etc., must also be normalized for the computer model. Nevertheless, assumptions about aircraft operations can be modeled, and existing noise can be measured and extrapolated to provide a reasonably accurate representation of the noise environment.

1987 Base Case

Ldn noise contours for 1987 are shown on Figure 14. The 65 Ldn contour is entirely on airport property. A small portion of the 55 Ldn contour extends off airport property to the north into an area which is designated on the County's Comprehensive Plan as Exclusive Farm Use, with 30-acre lot size minimums. This is compatible with existing and planned uses.

1992 Existing Runway Configuration

By 1992, annual aircraft operations are projected to increase to 15,400. Ldn noise contours were generated based upon these forecasts.

The 1992 Ldn noise contours with the existing runway configuration are shown on Figure 15. The 65 Ldn contour remains entirely on airport property. The 55 Ldn contour extends into land to the north designated as Exclusive Farm Use.

2007 Existing Runway Configuration

By 2007, aviation activity levels are forecasted to reach 24,500 operations annually. Noise contours for 2007 are shown on Drawing 17 and demonstrate that there is no encroachment of the 65 Ldn contour on noise-sensitive land uses. Again, the 65 Ldn contour remains entirely on airport property, with only the 55 Ldn contour extending into agricultural land to the north.

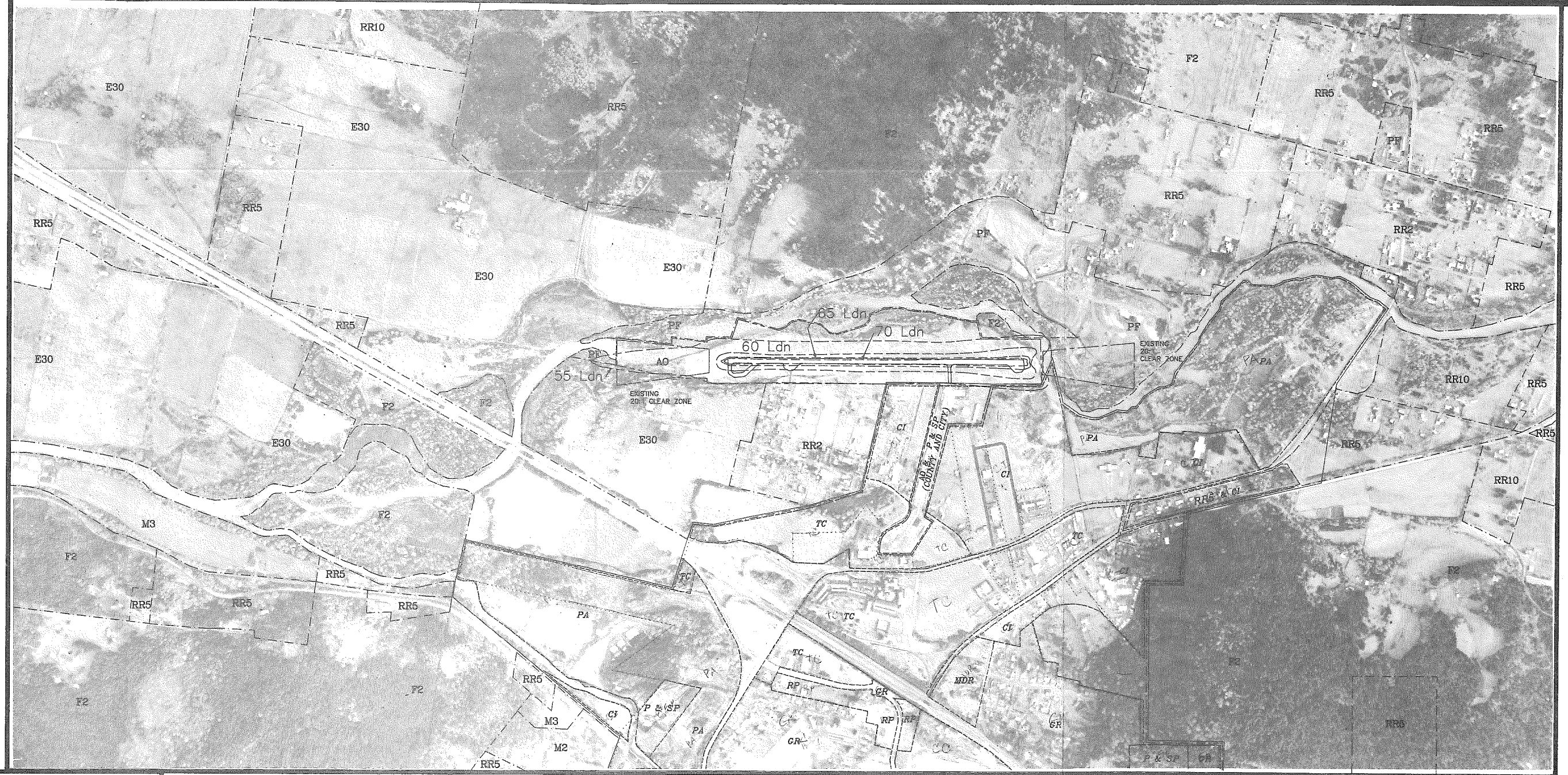
Discussion and Recommendations

It is clear from Figures 14, 15, 16, and 17 that projected growth in activity at Cottage Grove State Airport is not expected to result in greater noise impacts upon surrounding uses.

As demonstrated on Figure 14, it is apparent that significant adverse noise impacts do not exist with the current runway configuration and the existing level of aircraft operations. Noise levels are still within FAA standards of compatibility in 2007 with forecasted activity (Figure 17). Further, working-draft noise contours generated for 1992 with the proposed 190' runway extension shows no significant variation in noise impacts with the extension. The runway extension is a safety margin and will not affect the number, type or size of aircraft utilizing Cottage Grove State Airport.

The State Aeronautics Division desires an efficient airport operation which meets the area aviation demands and accrues the positive impacts of improving safety conditions and strengthening the economic base of the area, while working within the parameters of acceptable levels of environmental impact. The proposed development project will be undertaken to fulfill those goals and objectives.

Additionally, it is recommended that an ongoing airport advisory committee be maintained, including representatives of close-in residential areas, as well as users and tenants, to facilitate positive communication, education, and understanding.



DESIGNATIONS

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CR - GENERAL RESIDENTIAL
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RP - RESIDENTIAL PROFESSIONAL
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CC - COMMUNITY COMMERCIAL
TC - TOURIST COMMERCIAL
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R & CP - REGIONAL & COMMUNITY PARKS
P & SP - OTHER PUBLIC & SEMI-PUBLIC LANDS

LANE COUNTY ZONING DESIGNATIONS

E40 - EXCLUSIVE FARM USE 40 ACRE MINIMUM
E30 - EXCLUSIVE FARM USE 30 ACRE MINIMUM
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PF - PUBLIC FACILITIES
CR - RURAL COMMERCIAL
M3 - HEAVY INDUSTRIAL
RR2 - RURAL RESIDENTIAL 2 ACRE MINIMUM
RR5 - RURAL RESIDENTIAL 5 ACRE MINIMUM
RR10 - RURAL RESIDENTIAL 10 ACRE MINIMUM
SG - SAND AND GRAVEL PRODUCTS
A0 - AIRPORT OPERATIONS

The preparation of this document was financed in part by a planning grant from the FAA as provided under Section 505 of the Airport and Airways Improvement Act of 1982. This document does not necessarily reflect the views of the FAA until reviewed and signed.

LEGEND

CITY OF COTTAGE GROVE
LAND USE DESIGNATION LINE
LANE COUNTY ZONING
DESIGNATION LINE
CITY OF COTTAGE GROVE
CITY LIMITS
CITY OF COTTAGE GROVE
URBAN GROWTH BOUNDARY
EXISTING AIRPORT PROPERTY
LINE
NOISE CONTOUR LINE

LAND USE COMPATIBILITY* WITH YEARLY DAY-NIGHT AVERAGE SOUND LEVELS

LAND USE	Yearly Day-Night Average Sound Level (L _{dn}) in Decibels						
	Below 65	65-70	70-75	75-80	80-85	Over 85	
RESIDENTIAL							
Residential, other than mobile homes and transient lodgings	Y	N(1)	N(1)	N	N	N	
Mobile home parks	Y	N	N	N	N	N	
Transient lodgings	Y	N(1)	N(1)	N	N	N	
PUBLIC USE							
Schools	Y	N(1)	N(1)	N	N	N	
Hospitals and nursing homes	Y	Y	Y	N	N	N	
Churches, auditoriums, and concert halls	Y	Y	Y	N	N	N	
Governmental services	Y	Y	Y	Y	Y	Y	
Transportation	Y	Y	Y	Y	Y	Y	
Parking	Y	Y	Y	Y	Y	Y	
COMMERCIAL USE							
Offices, business and professional	Y	Y	Y	Y	Y	Y	
Wholesale and retail-building materials, hardware and farm equipment	Y	Y	Y	Y	Y	Y	
Retail trade-general	Y	Y	Y	Y	Y	Y	
Utilities	Y	Y	Y	Y	Y	Y	
Communication	Y	Y	Y	Y	Y	Y	
MANUFACTURING AND PRODUCTION							
Manufacturing, general	Y	Y	Y	Y	Y	Y	
Photographic and optical	Y	Y	Y	Y	Y	Y	
Agriculture (except livestock) and forestry	Y	Y	Y	Y	Y	Y	
Livestock farming and breeding	Y	Y	Y	Y	Y	Y	
Mining and fishing, resource production and extraction	Y	Y	Y	Y	Y	Y	
RECREATIONAL							
Outdoor sports arenas and spectator sports	Y	Y	Y	Y	Y	Y	
Outdoor music shells, amphitheaters	Y	Y	Y	Y	Y	Y	
Nature exhibits and zoos	Y	Y	Y	Y	Y	Y	
Amusements, parks, resorts and camps	Y	Y	Y	Y	Y	Y	
Golf courses, riding stables and water recreation	Y	Y	Y	Y	Y	Y	

Numbers in parentheses refer to notes.

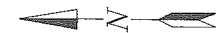
Notes for Table:

- Where the community determines that residential or school uses must be allowed, measures to achieve outdoor to indoor Noise Level Reduction (NLR) of at least 25 dB to 30 dB should be incorporated into building codes and be considered in individual approvals. Normal residential construction can be expected to provide a NLR of 20 dB, thus, the reduction requirements are often stated as 5, 10, or 15 dB over standard construction and normally assume mechanical ventilation and closed windows year round. However, the use of NLR criteria will not eliminate noise problems.
- Measures to achieve NLR of 25 dB must be incorporated into the design and construction of portions of these buildings where the public is received, office areas, noise sensitive areas or where the normal noise level is low.
- Measures to achieve NLR of 30 dB must be incorporated into the design and construction of portions of these buildings where the public is received, office areas, noise sensitive areas, or where the normal noise level is low.
- Measures to achieve NLR of 35 dB must be incorporated into the design and construction of portions of these buildings where the public is received, office areas, noise sensitive areas or where the normal noise level is low.
- Land use compatible provided special sound reinforcement systems are installed.
- Residential buildings require an NLR of 25.
- Residential buildings require an NLR of 30.
- Residential buildings not permitted.

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KEY TO TABLE

SLUCM
Y (Yes)
N (No)
NLR
25, 30, or 35
Standard Land Use Coding Manual
Land Use and related structures compatible without restrictions.
Land Use and related structures are not compatible and should be prohibited.
Noise Level Reduction (outdoor to indoor) to be achieved through incorporation of noise attenuation into the design and construction of the structure.
Land uses and related structures generally compatible; measures to achieve NLR of 25, 30, or 35 dB must be incorporated into the design and construction of structures.



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DIVISION OF AERONAUTICS

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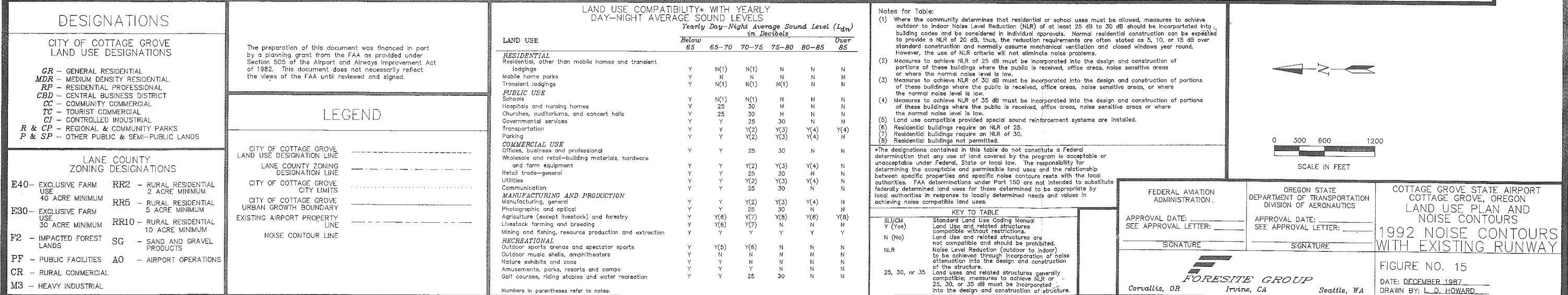
COTTAGE GROVE STATE AIRPORT
COTTAGE GROVE, OREGON
LAND USE PLAN AND
NOISE CONTOURS

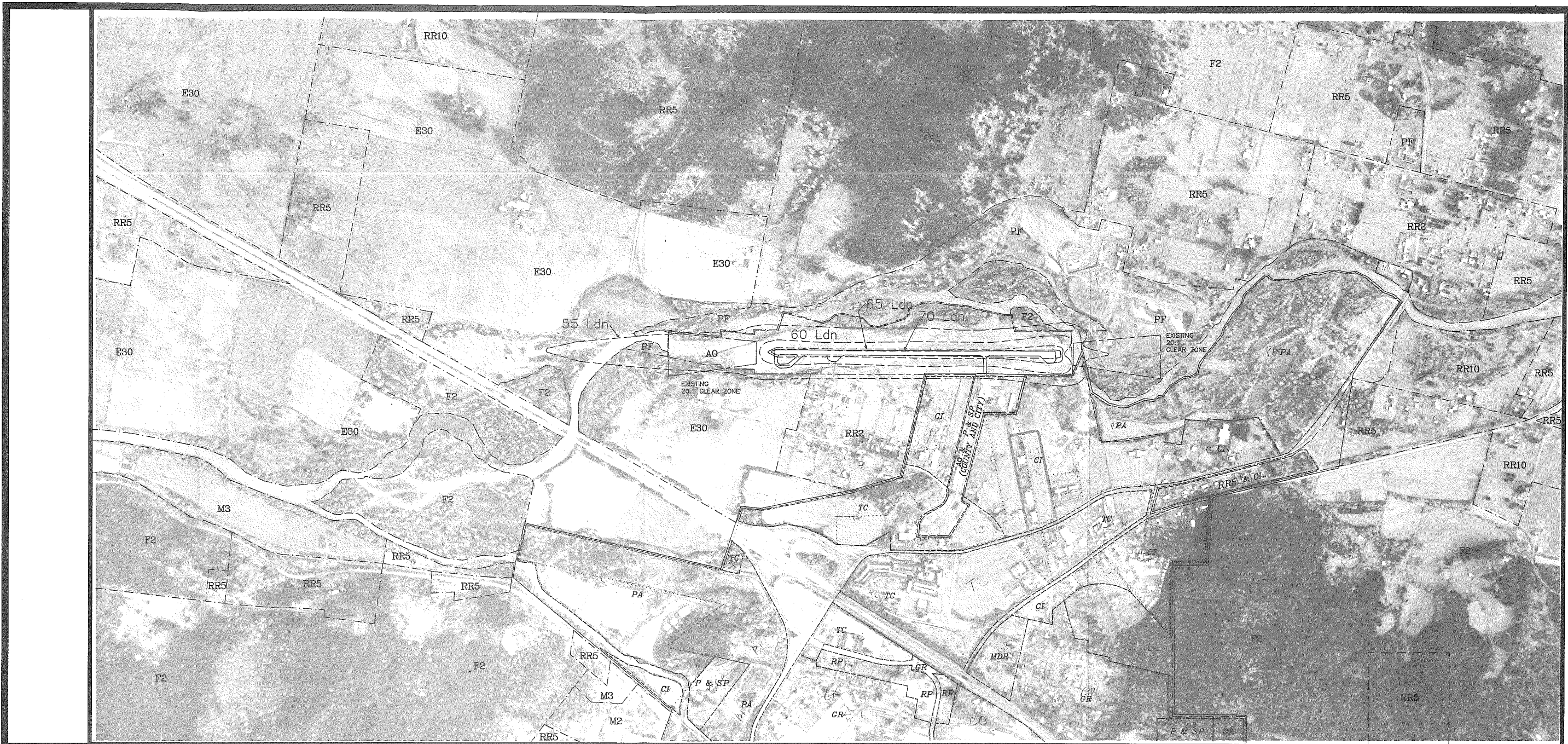
1987 NOISE CONTOURS
WITH EXISTING RUNWAY

FIGURE NO. 14

DATE: DECEMBER 1987
DRAWN BY: L. D. HOWARD

FORESITE GROUP
Corvallis, OR Irvine, CA Seattle, WA





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RR10 - RURAL RESIDENTIAL 10 ACRE MINIMUM
SG - SAND AND GRAVEL PRODUCTS
AO - AIRPORT OPERATIONS

LEGEND

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CITY OF COTTAGE GROVE
LAND USE DESIGNATION LINE
LANE COUNTY ZONING
DESIGNATION LINE
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CITY LIMITS
CITY OF COTTAGE GROVE
URBAN GROWTH BOUNDARY
EXISTING AIRPORT PROPERTY
LINE
NOISE CONTOUR LINE

LAND USE COMPATIBILITY* WITH YEARLY DAY-NIGHT AVERAGE SOUND LEVELS

LAND USE	Yearly Day-Night Average Sound Level (Ldn) in Decibels						
	Below 65	65-70	70-75	75-80	80-85	Over 85	
RESIDENTIAL							
Residential, other than mobile homes and transient lodgings	Y	N(1)	N(1)	N	N	N	
Mobile home parks	Y	N	N	N	N	N	
Transient lodgings	Y	N(1)	N(1)	N(1)	N	N	
PUBLIC USE							
Schools	Y	N(1)	N(1)	N	N	N	
Hospitals and nursing homes	Y	25	30	N	N	N	
Churches, auditoriums, and concert halls	Y	25	30	N	N	N	
Governmental services	Y	Y	25	30	N	N	
Transportation	Y	Y	Y(2)	Y(3)	Y(4)	Y(4)	
Parking	Y	Y	Y(2)	Y(3)	Y(4)	N	
COMMERCIAL USE							
Offices, business and professional	Y	Y	25	30	N	N	
Wholesale and retail-building materials, hardware and farm equipment	Y	Y	Y(2)	Y(3)	Y(4)	N	
Retail trade-general	Y	Y	25	30	N	N	
Utilities	Y	Y	Y(2)	Y(3)	Y(4)	N	
Communication	Y	Y	25	30	N	N	
MANUFACTURING AND PRODUCTION							
Manufacturing, general	Y	Y	Y(2)	Y(3)	Y(4)	N	
Photographic and optical	Y	Y	25	30	N	N	
Agriculture (except livestock) and forestry	Y	Y(6)	Y(7)	Y(8)	Y(8)	Y(8)	
Livestock farming and breeding	Y	Y(6)	Y(7)	N	N	N	
Mining and fishing, resource production and extraction	Y	Y	Y	Y	Y	Y	
RECREATIONAL							
Outdoor sports arenas and spectator sports	Y	Y(5)	Y(6)	N	N	N	
Outdoor music shells, amphitheaters	Y	N	N	N	N	N	
Nature exhibits and zoos	Y	Y	N	N	N	N	
Amusements, parks, resorts and camps	Y	Y	Y	N	N	N	
Golf courses, riding stables and water recreation	Y	Y	25	30	N	N	

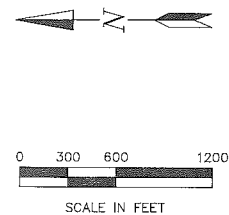
Numbers in parentheses refer to notes.

Notes for Table:

- Where the community determines that residential or school uses must be allowed, measures to achieve outdoor to indoor Noise Level Reduction (NLR) of at least 25 dB to 30 dB should be incorporated into building codes and be considered in individual approvals. Normal residential construction can be expected to provide a NLR of 20 dB, thus, the reduction requirements are often stated as 5, 10, or 15 dB over standard construction and normally assume mechanical ventilation and closed windows year round. However, the use of NLR criteria will not eliminate noise problems.
- Measures to achieve NLR of 25 dB must be incorporated into the design and construction of portions of these buildings where the public is received, office areas, noise sensitive areas or where the normal noise level is low.
- Measures to achieve NLR of 30 dB must be incorporated into the design and construction of portions of these buildings where the public is received, office areas, noise sensitive areas, or where the normal noise level is low.
- Measures to achieve NLR of 35 dB must be incorporated into the design and construction of portions of these buildings where the public is received, office areas, noise sensitive areas or where the normal noise level is low.
- Land use compatible provided special sound reinforcement systems are installed.
- Residential buildings require an NLR of 25.
- Residential buildings require an NLR of 30.
- Residential buildings not permitted.

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KEY TO TABLE	
SLCUM	Standard Land Use Coding Manual
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N (No)	Land use and related structures are not compatible and should be prohibited.
NLR	Noise Level Reduction (outdoor to indoor) to be achieved through incorporation of noise attenuation into the design and construction of the structure.
25, 30, or 35	Land uses and related structures generally compatible; measures to achieve NLR of 25, 30, or 35 dB must be incorporated into the design and construction of structure.



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DEPARTMENT OF TRANSPORTATION
DIVISION OF AERONAUTICS

APPROVAL DATE: _____
SEE APPROVAL LETTER: _____

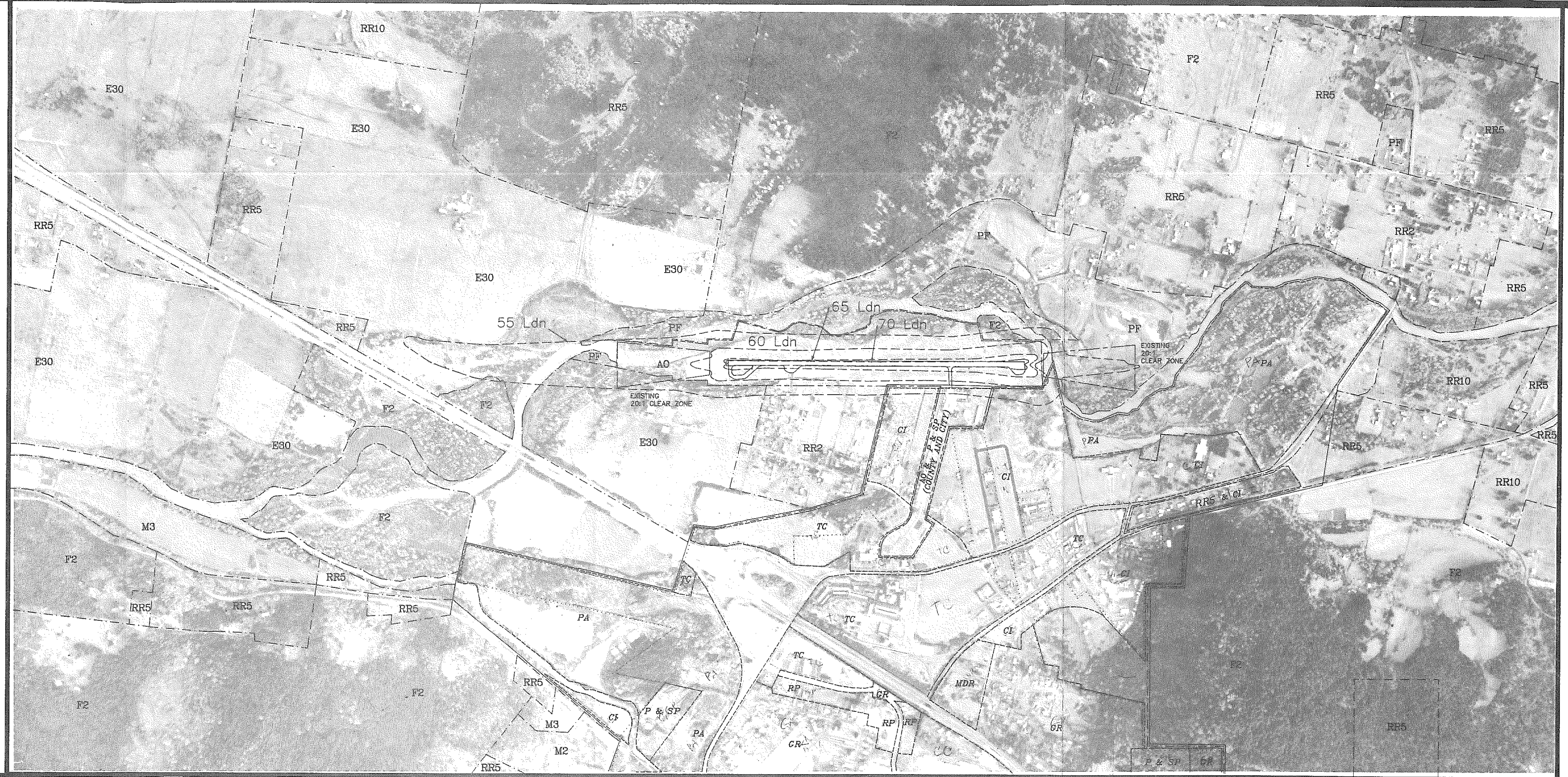
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COTTAGE GROVE STATE AIRPORT
COTTAGE GROVE, OREGON
LAND USE PLAN AND
NOISE CONTOURS
1997 NOISE CONTOURS
WITH EXISTING RUNWAY

FIGURE NO. 16

DATE: DECEMBER 1987
DRAWN BY: L.D. HOWARD

FORESITE GROUP
Corvallis, OR Irvine, CA Seattle, WA



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2 ACRE MINIMUM
RR5 - RURAL RESIDENTIAL
5 ACRE MINIMUM
RR10 - RURAL RESIDENTIAL
10 ACRE MINIMUM
SQ - SAND AND GRAVEL PRODUCTS
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LEGEND

CITY OF COTTAGE GROVE
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LANE COUNTY ZONING
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CITY LIMITS
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LAND USE	Yearly Day-Night Average Sound Level (L _{dn}) in Decibels						
	Below 65	65-70	70-75	75-80	80-85	Over 85	
RESIDENTIAL							
Residential, other than mobile homes and transient lodgings	Y	N(1)	N(1)	N	N	N	
Mobile home parks	Y	N	N	N	N	N	
Transient lodgings	Y	N(1)	N(1)	N(1)	N	N	
PUBLIC USE							
Schools	Y	N(1)	N(1)	N	N	N	
Hospitals and nursing homes	Y	25	30	N	N	N	
Churches, auditoriums, and concert halls	Y	25	30	N	N	N	
Governmental services	Y	Y	25	30	N	N	
Transportation	Y	Y	Y(2)	Y(3)	Y(4)	Y(4)	
Parking	Y	Y	Y(2)	Y(3)	Y(4)	N	
COMMERCIAL USE							
Offices, business and professional	Y	Y	25	30	N	N	
Wholesale and retail-building materials, hardware and farm equipment	Y	Y	Y(2)	Y(3)	Y(4)	N	
Retail trade-general	Y	Y	25	30	N	N	
Utilities	Y	Y	Y(2)	Y(3)	Y(4)	N	
Communication	Y	Y	25	30	N	N	
MANUFACTURING AND PRODUCTION							
Manufacturing, general	Y	Y	Y(2)	Y(3)	Y(4)	N	
Photographic and optical	Y	Y	25	30	N	N	
Agriculture (except livestock) and forestry	Y	Y(6)	Y(7)	Y(8)	Y(8)	Y(8)	
Livestock farming and breeding	Y	Y(6)	Y(7)	N	N	N	
Mining and fishing, resource production and extraction	Y	Y	Y	Y	Y	Y	
RECREATIONAL							
Outdoor sports arenas and spectator sports	Y	Y(5)	Y(5)	N	N	N	
Manufacturing, general	Y	N	N	N	N	N	
Nature exhibits and zoos	Y	Y	N	N	N	N	
Amusements, parks, resorts and camps	Y	Y	Y	N	N	N	
Golf courses, riding stables and water recreation	Y	Y	25	30	N	N	

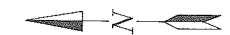
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Notes for Table:

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- Measures to achieve NLR of 25 dB must be incorporated into the design and construction of portions of these buildings where the public is received, office areas, noise sensitive areas, or where the normal noise level is low.
- Measures to achieve NLR of 30 dB must be incorporated into the design and construction of portions of these buildings where the public is received, office areas, noise sensitive areas, or where the normal noise level is low.
- Measures to achieve NLR of 35 dB must be incorporated into the design and construction of portions of these buildings where the public is received, office areas, noise sensitive areas, or where the normal noise level is low.
- Land use compatible provided special sound reinforcement systems are installed.
- Residential buildings require an NLR of 25.
- Residential buildings require an NLR of 30.
- Residential buildings not permitted.

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Corvallis, OR

OREGON STATE
DEPARTMENT OF TRANSPORTATION
DIVISION OF AERONAUTICS

APPROVAL DATE: _____
SEE APPROVAL LETTER: _____

SIGNATURE _____

Irvine, CA

Seattle, WA

COTTAGE GROVE STATE AIRPORT
COTTAGE GROVE, OREGON
LAND USE PLAN AND
NOISE CONTOURS
2007 NOISE CONTOURS
WITH EXISTING RUNWAY

FIGURE NO. 17

DATE: DECEMBER 1987
DRAWN BY: L. D. HOWARD

FORESITE GROUP

COMPATIBLE LAND USE

The compatibility of existing and planned uses in the vicinity of an airport is generally associated with the level of noise and safety impacts related to the airport. Compatibility or noncompatibility of land use is determined by comparing the Ldn noise contour with existing and potential land uses. The FAA has developed guidelines for land-use compatibility based on noise levels and the nature of the land use being impacted.² Commercial, industrial, and most public uses are considered compatible with airport operations, as long as they are consistent with performance standards of Federal Aviation Regulation (FAR) Part 77 relative to height and safety. Residential use is compatible in areas outside the 65 Ldn noise contour. Table 9B provides the land-use compatibility guidelines.

The Cottage Grove State Airport is located in Lane County, Oregon, east of the City of Cottage Grove. The airport lies outside the city limits, but inside the Urban Growth Boundary (UGB). Lane County has adopted an Airport Safety Combining Zone (/AS-RCP) that is designed to "safeguard land uses adjacent to the airport from noise and hazards associated with aircraft operations and to protect existing use and potential expansion of the airport itself from incompatible development."³ The zone requires conformance with Federal Aviation Administration Regulations, Part 77. The airport is planned for annexation into the Cottage Grove city limits in the near future.

The parcels in Lane County on the northern boundary of the airport are zoned Exclusive Farm Use (E-30) with a 30-acre minimum parcel size. A rural residential subdivision (RR-2) is located on the west side of the airport. Parcels to the east, on the other side of the Row River from the airport, are zoned Impacted Forest Lands (F-2) and Public Facility (PF). The area inside the Cottage Grove UGB adjacent to the airport runway on the west and south is zoned Industrial.

Six residences located in the RR-2 zone are within 400' of the runway centerline. As shown on Figures 14 through 17, however, the 65 Ldn noise contour remains entirely on airport property throughout the 20-year planning period. Size and types of aircraft using CGSA will not be affected by the 190' northwesterly extension of Runway 15-33. Working-draft noise contours generated for the runway extension show no significant change in impacts and that the 65 Ldn contour will remain on airport property.

² Airport Noise Compatibility Guidelines, FAR Part 150, Federal Aviation Administration.

³ Lane County Rural Comprehensive Plan 16.246(1), adopted 1984.

Table 9B

**LAND-USE COMPATIBILITY* WITH YEARLY DAY-NIGHT AVERAGE SOUND LEVELS
Federal Guidelines**

	Yearly Day-Night Average Sound Level (Ldn)					
	in Decibels					
	Below 65	65-70	70-75	75-80	80-85	Over 85
<u>Residential</u>						
Residential, other than mobile homes and transient lodgings	Y	N(1)	N(1)	N	N	N
Mobile home parks	Y	N	N	N	N	N
Transient lodgings	Y	N(1)	N(1)	N(1)	N	N
<u>Public Use</u>						
Schools	Y	N(1)	N(1)	N	N	N
Hospitals and nursing homes	Y	25	30	N	N	N
Churches, auditoriums, and concert halls	Y	25	30	N	N	N
Governmental services	Y	Y	25	30	N	N
Transportation	Y	Y	Y(2)	Y(3)	Y(4)	Y(4)
Parking	Y	Y	Y(2)	Y(3)	Y(4)	N
<u>Commercial Use</u>						
Offices, business and professional	Y	Y	25	30	N	N
Wholesale and retail-building materials, hardware and farm equipment	Y	Y	Y(2)	Y(3)	Y(4)	N
Retail trade-general	Y	Y	25	30	N	N
Utilities	Y	Y	Y(2)	Y(3)	Y(4)	N
Communication	Y	Y	25	30	N	N
<u>Manufacturing And Production</u>						
Manufacturing, general	Y	Y	Y(2)	Y(3)	Y(4)	N
Photographic and optical	Y	Y	25	30	N	N
Agriculture (except livestock) and forestry	Y	Y(6)	Y(7)	Y(8)	Y(8)	Y(8)
Livestock farming and breeding	Y	Y(6)	Y(7)	N	N	N
Mining and fishing, resource production and extraction	Y	Y	Y	Y	Y	Y
<u>Recreational</u>						
Outdoor sports arenas and spectator sports	Y	Y(5)	Y(5)	N	N	N
Other music shells, amphitheaters	Y	N	N	N	N	N
Nature exhibits and zoos	Y	Y	N	N	N	N
Amusement parks, resorts, and camps	Y	Y	Y	N	N	N
Golf courses, riding stables and water recreation	Y	Y	25	30	N	N

*The designations contained in this table do not constitute a Federal determination that any use of land covered by the program is acceptable or unacceptable under Federal, State, or local law. The responsibility for

Table 9B (Continued)

determining the acceptable and permissible land uses and the relationship between specific properties and specific noise contours rests with the local authorities. FAA determinations under Part 150 are not intended to substitute federally determined land uses for those determined to be appropriate by local authorities in response to locally determined needs and values in achieving noise compatible land uses.

KEY TO TABLE 1

SLUCM	Standard Land Use Coding Manual.
Y (Yes)	Land Use and related structures compatible without restrictions.
N (No)	Land Use and related structures are not compatible and should be prohibited.
NLR	Noise Level Reduction (outdoor to indoor) to be achieved through incorporation of noise attenuation into the design and construction of the structure.
25, 30, or 35	Land used and related structures generally compatible; measures to achieve NLR of 25, 30, or 35 dB must be incorporated into design and construction of structure.

NOTES FOR TABLE 1

- (1) Where the community determines that residential or school uses must be allowed, measures to achieve outdoor to indoor Noise Level Reduction (NLR) of at least 25 dB and 30 dB should be incorporated into building codes and be considered in individual approvals. Normal residential construction can be expected to provide a NLR of 20 dB, thus, the reduction requirements are often stated as 5, 10, or 15 dB over standard construction and normally assume mechanical ventilation and closed windows year round. However, the use of NLR criteria will not eliminate outdoor noise problems.
- (2) Measures to achieve NLR of 25 dB must be incorporated into the design and construction of portions of these buildings where the public is received, office areas, noise sensitive areas or where the normal noise level is low.
- (3) Measures to achieve NLR of 30 dB must be incorporated into the design and construction of portions of these buildings where the public is received, office areas, noise sensitive areas or where the normal noise level is low.
- (4) Measures to achieve NLR of 35 dB must be incorporated into the design and construction of portions of these buildings where the public is received, office areas, noise sensitive areas or where the normal noise level is low.
- (5) Land use compatible provided special sound reinforcement systems are installed.
- (6) Residential buildings require an NLR of 25
- (7) Residential buildings require an NLR of 35
- (8) Residential buildings not permitted.

Land acquisition is discussed in Chapter 7 of this Master Plan Report and is proposed in the terminal/hangar area. Acquisition is also recommended for the expanded clear zone for the 190' runway extension. Purchase, as opposed to avigation easement, is proposed because full control over existing and future obstructions is most viable for safe, cost-efficient airport operation.

It is recommended that both Lane County and the City of Cottage Grove incorporate this Master Plan into the periodic review of their respective Comprehensive Plans.

SOCIAL IMPACTS

This category relates primarily to the relocation of residences and businesses, or other forms of community disruption. While acquisition of property is proposed for the Preferred Alternative, no existing dwellings or businesses are affected; thus, no impacts are anticipated in this area.

Rerouting of traffic and/or traffic delays may occur in the vicinity of the airport during the construction phase. These impacts, however, are expected to be minimal and will be temporary in nature, thus are not considered significant.

INDUCED SOCIOECONOMIC IMPACTS

This category refers to impacts, such as shifts in business and economic activity demands on public services, or patterns of population growth. Because of the scope of the proposed improvements, it is anticipated that there will be no adverse impacts relative to these factors.

The construction phase of the project will yield a positive, temporary impact with the creation of jobs. The proposed Aviation Museum, with the associated employment, tourists, and influx of dollars into the community, is an example of an induced, positive impact related to upgrading Cottage Grove State Airport.

AIR QUALITY

Cottage Grove is located just south of the confluence of the Coast Fork of the Willamette and Row Rivers, in the Upper Willamette Valley airshed. The valley is bordered by the Coast Range on the west, with an average elevation of 3,400 feet mean sea level (MSL), and on the east by the Cascade Range, which averages, 5,500 feet MSL. The ranges converge at both ends of the valley, resulting in periods of atmospheric stagnation, primarily in the late summer and early fall. During these

periods of stagnant air, an inversion layer of warm air traps air pollution at low elevations. The primary source of pollution in the Cottage Grove area is smoke and particulates from residential wood heat and forest slash burning, and contamination from vehicular traffic associated with city and highway use. The levels of these pollutants do not exceed Federal or State air quality standards.

The air quality in the Cottage Grove area is relatively high, in part due to the absence of heavy polluting industries and large population concentrations.

The Cottage Grove State Airport is under the jurisdiction of the Lane Regional Air Pollution Authority located in Springfield, Oregon. A particulate monitor located at the Harrison School in Cottage Grove is the air quality monitoring station closest to the airport. The Lane Regional Air Pollution Authority also monitors ozone levels and meteorological data at stations in Saginaw, 5 miles north of the airport. Levels of all pollutants monitored by the Lane Regional Air Pollution Authority are well below standards set by the Clean Air Act and Oregon Department of Environmental Quality.

Although aviation activity does contribute to air pollution, projected aviation operations and associated automobile traffic will not produce a significant impact on air quality, according to FAA standards.⁴ Construction of proposed facilities could result in increased dust; however, this effect on air quality would be minimal, short term, and localized.

WATER QUALITY

The Oregon Water Resources Department and Department of Environmental Quality report no known surface or ground water quality problems in the airport planning area. Much of the ground water in the Cottage Grove area is contaminated by arsenic; however, this problem has not been identified at the airport.

The airport is served by the Cottage Grove Municipal Water System. Sewer is currently handled by a septic system that, to date, has been adequate. The city sewer line is extended to the airport property boundary. The City of Cottage Grove has a policy to annex properties before permitting hook-up to city sewer.

The Row River flows along the easterly boundary of the airport property. Construction of the runway extension and parallel taxiway will involve some ground disturbance; however, runoff into the Row River will be controlled through sound application designed to minimize erosion and sedimentation.

⁴ Airport Environmental Handbook, Order 5050.4A, Department of Transportation, FAA, 1985.

SPECIAL LAND USES, DOT ACT, SECTION 4F

Section 4f of the Department of Transportation Act specifies that no project will be approved that requires use of any publicly owned park, recreation area, or wildlife refuge.

The East Regional Park, owned by the City of Cottage Grove, is located directly south of the airport. The park is zoned as a regional and community park. It is undeveloped, except for a bike trail and a BMX track. The Cottage Grove Comprehensive Plan recommends that this park limit recreational development to low intensity uses in keeping with the natural character of the area.

No alternative proposes acquisition of any of this city-owned park land, and the aviation activities and noise levels do not have a significant impact on present or planned uses in the park.

HISTORIC, ARCHITECTURAL, ARCHAEOLOGICAL, AND CULTURAL RESOURCES

The Oregon State Historical Preservation Office (SHPO) has not conducted a reconnaissance survey to determine if cultural or archaeological resources are present on the airport property. SHPO has informed the Consultant that known archaeological sites exist in several locations within two miles of the airport.

Should any potential significant archaeological or cultural resources be discovered during construction, all necessary steps shall be taken, in coordination with SHPO, for the evaluation and preservation, if needed, of such resources.

There are no historic or unique architectural sites present at Cottage Grove State Airport.

BIOTIC COMMUNITIES

The most significant biotic community on the airport property is the riparian zone of the Row River. Except for this fringe of natural vegetation along the river, the airport vegetation consists of natural and introduced grasses that are mowed, and not in a pristine condition.

The riparian area supports a diverse range of plants and wild life and, along with the Coast Fork of the Willamette, is recognized on the City Comprehensive Plan as the most important wildlife habitat in the Cottage Grove area. The Comprehensive Plan recommends that the Oregon State Board of Aeronautics should preserve its land east of the airport in a natural condition to ensure that incompatible uses will not develop east of the airport.

Although the confluence of the Row River and the Coast Fork of the Willamette, further north of the airport, is protected by Greenway designation, the Row River adjacent to the airport is not.

The Row River supports populations of cutthroat trout, steelhead, chinook, and coho salmon. There are no significant spawning areas in the vicinity of the airport. Mammals using the riparian and aquatic habitat include: muskrat, beaver, otter, mink, raccoon, and opossum. Open grassland provides habitat for mice and rodents which in turn are a source of food for resident and wintering raptor populations. Because of the relatively sparsely settled nature of the airport vicinity, deer are infrequent, transient users of the airport habitat.

There are no known heron nesting colonies in the vicinity of the airport. Nesting waterfowl, especially geese, use the river environment and gravel excavation ponds. No sensitive bird habitat is identified by the Lane County Comprehensive Plan.

The proposal for the Cottage Grove State Airport will impact only previously disturbed, man-dominated areas adjacent to and west of the existing runway. Therefore, it is assumed that there will be no significant impact on biotic communities. No riparian vegetation will be disturbed, and any construction will be designed to minimize potential for erosion into the Row River.

ENDANGERED AND THREATENED SPECIES

No threatened or endangered species, as identified by the U.S. Department of Fish and Wildlife, are located on or around the airport.

WETLANDS

The airport planning area contains no wetlands that are listed on the National Wetland Inventory of the U.S. Department of the Interior. Lane County's Comprehensive Plan has not identified any significant wetlands in the planning area.

WILD AND SCENIC RIVERS

There are no state or national wild or scenic rivers in the airport planning area.

SHORELINE MANAGEMENT

This element is not applicable to the Cottage Grove State Airport.

COASTAL BARRIERS

This element is not applicable to the Cottage Grove State Airport.

FLOODPLAIN

Approximately the northerly half of the existing runway is within the 100-year floodplain as defined by the Federal Insurance Administration's Flood Insurance Study for Lane County, Oregon Unincorporated Areas and accompanying Flood Insurance Rate Maps. Approximately 700 feet of the northerly end of the runway is located in the designated floodway of the Row River. The average base flood height for the runway is 632 feet above sea level.

The flooding of the Row River in 1964 was considered a 100-year frequency flood by the Army Corps of Engineers. The flow at that time near the airport was 5,900 cubic feet per second (cfs) and caused backwater from the river to reach the runway. There was no damage.

Dorena Dam, completed in 1949 and operated by the Corps of Engineers, is located 4 miles upstream from the airport. Its primary function is flood control. The reservoir behind the dam is managed to capture and gradually release water from the major storms occurring between November and February. Downstream, excessive flows generally occur only in the event of a major storm after the projected full pool date, May 21. This situation occurred in June 1984, when the overflow from the reservoir measured 5,507 cfs at the river gauge located between the Dorena Dam and the Cottage Grove Airport.

Lane County has adopted a Floodplain Combining Overlay Zone and Implementing Ordinance, Chapter 16.244 of the Lane County Zoning Ordinance. The Ordinance is designed to manage the floodplain to promote the public health safety and general welfare, and to minimize public and private losses due to flood conditions. The ordinance requires a Floodplain Special Use Permit before any construction or development begins within any area of flood hazard. In addition, construction or fill in the floodway requires certification by an Oregon-registered licensed engineer that the construction when combined with all other existing and anticipated improvements will not result in any increase in flood levels during the occurrence of a base flood.

The 190' runway extension and new taxiway on the west side of, and parallel to, the runway will require County Floodplain Special Use permits and a certified engineering study. New construction to the existing runway will require appropriate engineering and construction to minimize erosion and sedimentation and to preserve natural drainage.

The Preferred Alternative will not result in an increased probability for loss of property or human life; all structures and aircraft tie-downs are outside of the 100-year floodplain. Because of the upstream flood control dam, flood events are rare and of short duration, usually less than 48 hours.

No damage or interruption of airport operations has occurred because of flood conditions. The runway extension and parallel taxiway will not increase probability for damage to airport facilities nor will it cause an adverse impact on natural and beneficial floodplain values.

FARMLAND

The Farmland Protection Policy Act (FPPA) requires consideration of adverse effects of Federal Programs on the preservation of farmland. The soils primarily occurring in the airport planning area are Camas Gravelly Sandy Loam, Chapman Loam, Fluvents, and Salem Gravelly Silt Loam, as classified by the U.S.D.A. Soil Conservation Service (SCS). Chapman Loam is identified as a prime agricultural soil. The land within the Cottage Grove Urban Growth Boundary adjacent to the airport property is zoned Industrial. In Lane County, the land adjacent to the airport is zoned Rural Residential and Exclusive Farm Use.

The great majority of land which is recommended for acquisition is, by virtue of its zoning designation, already committed to urban uses. The four acres recommended for acquisition in the clear zone of the extended runway is made up of Fluvents soils, which are not classified as prime agricultural soils. No impacts are, therefore, anticipated upon farmland.

ENERGY SUPPLY AND NATURAL RESOURCES

The preferred development plan will require consumption of fuel and aggregate resources. Consumption of fuel and other natural resources will not substantially increase as a result of the forecasted number of aircraft operations.

LIGHT EMISSIONS AND GLARE

The existing airport lighting and glare do not negatively impact any residences or businesses. The Preferred Alternative involves a non-directional beacon (NDB) and medium intensity Lighting (MIRL) within the 1993-1997 time frame. An area which could potentially be impacted by light emissions is the rural subdivision to the west of the airport. Care should be taken in the installation of lighting that nearby residences are protected from light emissions. Shielding and directing of lighting can be implemented, if necessary to ensure compatibility with adjacent uses.

SOLID WASTE

Lane County operates a solid waste transfer station approximately 10,000 feet (3,050 meters) east of the south end of the runway and across the Row River. This site is northeast of a former sanitary landfill that was closed in July 1986. Airport records indicate a past problem with bird strikes that may have been associated with the landfill. The transfer station is open to the public, Wednesday through Saturday from 8 a.m. to 6 p.m. Large metal containers are used to store the solid waste. The containers are covered with wire mesh screen when the site is closed. Full containers are removed to a sanitary landfill site 12 miles north of the airport.

The new transfer site should cause fewer problems with bird strikes than the previous sanitary landfill because the solid waste is covered and the site is no longer in a direct line of flight with the runway.

CONSTRUCTION IMPACTS

Because the proposed development is still in the conceptual state, preliminary engineering work has not been initiated; thus, actual construction details are not known. However, certain conditions are relevant to potential impacts, regardless of the specific development plan.

Construction activities will impact noise levels during working hours and air quality, due to dust. Engineering design and construction techniques will take into consideration noise and air quality impacts, as well as potential water quality impacts from use of petroleum products such as sealants and pavement. The following controls are recommended:

- o Construction to occur in conformance with FAA Advisory Circular 150/5370-10, "Standards for Specifying Construction on Airports."
- o Construction to occur in conformance with Lane County or City of Cottage Grove regulations, whichever are applicable at the time.
- o Where a disparity exists between FAA and County or City requirements, the more restrictive requirement should apply.
- o Sprinkling to be implemented to minimize dust.
- o Construction hours to be controlled and sound-suppressing equipment to be utilized.
- o Cleared areas to be replanted as soon as feasible.
- o Short-term erosion control to be provided.

APPENDICES

BASED AIRCRAFT AND AIRPORT USERS QUESTIONNAIRE SUMMARY

Appendix A

to

Airport Master Plan

for

Cottage Grove State Airport
Cottage Grove, Oregon

APPENDIX A
 BASED AIRCRAFT AND AIRPORT USERS QUESTIONNAIRE
 Cottage Grove State Airport Master Plan
 Summary

Question 1:

Company and Aircraft Information.

<u>Ownership per Aircraft</u>	<u>Number</u>	<u>Percent of Respondents</u>
Company	6	50%
Private	6	50%

Question 2:

Where are Aircraft Based?

a. Aircraft Based at -	<u>Number</u>	<u>% of Total</u>
Cottage Grove-	11	92%
Sutherlin -	1	8%

b. Aircraft Type

Single	-	12	100%
Twin	-	0	0%

Question 3:

How are Aircraft used?

Category of Usage

Corporate	-	25%
Private Bus.	-	14%
Personal/Rec.	-	53%
Instructional-		8%

Question 4:

Reasons for use of Airport?

- a. Service and set-up product line
- b. Facilities
- c. Accessibility
- d. Available storage
- e. Convenience to homes and offices
- f. Antique Fly-ins
- g. Storage

Question 5:

Percent of trips outside Oregon -

Private	-	19.0% average
Corporate	-	49.0% average

Question 6:

Closure of Cottage Grove State Airport - what would happen?

a. Relocate Aircraft to:

Creswell	47%	Roseburg	06%
Daniels	17%	Sutherlin	06%
Eugene	08%	Walker	06%

b. Sell Aircraft - 08%

Question 7:

Is the Airport serving the needs of General Aviation?

Yes	-	67%
No	-	33%

Reasons for No:

- a) Need for runway extension
- b) Need for parallel taxiway
- c) Additional hangars needed
- d) Need for rental cars

Question 8:

What new facilities are needed?

<u>Facility</u>	<u>Percent Respondents</u>
Parallel Taxiway	92%
Aircraft Parking	33%
Additional Repair Service	25%
Hangars	17%
Longer Runway	8%
Other (VASI)	8%
Rental Cars	8%

Question 9:

Is a longer runway needed?

Yes	-	58%
No	-	42%

Critical Aircraft - Corporate Jets

Question 10:

a. What Corporate facilities are in Lane County?

Headquarters	-	25%
Manufacturing	-	8%

b. In which municipality are the facilities located?

Cottage Grove	-	16%
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Question 11:

Number of company employees?

Companies	-	42%
Total Employees	-	63
Employee Range	-	5 to 30
Average Employees	-	12.6

Question 12:

Additional Comments:

- o Operations nearing peak, need new parallel taxiway
- o Airport growth equals Cottage Grove growth
- o Well-run airport; need new taxiway
- o Need new taxiway, return VASI, and second repair service
- o Airport is source of dollars to community
- o Cottage Grove State Airport as alternate for Eugene
- o Need parallel taxiway

BASED AIRCRAFT AND AIRPORT USERS QUESTIONNAIRE

Appendix B

to

Airport Master Plan

for

Cottage Grove State Airport
Cottage Grove, Oregon

Appendix B
FORESITE GROUP
for
Oregon Aeronautics Division

COTTAGE GROVE STATE AIRPORT
BASED AIRCRAFT AND AIRPORT USERS QUESTIONNAIRE
June 1987

Note: The following information is to be used for statistical purposes only and will remain confidential. Please return to J. Hawley, FORESITE Group, P.O. Box 1836, Corvallis, OR 97339, as soon as possible, but before June 30, 1987. If you have additional information or questions please call Jim Hawley of FORESITE Group at 758-4131.

1. In order that we may fully describe the benefit and use of the Cottage Grove State Airport, please provide the following information:

For company-owned/used aircraft:

Company name _____
Company address _____
Title of person completing
the questionnaire _____

Aircraft based:

<u>Make</u>	<u>Model</u>	<u>Number</u>
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For privately-owned/used aircraft:

Address of person basing aircraft:

Aircraft based:

<u>Make</u>	<u>Model</u>	<u>Number</u>
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2. At which airport is the above aircraft based?
3. How are the aircraft used by you or your company?

<u>Usage Category</u>	<u>% of Use</u>
____ Corporate Business	
____ Private Business	
____ Government	
____ Charter/Commercial	
____ Personal/Recreational	
____ Instructional	
____ Aerial Application	
____ TOTAL	____ 100%

4. Describe briefly how the use of Cottage Grove State Airport benefits your activities or those of your company.
5. Please estimate the percentage of trips made by these aircraft to or from points outside the State of Oregon.
- _____ % of trips to/from points outside Oregon.
6. If the Cottage Grove State Airport were forced to close, how would your aircraft operations be affected? Please check one.
- _____ Move present aircraft to _____ airport.
 _____ Sell aircraft and cease flying.
 _____ Sell aircraft and rent or charter aircraft as required from _____ airport
 _____ Sell aircraft and rely on scheduled airlines.
7. Is the Cottage Grove State Airport adequately serving the needs of General aviation: Yes _____ No _____ (Explain)
8. What new facilities or services do you feel the Cottage Grove State airport should have: _____
9. The possible need for a runway extension is one the specific issues being addressed in the Airport Master Plan Study. Do you believe a longer runway is needed?
- a. What is the critical aircraft?
- b. Number of operations anticipated annually at Cottage Grove State Airport if runway is extended?

FOR TENANTS OPERATING AIRCRAFT FOR CORPORATE USE. (Others skip to question 12).

10. What type of facilities does your company have in the Lane County Area, and in which municipalities are they located?

	<u>Municipality</u>
_____ Headquarters or Main Office	_____
_____ Regional or Divisional Offices	_____
_____ Manufacturing Plant (s)	_____
_____ Distribution Center	_____
_____ Research & Development	_____
_____ Facilities	_____
_____ Other (explain) _____	_____
_____	_____

11. How many total employees are employed by your company in Cottage Grove County Area office, plants, or other facilities.
_____employees
12. Please share any additional comments you may have. (Use back of page, if necessary.)

GLOSSARY

Appendix C

to

Airport Master Plan

**Cottage Grove State Airport
Cottage Grove, Oregon**

GLOSSARY

GLOSSARY
FOR
AIRPORT PLANNING

The following pages identify and define the terminology most frequently used in the various aspects of airport planning. It is intended to guide the layperson through the complex vocabulary of a highly technical field. The definitions contained herein were drawn from FAR Part 1, Definitions and Abbreviations, FAA Order 1000.15A, FAA Glossary, the FAA Airmen's Information Manual (AIM), and publications from other public and private sources, including FORESITE Group, Inc. This glossary is divided into the following subject areas:

<u>Page</u>	<u>Section</u>
1-1	ABBREVIATIONS
2-1	AIRPORT MASTER DEVELOPMENT PLANNING
3-1	AIRPORT NOISE, ENVIRONMENTAL, AND LAND-USE PLANNING
4-1	AIRPORT NAVIGATIONAL AIDS, AIRSPACE, AND AIR TRAFFIC CONTROL
5-1	FORECASTING AND STATISTICAL METHODS

ABBREVIATIONS

<u>AC</u>	- Advisory Circular
<u>ADO</u>	- Airports District Office (FAA)
<u>AFSS</u>	- Automated Flight Service Station
<u>AGL</u>	- Above Ground Level
<u>AIP</u>	- Airport Improvement Program
<u>ALP</u>	- Airport Layout Plan
<u>ALS</u>	- Approach Lighting System
<u>ARP</u>	- Airport Reference Point
<u>ARSA</u>	- Airport Radar Service Area
<u>ASNA</u>	- Aviation Safety and Noise Abatement Act of 1979
<u>ASR</u>	- Airport Surveillance Radar
<u>ASV</u>	- Annual Service Volume
<u>ATC</u>	- Air Traffic Control
<u>ATCT</u>	- Airport Traffic Control Tower
<u>BRL</u>	- Building Restriction Line
<u>CFR</u>	- Crash/Fire/Rescue
<u>DOT</u>	- Department of Transportation
<u>EA</u>	- Environmental Assessment
<u>EIS</u>	- Environmental Impact Statement
<u>EPA</u>	- Environmental Protection Agency
<u>FAA</u>	- Federal Aviation Administration
<u>FAR</u>	- Federal Aviation Regulations
<u>FBO</u>	- Fixed Base Operator
<u>GA</u>	- General Aviation
<u>GS</u>	- Slide Slope
<u>HIRL</u>	- High-Intensity Runway Lighting
<u>HUD</u>	- Housing and Urban Development
<u>IFR</u>	- Instrument Flight Rules
<u>ILS</u>	- Instrument Landing System
<u>INM</u>	- Integrated Noise Model
<u>LDA</u>	- Localizer Directional Aid
<u>Ldn</u>	- Day-Night Average Sound Level
<u>LIRL</u>	- Low-Intensity Runway Lighting
<u>LOC</u>	- ILS Localizer
<u>MIRL</u>	- Medium-Intensity Runway Lighting
<u>MITL</u>	- Medium-Intensity Taxiway Lighting
<u>MLS</u>	- Microwave Landing System
<u>MSL</u>	- Mean Sea Level
<u>NAVAID</u>	- Air Navigation Facility/Navigational Aid
<u>NDB</u>	- Nondirectional Beacon
<u>NEPA</u>	- National Environmental Policy Act of 1969
<u>NPIAS</u>	- National Plan of Integrated Airport Systems
<u>PAPI</u>	- Precision Approach Path Indicator
<u>r/R²</u>	- Correlation Coefficient/Determination Coefficient
<u>REIL</u>	- Runway End Identifier Lights
<u>SEPA</u>	- State Environmental Policy Act
<u>STOL</u>	- Short Takeoff and Landing
<u>TCA</u>	- Terminal Control Area
<u>TRACON</u>	- Terminal Radar Approach Control
<u>TRSA</u>	- Terminal Radar Service Area
<u>VASI</u>	- Visual Approach Slope Indicator
<u>VFR</u>	- Visual Flight Rules
<u>VOR</u>	- Very High Frequency Omnidirectional Range

AIRPORT MASTER DEVELOPMENT PLANNING

ACTIVE AIRCRAFT - Aircraft registered with the FAA and reported or estimated to have been flown during the preceding calendar year.

AIP - The Airport Improvement Program of the Airport and Airways Improvement Act of 1982. Under this program, the FAA provides funding assistance for the design and development of airports and airport facilities.

AIR CARRIER - A commercial operator engaging in the carriage of persons or property in air commerce for compensation or hire. Air carriers are certificated in accordance with FAR Parts 121 and 127, and generally operate aircraft having a seating capacity of more than 30 passengers or a maximum payload capacity of more than 7,500 pounds.

AIR TAXI - Scheduled and/or nonscheduled aircraft operations carrying passengers and/or cargo for compensation. The capacity of air taxi aircraft is limited by Part 135 of the Federal Aviation Regulations.

AIR TRAFFIC - Aircraft operating in the air or on an airport surface, excluding loading ramps and parking areas.

AIRCRAFT APPROACH CATEGORY - A grouping of aircraft based on a speed of 1.3 times the stall speed in the landing configuration at maximum gross landing weight. An aircraft shall fit in only one category. If it is necessary to maneuver at speeds in excess of the upper limit of a speed range for a category, the minima for the next higher category should be used. The categories are as follows:

- Category A - Speed less than 91 knots.
- Category B - Speed 91 knots or more but less than 121 knots.
- Category C - Speed 121 knots or more but less than 141 knots.
- Category D - Speed 141 knots or more but less than 166 knots.
- Category E - Speed 166 knots or more.

AIRCRAFT MIX - The number of aircraft movements categorized by capacity group or operational group, and specified as a percentage of the total aircraft movements.

AIRCRAFT OPERATION - An aircraft takeoff or landing.

AIRPLANE DESIGN GROUP (PHYSICAL CHARACTERISTICS) - The airplane design group subdivides airplanes by wingspan. The airplane design group concept links an airport's dimensional standards to aircraft approach categories or to airplane design groups or to runway instrumentation configurations. The airplane design groups are:

1. Airplane Design Group I: Wingspan up to but not including 49 feet.
2. Airplane Design Group II: Wingspan 49 feet up to but not including 79 feet.
3. Airplane Design Group III: Wingspan 79 feet up to but not including 118 feet.
4. Airplane Design Group IV: Wingspan 118 feet up to but not including 171 feet.
5. Airplane Design Group V: Wingspan 171 feet up to but not including 197 feet.
6. Airplane Design Group VI: Wingspan 197 feet up to but not including 262 feet.

AIRPORT - An area of land or water used or intended to be used for landing and takeoff of aircraft; includes buildings and facilities, if any. Airports are designed in the following types:

1. Basic Utility - Stage I: This type of airport serves about 75 percent of the single-engine and small twin-engine airplanes used for personal and business purposes. Precision approach operations are not usually anticipated. This airport is designed to serve small airplanes in Airplane Design Group I.
2. Basic Utility - Stage II: This type of airport serves all the airplanes of Stage I, plus some small business and air taxi-type twin-engine airplanes. Precision approach operations are not usually anticipated. This airport is designed to serve small airplanes in Airplane Design Groups I and II.
3. General Utility - Stage I: This type of airport serves all small airplanes. Precision approach operations are not usually anticipated. This airport is designed to serve those aircraft which can also be served at Basic Utility Airports.
4. General Utility - Stage II: This type of airport serves larger airplanes in Aircraft Approach Categories A and B and Airplane Design Groups I and II, and usually has the capability for precision approach operations. General Utility Airports may also be designed to serve Aircraft Approach Category A airplanes in Airplane Design Group III.

5. Transport - This type of can airport serve aircraft in Approach Categories C and D, and usually those in Airplane Design Group III and higher. Transport runways usually have the capability for precision approach operations.

AIRPORT ELEVATION/FIELD ELEVATION - The highest point of an airport's usable runways measured in feet from mean sea level.

AIRPORT HAZARD - Any structure or natural object located on or in the vicinity of a public airport, or any use of land near such airport, that obstructs the airspace required for the flight of aircraft in landing or taking off at the airport or is otherwise hazardous to aircraft landing, taking off, or taxiing at the airport.

AIRPORT LAYOUT PLAN (ALP) - A graphic presentation, to scale, of existing and proposed airport facilities, their location on the airport, and the pertinent clearance and dimensional information required to show conformance with applicable standards. To be eligible for AIP funding assistance, an airport must have an FAA-approved Airport Layout Plan.

AIRPORT LIGHTING - Various lighting aids that may be installed on an airport. Types of airport lighting include:

1. Approach Light System/ALS: An airport lighting facility which provides visual guidance to landing aircraft by radiating light beams in a directional pattern by which the pilot aligns with the extended centerline of the runway on his final approach for landing.

Condenser-Discharge Sequential Flashing Lights/Sequenced Flashing Lights may be installed in conjunction with the ALS at some airports. Types of Approach Light Systems are:

- a. ALSF-I: Approach Light System with Sequenced Flashing Lights in ILS Cat-I configuration.
- b. ALSF-II: Approach Light System with Sequenced Flashing Lights in ILS Cat-II configuration.
- c. SSALF: Simplified Short Approach Light System with Sequenced Flashing Lights.
- d. SSALR: Simplified Short Approach Light System with Runway Alignment Indicator Lights.
- e. MALSF: Medium Intensity Approach Light System with Sequenced Flashing Lights.
- f. MALSR: Medium Intensity Approach Light System with Runway Alignment Indicator Lights.
- g. LDIN: Sequenced Flashing Lead-in Lights.
- h. RAIL: Runway Alignment Indicator Lights (Sequenced Flashing Lights which are installed only in combination with other light systems).

1. ODALS: Omnidirectional Approach Lighting System consists of seven omnidirectional flashing lights located in the approach area of a nonprecision runway.
2. Runway Lights/Runway Edge Lights: Lights having a prescribed angle of emission used to define the lateral limits of a runway. Runway lights are uniformly spaced at intervals of approximately 200 feet, and the intensity may be controlled or preset.
3. Touchdown Zone Lighting: Two rows of transverse light bars located symmetrically about the runway centerline normally at 100 foot intervals. The basic system extends 3,000 feet along the runway.
4. Runway Centerline Lighting: Flush centerline lights spaced at 50-foot intervals beginning 75 feet from the landing threshold and extending to within 75 feet of the opposite end of the runway.
5. Threshold Lights: Fixed green lights arranged symmetrically left and right of the runway centerline, identifying the runway threshold.
6. Runway End Identifier Lights/REIL: Two synchronized flashing lights, one on each side of the runway threshold, which provide rapid and positive identification of the approach end of a particular runway.
7. Visual Approach Slope Indicator/VASI: An airport lighting facility providing vertical visual approach slope guidance to aircraft during approach to landing by radiating a directional pattern of high intensity red and white focused light beams which indicate to the pilot that he is "on path" if he sees red/white, "above path" if white/white, and "below path" if red/red.
8. Precision Approach Path Indicator/PAPI: (same function and description as for VASI but different configuration).
9. Boundary Lights: Lights defining the perimeter of an airport or landing area.

AIRPORT MASTER PLAN - An assembly of appropriate documents and drawings covering the development of a specific airport from a physical, economical, social, and political jurisdictional perspective. The airport layout plan is a part of this plan.

AIRPORT MASTER PLANNING - The development for planning purposes of information and guidance to determine the extent, type, and nature of development needed at a specific airport. It may include the preparation of an airport layout plan and feasibility studies, and the conduct of such other studies, surveys, and planning actions as may be necessary to determine the short-, intermediate-, and long-range aeronautical demands required to be met by a particular airport as a part of a system of airports.

AIRPORT REFERENCE POINT (ARP) - An ARP is a point having equal relationship to all existing and proposed landing and takeoff areas which is used to locate the airport geographically.

AIRPORT ROTATING BEACON - A visual NAVAID operated at many airports. At civil airports, alternating white and green flashes indicate the location of the airport.

AIRPORT TRAFFIC AREA - Unless otherwise specifically designated in FAR Part 93, that airspace within a horizontal radius of 5 statute miles from the geographical center of any airport at which a control tower is operating, extending from the surface up to, but not including, an altitude of 3,000 feet above the elevation of the airport. Unless otherwise authorized or required by ATC, no person may operate an aircraft within an airport traffic area except for the purpose of landing at, or taking off from, an airport within that area.

AIRSIDE - Portion of the airport directly related to the arrival and departure of aircraft, including such airfield facilities as runways, taxiways, navigational aids, marking, and lighting. (See LANDSIDE.)

ANNUAL SERVICE VOLUME (ASV) - ASV is a reasonable estimate of an airport's annual capacity. It accounts for differences in runway use, aircraft mix, weather conditions, etc., that would be encountered over a year's time.

APPROACH AND CLEAR ZONES PLAN - The Approach and Clear Zones Plan is compiled from the criteria in FAR Part 77, Objects Affecting Navigable Airspace. It shows the area affected by the Airport Obstructions Zoning Ordinance and includes layout of runways, airport boundary, elevations, and area topography. Applicable height limitation areas are shown in detail.

APPROACH SLOPES - The ratios of horizontal to vertical distance indicating the degree of inclination of the Approach Surface. The various ratios include:

- | | |
|------|---|
| 20:1 | For all utility and visual runways extended from the primary surface a distance of 5,000 feet. |
| 34:1 | For all nonprecision instrument runways other than utility extended from the primary surface for a distance of 10,000 feet. |

50:1/40:1 For all precision instrument runways extending from the primary surface for a distance of 10,000 feet at an approach slope of 50:1 and an additional 40,000 feet beyond this at a 40:1 Approach Slope.

APPROACH SURFACE - An element of the airport imaginary surfaces longitudinally centered on the extended runway centerline, extending outward and upward from the end of the primary surface at a designated slope.

APRON/RAMP - An area designated for aircraft use, other than taxiways and runways. Example uses for an apron include loading and unloading, parking, maintenance, refueling, before take-off engine runup, and as a temporary traffic holding area.

BASED AIRCRAFT - An aircraft permanently stationed at an airport.

BUILDING RESTRICTION LINE (BRL) - A line shown on the Airport Layout Plan beyond which airport buildings must not be positioned in order to limit their proximity to aircraft movement areas.

CAPACITY - Capacity (throughput capacity) is a measure of the maximum number of aircraft operations which can be accommodated on the airport or airport component in an hour. Since the capacity of an airport component is independent of the capacity of other airport components, it can be calculated separately.

CEILING - The height above the earth's surface of the lowest layer of clouds or other phenomena which obscure vision.

C/F/R - Crash/Fire/Rescue facilities including vehicles, personnel, and buildings.

CLEAR ZONE - An area extending from the end of a runway which is kept clear of obstructions to approaching and departing aircraft.

COMMERCIAL SERVICE - Commercial service airports are public-use airports which receive scheduled passenger service aircraft and which annually enplane 2,500 or more passengers.

COMMUTER AIR CARRIER - An air carrier, certificated in accordance with FAR Part 135, which operates aircraft with a maximum of 60 seats and provides at least five scheduled round trips per week between two or more points, or which carries mail.

COMPASS ROSE - A circle, graduated in degrees, printed on some charts or marked on the ground at an airport. It is used as a reference to either true or magnetic direction.

CONVENTIONAL HANGAR - A large building used to store more than one aircraft and/or to conduct aircraft maintenance.

CRITICAL AIRCRAFT - In airport design, the aircraft which controls one or more design items such as runway length, pavement strength, lateral separation, etc., for a particular airport. The same aircraft may not be critical to all design items.

CROSSWIND RUNWAY - A runway additional to the primary runway to provide for wind coverage not adequately provided by the primary runway.

DISPLACED THRESHOLD - A threshold located at a point on the runway other than at the runway end. Except for the approach standards defined in FAR Part 77, approach surfaces are associated with the threshold location.

FIXED BASE OPERATOR (FBO) - A private firm providing airport services such as fuel sales, aircraft maintenance, aircraft rental, and flight instruction.

FLIGHT SERVICE STATION (FSS) - A facility operated by the FAA to provide flight service assistance.

GENERAL AVIATION (GA) - The portion of civil aviation which includes all facets of aviation except scheduled air carriers.

GENERAL AVIATION AIRPORT - General Aviation airports are either publicly or privately owned airports which serve general aviation aircraft users.

HELIPAD - A small, designated area, usually with a prepared surface, on a heliport, airport, landing/takeoff area, apron/ramp, or movement area used for takeoff, landing, or parking of helicopters.

HELIPORT - An area of land, water, or structure used or intended to be used for the landing and takeoff of helicopters, including buildings and facilities, if any.

IMAGINARY SURFACES - Those areas established in relation to the airport and to each runway consistent with FAR Part 77 in which any object extending above these imaginary surfaces is, by definition, an obstruction.

ITINERANT OPERATIONS - All operations at an airport which are not local operations. (See LOCAL OPERATIONS.)

LANDING GEAR - That part of an aircraft which is required for landing. Gear may be configured as Single-Wheel Gear (SWG or SW), Dual-Wheel Gear (DWG or DW), or Dual-Tandem-Wheel Gear (DTWG or DTW).

LANDSIDE - Portions of the airport interfacing with or supporting the airfield functions, including such facilities as terminal area buildings, aircraft parking apron, automobile parking area, fuel storage, air cargo, and ground access. (See AIRSIDE.)

LARGE AIRCRAFT - Aircraft of more than 12,500 pounds maximum certificated takeoff weight.

LOCAL OPERATIONS - Operations by aircraft flying in the traffic pattern or within sight of the control tower, aircraft known to be arriving or departing from flight in local practice areas, or aircraft executing practice instrument approaches at the airport.

MOVEMENT AREA - The runways, taxiways, and other areas of an airport/heliport which are utilized for taxiing/hover taxiing, takeoff, and landing of aircraft, exclusive of loading ramps and parking areas. At those airports/heliports with a tower, specific approval for entry onto the movement area must be obtained from ATC.

MULTI-ENGINE AIRCRAFT - Reciprocating-powered, fixed-wing aircraft having more than one engine and categorized as weighing more than or less than 12,500 pounds maximum gross weight.

NATIONAL PLAN OF INTEGRATED AIRPORT SYSTEMS (NPIAS) - A plan prepared annually by the FAA which identifies, for the Congress and the public, the composition of a national system of airports together with the airport development necessary to anticipate and meet the present and future needs of civil aeronautics, to meet requirements in support of the national defense, and to meet the special needs of the Postal Service. The plan includes both new facilities and qualitative improvements to existing airports to increase their capacity, safety, technological capability, etc.

NONPRECISION INSTRUMENT RUNWAY - A runway having an existing or planned instrument approach procedure from which a straight-in landing is approved but no electronic glide slope information is available and for which no precision approach facilities are planned.

OBSTRUCTION - An object which penetrates an imaginary surface described in the FAA's Federal Aviation Regulations (FAR), Part 77.

OBSTRUCTION LIGHTS - Lights, often red and white, mounted on a surface structure or on natural terrain, to warn pilots of an obstruction.

PARALLEL RUNWAYS - Two or more runways at the same airport whose centerlines are parallel. Designated both by runway number and L (left) or R (right), or, if three parallel runways exist, L (left), C (center), and R (right).

PRECISION INSTRUMENT RUNWAY - A runway having an existing or planned instrument approach that is essentially aligned with the runway centerline and that has electronic glide slope information for guidance of the descent of the aircraft to the touchdown point on the runway.

PRIMARY SERVICE - Primary service airports are public-use airports which receive scheduled passenger service aircraft and which annually enplane one one-hundredth percent (0.01%) or more of the combined total domestic passenger enplanements for all United States air carriers.

PRIMARY SURFACE - A primary surface is longitudinally centered on a runway. When the runway has a specially prepared hard surface, the primary surface extends 200 feet beyond each end of that runway. When the runway has no specially prepared hard surface, or planned hard surface, the primary surface terminates at each end of the runway. The width of a primary surface ranges from 250 feet to 1,000 feet, depending on the existing or planned approach system. The elevation of any point on the primary surface is the same as the elevation of the nearest point on the runway centerline.

REGIONAL AIRLINE - An airline providing regularly scheduled passenger or cargo service with aircraft usually seating less than 60 passengers or cargo aircraft with 18,000-pound payload or less. Special provisions, however, enable regional airlines to operate any size aircraft under certain conditions.

REGISTERED GENERAL AVIATION AIRCRAFT - A general aviation aircraft registered with FAA.

RELIEVER - Reliever airports are general aviation airports which have the function of relieving congestion at a primary service airport and which provide the general aviation user with an alternate for access to the overall community. Reliever airports receive higher priority for funding assistance than other general aviation airports.

RELOCATED THRESHOLD - A permanent threshold located at the relocated runway end.

RUNWAY - A defined rectangular area, on a land airport prepared for the landing and takeoff run of an aircraft along its length. Runways are normally numbered in relation to their magnetic direction rounded off to the nearest 10 degrees, e.g., Runway 01, Runway 26. (See PARALLEL RUNWAYS.)

RUNWAY GRADIENT (EFFECTIVE) - The average gradient consisting of the difference in elevation of the two ends of the runway divided by the runway length may be used, provided that no intervening point on the runway profile lies more than five feet above or below a straight line joining the two ends of the runway. In excess of five feet, the runway profile will be segmented and aircraft data will be applied for each segment separately.

RUNWAY LENGTH - LANDING - The measured length from the threshold to the end of the runway.

RUNWAY LENGTH - PHYSICAL - The actual measured length of the runway.

RUNWAY LENGTH - TAKEOFF - The measured length from where the takeoff is designed to begin to the end of the runway.

RUNWAY LIGHTING SYSTEM - A system of lights running the length of a runway that may be either high intensity (HIRL), medium intensity (MIRL), or low intensity (LIRL).

RUNWAY SAFETY AREA - An area extending 200 feet beyond each end of a runway and at a lateral distance that is determined by the runway classification and type of approach, which is intended to accommodate the occasional passage of aircraft as well as emergency and snow removal vehicles.

SEGMENTED CIRCLE - A system of visual indicators designed to provide traffic pattern information at an airport without an operating control tower.

SMALL AIRCRAFT - Aircraft of 12,500 pounds or less, maximum certificated takeoff weight.

STOPWAY - An area beyond the takeoff runway, no less wide than the runway and centered upon the extended centerline of the runway, able to support an airplane during an aborted takeoff without causing structural damage to the airplane, and designated by the airport authorities for use in decelerating an airplane during an aborted takeoff.

T-HANGAR - A T-shaped aircraft hangar that provides shelter for a single plane.

TAXI - The movement of an airplane under its own power on the surface of an airport. Also, it describes the surface movement of helicopters equipped with wheels.

TAXILANE - The portion of the aircraft parking area used for access between taxiways, aircraft parking positions, hangars, storage facilities, etc. A taxilane is outside the movement area.

TAXIWAY - A defined path, from one part of an airport to another, selected or prepared for the taxiing of aircraft.

TERMINAL AREA - The area used or intended to be used for such facilities as terminal and cargo buildings, gates, hangars, shops, other service buildings, automobile parking, airport motels, restaurants, garages, and automobile service.

THRESHOLD - The beginning of that portion of the runway available and suitable for the landing of airplanes.

TIE-DOWN AREA - A parking area for securing aircraft; can be for overnight (transient operator) or permanent use (in lieu of a hangar).

TOUCH-AND-GO/TOUCH-AND-GO LANDING - An operation by an aircraft that lands and departs on a runway without stopping or exiting the runway.

TRANSIENT - Operations or other activity performed by aircraft not based at the airport.

TRANSITIONAL SURFACE - An element of the imaginary surfaces extending outward and upward at right angles to the runway centerline and runway centerline extended at a slope of 7:1 from the sides of the primary and approach surfaces to where they intersect the horizontal and conical surfaces.

ULTRALIGHT VEHICLE - An aeronautical vehicle operated for sport or recreational purposes which does not require FAA registration, an airworthiness certificate, nor pilot certification. They are primarily single-occupant vehicles, although some two-place vehicles are authorized for training purposes. Operation of an ultralight vehicle in certain airspace requires authorization from ATC.

UTILITY RUNWAY - A runway intended to be used by propeller-driven aircraft of 12,500 pounds maximum gross weight or less.

VISUAL APPROACH RUNWAY - A runway intended for visual approaches only, with no straight-in instrument approach procedure either existing or planned for that runway.

WIND COVERAGE - The percent of time for which aeronautical operations are considered safe due to acceptable crosswind components.

WIND ROSE - A graphic depiction of historical prevailing wind patterns by speed and direction at a given location. A series of concentric circles cut by radial lines indicates the average percentage of time during the observation period that winds were occurring at successive wind speed groupings and by true direction. Wind rose data are used primarily for determining optimal runway alignment for wind coverage.

AIRPORT NOISE, ENVIRONMENTAL AND LAND-USE PLANNING

AIRPORT APPROACH SAFETY ZONE - An element of either an Airport Impact Zone or an Airport Overlay Zone which consists of a portion of the Airport Approach Surface as defined in FAR Part 77. The actual boundaries and land-use provisions are determined by the local jurisdiction.

AIRPORT DEVELOPMENT ZONE - A zone which replaces the existing zoning for the airport property encompassing the land presently owned by the airport and, if feasible, areas identified for future purchase, clear zones, and areas with noise levels greater than Ldn 70.

AIRPORT ENVIRONS - The land uses and people in the area surrounding an airport which can be directly affected by the operation of the airport.

AIRPORT IMPACT ZONE - A zone used to place land-use conditions on land impacted by airport operations. It establishes a new zone and provisions which replace an existing zone and standards.

AIRPORT NOISE ABATEMENT PROGRAM - A program designed to mitigate noise impacts around an airport through changes in the manner in which aircraft are flown, or changes in the operation or layout of the airport.

AIRPORT NOISE COMPATIBILITY PROGRAM - That program reflected in documents (and revised documents) developed in accordance with Appendix B of FAR Part 150, including the measures proposed or taken by the airport operator to reduce existing noncompatible land uses and to prevent the introduction of additional noncompatible land uses within the area.

AIRPORT OBSTRUCTION ZONING ORDINANCE - A local height restriction ordinance which follows FAR Part 77, implements a local community's comprehensive plan, and provides specific height standards for the area beneath the airport imaginary surfaces.

AIRPORT OVERLAY ZONE - A zone intended to place additional land use conditions on land impacted by the airport while retaining the existing underlying zone.

AMBIENT NOISE - All encompassing noise associated with a given environment, being usually a composite of sounds from many sources near and far.

AVIATION NOISE ABATEMENT POLICY (ANAP) - Policy adopted jointly by the Secretary of Transportation and the FAA on November 18, 1976, delineating the responsibilities of the FAA, airport operators, and local communities in achieving reductions in airport noise.

AVIATION SAFETY AND NOISE ABATEMENT ACT OF 1979 (ASNA) - Established airport noise compatibility planning policies incorporated into FAR Part 150. Amended by the Airport and Airway Improvement Act of 1982. (See FAR PART 150.)

AVIGATION AND HAZARD EASEMENT An easement which provides right of flight at any altitude above the approach surface, prevents any obstruction above the approach surface, provides a right to cause noise vibrations, prohibits the creation of electrical interferences, and grants right-of-way entry to remove trees or structures above the approach surface.

COMPREHENSIVE PLAN - A set of public decisions dealing with how the land, air, and water resources of an area are to be used. The plan provides for all resources, uses, public facilities, and services in an area. It also incorporates the plans and programs of the various governmental units into a single management tool for the planning area.

CONDITIONAL USE - A land-use regulatory procedure in which an applicant must adhere to "standards for approval" as established by local officials. A conditional-use procedure allows extensive public review of any development being considered.

EASEMENT - The legal right held by one party to make use of the land of another for a limited purpose.

ENVIRONMENTAL ASSESSMENT (EA) - A concise public document, prepared under the guidelines of the National Environmental Policy Act of 1969, and for which a federal agency is responsible that serves to:

1. Briefly provide sufficient evidence and analysis for determining whether to prepare an environmental impact statement or a finding of no significant impact.
2. Aid an agency's compliance with the Act when no environmental impact statement is necessary.
3. Facilitate preparation of a statement when one is necessary.

It includes brief discussions of the need for the proposal, of alternatives as required, of the environmental impacts of the proposed action and alternatives, and a listing of agencies and persons consulted.

FEDERAL AVIATION REGULATIONS (FAR), PART 36 - FAR Part 36 contains noise certification standards for most airplane types, generally requiring newly designed and manufactured aircraft to be significantly quieter than older aircraft.

FEDERAL AVIATION REGULATIONS (FAR), PART 150 - Implements portions of Title I of the Aviation Safety and Noise Abatement Act. It specifically establishes a single system for the measurement of airport (and background) noise, a single system for determining the exposure of individuals to airport noise, and a standardized airport noise compatibility planning program.

HOLD HARMLESS AGREEMENT - An agreement which holds airport sponsors or jurisdictions harmless from alleged damages resulting from airport operations. Such agreements are recorded in deeds or permits as a condition of approval of a regulatory land-use decision.

INTEGRATED NOISE MODEL (INM) - The FAA's Integrated Noise Model is the standard prediction analysis tool to which all computer-based airport noise exposure models are compared. The INM calculates the total impact of aircraft noise at or around airports. This noise exposure level can be presented in contours of equal noise exposure for any one of the following noise measures, Noise Exposure Forecast (NEF), Equivalent Sound Level (Leq), Day-Night Average Sound Level (Ldn), and Community Noise Equivalent Level (CNEL); however, only the Ldn is approved for use with FAR Part 150.

LAND USE - The present or planned utilization of a given parcel of land. Such land uses are normally indicated or delineated on a land-use map. Land-use maps may indicate usages for any given time period past, present, or future, and such period should always be indicated.

LIMITED AVIGATION EASEMENT - An easement which provides right of flight above approach slope surfaces, prohibits any obstruction penetrating the approach slope surface, and provides right of entry to remove any structure or growth penetrating the approach slope surface.

NATIONAL ENVIRONMENTAL POLICY ACT OF 1969 (NEPA) - The basic national charter for protection of the environment, establishing policy, setting goals, and providing means for carrying out policy. NEPA encourages and facilitates public involvement in decisions which affect the quality of the human environment, and provides a process to identify and assess the reasonable alternatives to proposed actions that will avoid or minimize adverse effects of these actions. Specifically, it mandates when and how to prepare an environmental assessment study or environmental impact statement. Airport planning/development compliance with NEPA is controlled by FAA Order 1050.1D, Policies and Procedures for Considering Environmental Impacts.

NOISE CONTOURS - A noise impact boundary line connecting places on a map where the level of sound is the same. Some of the terminology and methods used in developing noise contours include:

1. **A-Weighted Sound Level (dBA):** Commonly used sound measurement which approximates the manner in which the human ear responds to sounds.
2. **Composite Noise Rating (CNR):** A measure, taken over a 24-hour period, of the noise environment produced by aircraft operations. The CNR is calculated from aircraft noise and is expressed in terms of the maximum perceived noise level (PNL) and the number of operations in daytime and nighttime periods.
3. **Day-Night Average Sound Level (Ldn):** Equivalent noise level produced by airport/aircraft operations during a 24-hour time period, with a 10-decibel penalty applied to the level measured during the nighttime hours of 10 pm to 7 am.
4. **Equivalent Sound Level (Leq):** The constant sound level which, in a given situation and time period, conveys the same sound energy as does the actual time-varying sound in the same period. The equivalent sound level is the same as the average sound level.
5. **Noise Exposure Forecast (NEF):** A measure of the noise environment over a 24-hour period. It is based upon summation of individual noise events over the 24-hour period, with adjustments applied for nighttime noises.

NOISE EXPOSURE AREA/ZONE - An element of the airport impact zone or airport overlay zone used to identify areas sensitive to aircraft noise. Included in the corridor may be:

Severe Noise Impact (Ldn 75+)
Substantial Noise Impact (Ldn 65-75)
Moderate Noise Impact (Ldn 55-65)

NOISE EXPOSURE MAP - A scaled, geographic depiction of an airport, its noise contours, and its surrounding area developed in accordance with FAR Part 150, including the accompanying documentation setting forth the required descriptions of projected aircraft operations at that airport in the current year and during the fifth calendar year beginning after submission of the map, together with the ways, if any, those operations for each of those years will affect the map (including noise contours and the forecast land uses).

NOISE IMPACT - A condition that exists when the noise levels that occur in an area exceed a level identified as appropriate for the activities in that area.

PROPONENT - Any person who proposes to erect or construct any object or structure that exceeds certain minimum altitudes that may be a potential hazard to air navigation and who may be responsible for lighting and marking such object or structure.

NOISE LEVEL REDUCTION (NLR) - The amount of noise level reduction achieved through incorporation of noise attenuation (between outdoor and indoor levels) in the design and construction of a structure.

URBANIZED LAND - Lands within the urban growth boundary (UGB) which are (a) determined to be necessary and suitable for future urban areas, (b) served by urban services and facilities, and (c) needed for the expansion of an urban area.

ZONING - The demarcation of a jurisdiction by ordinance into zones and the establishment of regulations to govern the use of the land and the location, height, use, and coverage of structures within each zone.

AIR NAVIGATIONAL AIDS, AIRSPACE, AND AIR TRAFFIC CONTROL

AIR NAVIGATION FACILITY (NAVAID) - Any structure or mechanism used by pilots for guiding or controlling flight in the air, or the landing and takeoff of an aircraft.

AIR ROUTE TRAFFIC CONTROL CENTER (ARTCC) - A facility established to provide positive air traffic control service to aircraft operating on IFR flight plans within controlled airspace. Controls only traffic en route between airports.

AIRPORT SURVEILLANCE RADAR (ASR) - Approach control radar used to detect and display an aircraft's position in the terminal area. ASR provides range and azimuth information but does not provide elevation data. Coverage of the ASR can extend up to 60 miles.

AIRSPACE - Space above the ground in which aircraft travel; divided into corridors, routes, and restricted zones.

APPROVED INSTRUMENT APPROACH - An instrument approach approved for general use and publication by the FAA. It must meet design, accuracy, and equipment requirements set by the FAA, and is subject to periodic FAA flight checks.

AREA NAVIGATION (RNAV) - A method of navigation that permits aircraft operation on any desired course within the coverage of station-referenced navigation signals or within the limits of a self-contained system capability.

AUTOMATIC DIRECTION FINDER (ADF) - An aircraft radio navigation system which senses and indicates the direction to a nondirectional radio beacon (NDB) ground transmitter. Direction is indicated to the pilot as a magnetic bearing or as a relative bearing to the aircraft, depending on the type of indicator installed in the aircraft.

AZIMUTH - Horizontal direction or bearing; usually measured from the reference point of 0 degrees clockwise through 360 degrees.

BEARING - The horizontal direction to or from any point, usually measured clockwise from true north, magnetic north, or some other reference point, through 360 degrees.

CONTROLLED AIRSPACE - Airspace designated as a control zone, airport radar service area, terminal control area, transition area, control area, continental control area, or positive control area within which some or all aircraft may be subject to air traffic control. Types of controlled airspace in the vicinity of airports are as follows:

1. Control Zone: Controlled airspace which extends upward from the surface and terminates at the base of the continental control area. Control zones that do not underlie the continental control area have no upper limit. A control zone may include one or more airports and is normally a circular area within a radius of 5 statute miles and any extensions necessary to include instrument approach and departure paths.
2. Airport Radar Service Area (ARSA): Regulatory airspace surrounding designated airports wherein ATC provides radar vectoring and sequencing on a full-time basis for all IFR and VFR aircraft. The service provided in an ARSA includes: IFR/IFR - standard IFR separation; IFR/VFR - traffic advisories and conflict resolution; and VFR/VFR - traffic advisories and, as appropriate, safety advisories. The ARSAs are depicted on VFR aeronautical charts.
3. Terminal Control Area (TCA): Controlled airspace extending upward from the surface or higher to specified altitudes, within which all aircraft are subject to operating rules and pilot and equipment requirements specified in FAR Part 91. TCAs are depicted on aeronautical charts.
4. Transition Area: Controlled airspace extending upward from 700 feet or more above the surface of the earth when designated in conjunction with an airport for which an approved instrument approach procedure has been prescribed, or from 1,200 feet or more above the surface of the earth when designated in conjunction with airway route structures or segments.
5. Control Area: Airspace designated as Colored Federal Airways, VOR Federal Airways, control areas associated with jet routes outside the continental control area, additional control areas, control area extensions, and area low routes.

COURSE -

1. The intended direction of flight measured in degrees from north.
2. The ILS localizer signal pattern usually specified as front course or back course.

DECISION HEIGHT (DH) - With respect to the operation of aircraft, DH means the height at which a decision must be made during an ILS instrument approach to either continue the approach or to execute a missed approach.

DISTANCE MEASURING EQUIPMENT (DME) - A navigation ground station capable of receiving interrogations from aircraft and transmitting signals which allow time, speed, and distance computations to be made. The station is usually sited with VOR and, at times, an ILS.

FEDERAL AVIATION REGULATIONS (FAR), PART 77 - Part 77, **Objects Affecting Navigable Airspace**, establishes standards for determining obstructions to navigable airspace.

FINAL APPROACH COURSE - A straight-line extension of a localizer, a final approach radial/bearing, or a runway centerline, all without regard to distance.

GLIDE SLOPE (GS) - Provides vertical guidance for aircraft during approach and landing. The glide slope consists of the following:

1. Electric components emitting signals which provide vertical guidance by reference to airborne instruments during instrument approaches such as an ILS, or
2. Visual ground aids, such as VASI, which provide vertical guidance for a VFR approach or for the visual portion of an instrument approach and landing.

HEIGHT ABOVE TOUCHDOWN (HAT) - The height of the decision height or minimum descent altitude above the highest runway elevation in the touchdown zone (first 3,000 feet of the runway). HAT is published on instrument approach charts in conjunction with all straight-in minima. (See **DECISION HEIGHT, MINIMUM DESCENT ALTITUDE.**)

IFR CONDITIONS - Weather conditions below the minimum for flight under visual flight rules (VFR).

INSTRUMENT APPROACH - The act of making an approach to an airport solely by reference to instruments. To be counted in FAA statistics as an instrument approach, the aircraft must descend through clouds at some interval from the initial approach fix to the airport.

INSTRUMENT APPROACH AID - Any of several FAA-approved electronic aids designed to provide guidance to pilots making instrument approaches.

INSTRUMENT FLIGHT RULES (IFR) - Rules governing the procedures for conducting instrument flight. Pilots are required to follow these rules when operating in controlled airspace with a visibility of less than three miles and/or a ceiling lower than 1,000 feet.

INSTRUMENT LANDING SYSTEM (ILS) - The instrument landing system is designed to provide electronic instrument guidance to the pilot to permit exact alignment and angle of descent of a properly equipped aircraft on final approach for landing.

GLOSSARY - Airport Navigational Aids, Airspace, and
Air Traffic Control

INSTRUMENT OPERATION - A takeoff or landing of an aircraft which has an instrument flight clearance.

LIGHTING AND MARKING OF HAZARDS TO AIR NAVIGATION - Installation of appropriate lighting fixtures, painted markings, or other devices to such objects or structures that constitute hazards to air navigation.

LOCALIZER - The component of an ILS which provides course guidance to the runway.

LOCALIZER TYPE DIRECTIONAL AID (LDA) - A NAVAID used for nonprecision instrument approaches with utility and accuracy comparable to a localizer but which is not part of a complete ILS and is not aligned with the runway.

LORAN - An electronic navigational system by which hyperbolic lines of position are determined by measuring the difference in the time of reception of synchronized pulse signals from two fixed transmitters.

MICROWAVE LANDING SYSTEM (MLS) - An advanced form of precision approach equipment with improved accuracy and fewer siting problems than current ILS. An MLS also can permit curved path approaches to the runway instead of requiring a straight path as an ILS and PAR do.

MINIMUM DESCENT ALTITUDE (MDA) - The lowest altitude, expressed in feet above mean sea level, to which descent is authorized on final approach or during circle-to-land maneuvering in execution of a standard instrument approach procedure where no electronic glide slope is provided. (See NONPRECISION APPROACH PROCEDURE.)

MISSED APPROACH -

1. A maneuver conducted by a pilot when an instrument approach cannot be completed to a landing. The route of flight and altitude are shown on instrument approach procedure charts. A pilot executing a missed approach prior to the Missed Approach Point (MAP) must continue along the final approach to the MAP. The pilot may climb immediately to the altitude specified in the missed approach procedure.
2. A term used by the pilot to inform ATC that he is executing the missed approach.
3. At locations where ATC radar service is provide, the pilot should conform to radar vectors, when provided by ATC, in lieu of the published missed approach procedure.

NAVIGATIONAL AID (NAVAID) - Any visual or electronic device airborne or on the surface which provides point-to-point guidance information or position data to aircraft in flight.

NONDIRECTIONAL BEACON/RADIO BEACON (NDB) - An L/MF or UHF radio beacon transmitting nondirectional signals whereby the pilot of an aircraft equipped with direction-finding equipment can determine his bearing to or from the radio beacon and "home" on or track to or from the station. When the radio beacon is installed in conjunction with the instrument landing system marker, it is normally called a compass locator.

NONPRECISION APPROACH PROCEDURE/NONPRECISION APPROACH - A standard instrument approach procedure in which no electronic glide slope is provided; e.g., VOR, TACAN, NDB, LOC, LDA, SDF, or ASR.

NONPRECISION INSTRUMENT APPROACH AID - An electronic aid designed to provide an approach path for aligning an aircraft on its final approach to a runway. It lacks the high accuracy of the precision approach equipment and does not provide descent guidance. The VHF Omnidirectional Range (VOR) and the nondirectional beacon (NDB) are two examples of nonprecision instrument equipment.

PRECISION APPROACH PROCEDURE/PRECISION APPROACH - A standard instrument approach procedure in which an electronic glide slope is provided, e.g., ILS, MLS, and PAR.

PRECISION APPROACH RADAR/PAR - Radar equipment in some ATC facilities operated by the FAA, and/or the military services at joint-use civil/military locations and separate military installations, to detect and display azimuth, elevation, and range of aircraft on the final approach course to a runway. This equipment may be used to monitor certain non-radar approaches, but is primarily used to conduct a precision instrument approach (PAR) wherein the controller issues guidance instructions to the pilot based on the aircraft's position in relation to the final approach course (azimuth), the glide path (elevation), and the distance (range) from the touchdown point on the runway as displayed on the radar scope.

PRECISION INSTRUMENT APPROACH AID - An electronic aid designed to provide an approach path for exact alignment and descent guidance of an aircraft on final approach to a runway. Instrument Landing System (ILS), Precision Approach Radar (PAR), and Microwave Landing System (MLS) are the existing precision NAVAIDS.

RADIAL - A magnetic bearing extending from a VOR/VORTAC/TACAN navigational facility.

ROTATING BEACON - An airport aid allowing pilots the ability to locate an airport while flying under VFR conditions at night.

SPECIAL-USE AIRSPACE - Airspace of defined dimensions identified by an area on the surface of the earth wherein activities must be confined because of their nature and/or wherein limitations may be imposed upon aircraft operations that are not a part of those activities. Types of special-use airspace include:

1. Alert Area: Airspace which may contain a high volume of pilot training activities or an unusual type of aerial activity, neither of which is hazardous to aircraft. Alert areas are depicted on aeronautical charts for the information of nonparticipating pilots.
2. Controlled Firing Area: Airspace wherein activities are conducted under conditions so controlled as to eliminate hazards to nonparticipating aircraft and to ensure the safety of persons and property on the ground.
3. Military Operations Area (MOA): An MOA is an airspace assignment of defined vertical and lateral dimensions established outside positive control areas to separate/segregate certain military activities from IFR traffic and to identify for VFR traffic where these activities are conducted.
4. Prohibited Area: Designated airspace within which the flight of aircraft is prohibited.
5. Restricted Area: Airspace designated under FAR Part 73 within which the flight of aircraft, while not wholly prohibited, is subject to restriction. Most restricted areas are designated joint use and IFR/VFR operations in the area may be authorized by the controlling ATC facility when it is not being utilized by the using agency. Restricted areas are depicted on en route charts.
6. Warning Area: Airspace which may contain hazards to nonparticipating aircraft in international airspace.

SIMPLIFIED DIRECTIONAL FACILITY (SDF) - A NAVAID used for nonprecision instrument approaches. The final approach course is similar to that of an ILS localizer except that the SDF course may be offset from the runway, generally not more than 3 degrees, and the course may be wider than the localizer, resulting in a lower degree of accuracy.

TACTICAL AIR NAVIGATION (TACAN) - An ultra-high frequency electronic air navigation aid which provides suitably equipped aircraft a continuous indication of bearing and distance to the TACAN station. (See VORTAC.)

TERMINAL RADAR APPROACH CONTROL (TRACON) - An FAA traffic control facility using radar and air/ground communications to provide approach control services to aircraft arriving, departing, or transiting the airspace controlled by the facility. Service may be provided to both civil and military airports. A TRACON is similar to a RAPCON (USAF), RATCF (USN), and ARAC (Army).

TERMINAL RADAR SERVICE AREA (TRSA) - Airspace surrounding designated airports wherein ATC provides radar vectoring, sequencing, and separation on a full-time basis for all IFR and participating VFR aircraft. TRSAs are depicted on VFR aeronautical charts. Pilot participation is urged but is not mandatory.

TERMINAL VERY HIGH FREQUENCY OMNIDIRECTIONAL RANGE STATION (TVOR) - An electronic navigation aid that provides guidance, both for enroute flights on low altitude "Victor" airways and for non-precision approaches. (See also NONPRECISION APPROACH AID.)

TOWER/AIRPORT TRAFFIC CONTROL TOWER - A terminal facility that uses air/ground communications, visual signaling, and other devices to provide ATC services to aircraft operating in the vicinity of an airport or on the movement area. Authorizes aircraft to land or take off at the airport controlled by the tower or to transit the airport traffic area regardless of flight plan or weather conditions (IFR or VFR). A tower may also provide approach control services (radar or nonradar).

TRAFFIC PATTERN - The traffic flow that is prescribed for aircraft landing at, taxiing on, or taking off from an airport. The components of a typical traffic pattern are upwind leg, crosswind leg, downwind leg, base leg, and final approach.

1. Upwind Leg: A flight path parallel to the landing runway in the direction of landing.
2. Crosswind Leg: A flight path at right angles to the landing runway off its upwind end.
3. Downwind Leg: A flight path parallel to the landing runway in the direction opposite to landing. The downwind leg normally extends between the crosswind leg and the base leg.
4. Base Leg: A flight path at right angles to the landing runway off its approach end. The base leg normally extends from the downwind leg to the intersection of the extended runway centerline.
5. Final Approach: A flight path in the direction of landing along the extended runway centerline. The final approach normally extends from the base leg to the runway. An aircraft making a straight-in approach VFR is also considered to be on final approach.

UNICOM - A nongovernment communication facility which may provide airport information at certain airports. Locations and frequencies of UNICOMS are shown on aeronautical charts and publications.

VICTOR AIRWAYS - VOR Federal Airways are corridors extending between VOR or VORTAC navigational aids (or intersections) and normally include controlled airspace eight miles wide between 1,200 feet AGL and 18,000 feet MSL. They are depicted on aeronautical charts and are identified by a "V" (Victor) followed by the airway number.

VISUAL DESCENT POINT/VDP - A defined point on the final approach course of a nonprecision straight-in approach procedure from which normal descent from the MDA to the runway touchdown point may be commenced, provided the approach threshold of that runway, or approach lights, or other markings identifiable with the approach end of that runway are clearly visible to the pilot.

VISUAL FLIGHT RULES (VFR) - Rules that govern flight procedures under visual conditions. Also indicates a type of flight plan.

VOR/VERY HIGH FREQUENCY OMNIDIRECTIONAL RANGE STATION - A ground-based electronic navigation aid transmitting very high frequency navigation signals, 360 degrees in azimuth, oriented from magnetic north. Used as the basis for navigation in the National Airspace System. The VOR periodically identifies itself by Morse Code and may have an additional voice identification feature. Voice features may be used by ATC or FSS for transmitting instructions/information to pilots. (See Navigational Aid.)

VORTAC/VHF OMNIDIRECTIONAL RANGE/TACTICAL AIR NAVIGATION - A navigation aid providing VOR azimuth, TACAN azimuth, and TACAN distance measuring equipment (DME) at one site. (See Distance Measuring Equipment, Navigational Aid, TACAN, VOR.)

FORECASTING AND STATISTICAL METHODS

DEPENDENT VARIABLE - The variable that is of interest to the researcher; the variable that is to be forecast. In regression analysis, this is the variable on the left-hand side of the equation.

ECONOMETRIC METHODS - Regression correlation techniques applied to a great variety of forecasting problems to ascertain the relationships between the dependent variables and such explanatory and logically relevant economic variables as income, demographic variables such as population, and other market factors, such as usage impedance and intermodal competition. (See Regression Equation.)

INDEPENDENT VARIABLE - An indicator on the basis of which the dependent variable is projected. The Independent Variable may or may not cause the interval change in a dependent variable with which it is associated.

MARKET SHARE ANALYSIS - Proportions a large-scale activity down to a local level, assuming that the proportion of the large-scale activity which can be assigned to the local level is a regular and predictable quantity. Also known as "ratio" or "top-down" modeling, this method is commonly used to develop microforecasts from the exogenous sources of macroforecasts.

MULTIPLE REGRESSION - A regression model with more than one independent variable. (See REGRESSION EQUATION.)

r - The correlation coefficient. Measures the degree of association or covariance between the independent and dependent variable. Correlation does not equal causation. (See R^2 .)

R^2 - The coefficient of determination. Indicates the percentage variation in the dependent variable that is explained by variations in the causal variables.

REGRESSION EQUATION - A regression equation is the mathematical representation of a regression model. It states that one or more independent variables and a constant term are related to the dependent variable in an additive fashion. The relationship may be linear or one of several curvilinear types.

SIMPLE REGRESSION - Simple regression involves a single independent variable. It assumes a linear relationship between the independent variable and the dependent variable. That relationship is estimated using the method of "least squares" and a set of observed values.

STANDARD ERROR - A measure of the precision of a coefficient. It tells how reliably the relationship has been measured; the standard deviation for a relationship.

TIME SERIES DATA - Data that examine a decision unit at different points in time. Trend extrapolation examines a historical pattern of activity and assumes that those factors which determined the variation in activity level in the past will continue to exhibit similar relationships in the future.

